A simpler algorithm for integer multiplication below $n \log n$, with a machine-checked proof*

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Abstract

We give a simpler algorithm that multiplies two n-bit integers in $O(n(\log n)^{1-\kappa})$ bit operations on a deterministic multitape Turing machine, with $\kappa = 2^{-46}$, together with a machine-checked proof. The only previously known sub-$n \log n$ algorithm [OAI26] obtains its saving from two fixed finite linear networks but embeds them in a Harvey–van der Hoeven-style architecture [HvdH21]: Chinese remaindering, prime-length Gaussian resampling, and Bluestein–Nussbaumer ring convolutions. We keep the two networks and discard the rest. Our reduction runs on a single power-of-two grid: a fixed smooth kernel spreads the input along the higher axes, one complex tensor FFT is applied, and a two-term cyclic recurrence per axis undoes the spreading. A second ingredient, an exact three-layer identity for the butterfly layer of [OAI26], lets that layer permute address bits directly; this removes a quadratic layout bottleneck and improves the exponent from $\kappa = 2^{-182}$ to 2^{-46}. Every component is implemented in exact arithmetic and tested bit for bit, and the complexity bound and network certificates are verified in Lean 4.

*This work was carried out with substantial assistance from Gemini 4 agents, including proof development, auditing, and the Lean 4 formalization; see Section 1.4.

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1 Introduction

1.1 Background

Schönhage and Strassen [SS71] showed in 1971 that two n-bit integers can be multiplied in $O(n \log n \log\log n)$ steps on a deterministic multitape Turing machine, and conjectured that $\Omega(n \log n)$ steps are required. Fürer [Für07, Für09] replaced the $\log\log n$ factor by $2^{O(\log^* n)}$ (see also [DKSS13]), Harvey, van der Hoeven, and Lecerf [HvdHL16] sharpened the base of that exponent, and Harvey and van der Hoeven [HvdH21] achieved $O(n \log n)$. Their algorithm reduces one-dimensional integer convolution to multidimensional discrete Fourier transforms (DFTs) whose side lengths are powers of two. A Chinese remainder (CRT) map embeds the input onto cyclic axes of pairwise coprime prime lengths, and multidimensional Gaussian resampling interpolates from those prime lengths onto slightly larger powers of two.

A recent preprint [OAI26] breaks the $n \log n$ barrier in the multitape Turing machine model, obtaining running time $O(n(\log n)^{1-\kappa})$ with $\kappa = 2^{-182}$. Its power saving originates in two fixed finite linear networks at ground-set parameter $h = 100$ ($m = h^3 = 10^6$), each having W roles whose total rank s of recursive edge calls is strictly smaller than the trivial count Wm (Section 3.4). Unrolling these networks over m-ary digit decompositions yields two sub-linear-overhead tape primitives on arrays of bit volume V.

- The first interchanges two disjoint contiguous u-bit address fields in time $O(Vu^\tau)$ for a fixed $\tau < 1$ [OAI26, Lemma 12].
- The second applies a simultaneous normalized butterfly layer $H_0(u, v) = ((u + v)/2, (u - v)/2)$ to one selected bit in each of $D \le d$ consecutive K-bit address chunks, in time $O(Vd^{\lambda'})$ for a fixed $\lambda' < 1$ [OAI26, Prop. 14].

Around those two primitives, however, [OAI26] retains the full Harvey–van der Hoeven and Bluestein–Nussbaumer architecture [Blu70, NQ78, Nus80, HvdH21]. It combines CRT splitting onto distinct prime axes, Gaussian resampling to powers of two, and Bluestein's chirp reduction of the middle transform to polynomial-ring convolutions over $\mathbb{C}[y]/(y^r + 1)$. That surrounding stack has two major costs. Qualitatively, it introduces prime-modulus arithmetic for the CRT map, grid expansion, $O(d^2)$ adjacent axis-swap passes, and an error amplification proportional to the record count $M = R/r$ (for grid size R). Quantitatively, grouping record-address bits into K-bit chunks by chunk interchanges costs $O(V \cdot PK^{\tau-1})$ at precision P. This cost interacts with the butterfly layer's internal basis-change constraint to cap κ *quadratically* in the bit network's deficit $1 - \tau$.

In a companion manuscript [MS], we introduce a two-dimensional $O(n \log n)$ multiplier that avoids both CRT maps and coprime-length resampling. It interprets the Cooley–Tukey twiddle of $F_{t_1 r}$ as a fractional shift $j_2 \mapsto j_2 + j_1/t_1$ along the second axis and spreads with a real sinc-Gaussian kernel. By Poisson summation, this relates $F_{t_1 r}$ to the tensor DFT $F_{t_1, r}$ on the *same* power-of-two grid up to a two-alias cyclic deconvolution per column, with a separate alternating-sum identity required at the singular Nyquist column.

1.2 Main results

Our contributions are threefold.

(i) *A simpler algorithm* (Sections 4–7; Algorithm 1). The reduction from integer multiplication to the two finite-network primitives of [OAI26] becomes a five-step pipeline on a single power-of-two grid, with no Chinese remainder map, prime lengths, resampling, Bluestein chirp, polynomial-ring arithmetic, or Nyquist special case. The written proof uses nothing from [HvdH21] and only Lemmas 3–4 and Propositions 7–8 of [OAI26] as black boxes (Table 1).

(ii) *A new tool* (Section 5). Three-layer Walsh identities make the simultaneous butterfly layer an exact engine for permuting address-bit positions, which removes the chunk-layout bottleneck of [OAI26]. As a consequence, the saving exponent is limited linearly rather than quadratically by the network deficit and the guard, rising from 2^{-182} to 2^{-46} (Theorem 1; 2^{-56} already with the printed constants of [OAI26, Prop. 14]).

(iii) *A machine-checked proof and an executable artifact* (Theorem 2; Section 1.4 and Appendix E).

Algorithm 1 (at a glance). *Input:* two n-bit integers (if $n < n_*$ for the fixed cutoff n_* of (46), invoke the independent multiplier M directly). Set $b = \lfloor \log_2 n \rfloor$, $N_{\mathrm{dig}} = \lceil n/b \rceil$, least power of two $R = 2^k \geq 2N_{\mathrm{dig}}$, balanced powers of two $\mathbf{t} = (t_1, \ldots, t_d)$ with $t_1 \cdots t_d = R$ ($d = \lfloor k^{\varepsilon} \rfloor$, $t_d = r = 2^{\ell}$), $A_a = \prod_{i<a} t_i$ ($1 \leq a \leq d+1$), and precisions $p, \omega, P, q = P + 2$ via (40) and (24).

1. *Pack:* split the two inputs into radix-2^b digits $x_j^{(0)}, x_j^{(1)} \in [0, 2^b)$ ($0 \leq j < N_{\mathrm{dig}}$) and set $x_j = 2^{-b}(x_j^{(0)} + i x_j^{(1)})$, with $x_j = 0$ for $N_{\mathrm{dig}} \leq j < R$ (Proposition 7.2).
2. *Spread:* pre-scale to $\widetilde{\mathbb{C}}_q = 2^{-q}(\mathbb{Z} + i\mathbb{Z})$ and spread axes $a = d, \ldots, 2$ on the same grid by the kernel $g_a(s) = e^{-\pi i s/t_a} \operatorname{sinc}(s) e^{-\pi s^2/\alpha^2}$ ($\alpha = 2\sqrt{\omega \ln 2/\pi}$) shifted by $\delta_a(\mathbf{j}) = (\sum_{i<a} j_i A_i)/A_a$ (for $\mathbf{j} = (j_1, \ldots, j_d)$, $j = \sum_{i=1}^{d} j_i A_i$, $0 \leq j_i < t_i$) and folded modulo t_a, producing U (Lemma 4.2, Section 4.4).
3. *Tensor FFT:* evaluate $Y \approx F_{\mathbf{t}} U$ on $\widetilde{\mathbb{C}}_q$, executing record-axis rounds via butterfly layers and chunk/bit-reversal layouts $L, (LB)^{-1}$ via exact Walsh-layer positional permutations (Theorems 5.8 and 6.5).
4. *Deconvolve:* for $a = 2, \ldots, d$, solve $c_0 Z_f + c_1 Z_{(f - R/A_a) \bmod R} = Y_f$ on each cycle of length A_a and update $Y \leftarrow Z$ (where $f = \sum_{i=1}^{d} u_i R/A_{i+1}$ with $0 \leq u_i < t_i$, $c_\nu = \hat{g}_a(u_a/t_a - \nu)$ for $\nu \in \{0, 1\}$, and $\hat{g}_a(\xi) = \int_{\mathbb{R}} g_a(s) e^{-2\pi i \xi s}\, ds$), then rescale to $\hat{y} \approx F_R x$ (Lemmas 4.4–4.6, Theorem 4.7).
5. *Multiply and recover:* form $\hat{w} = \overline{\mathcal{Q}_{p+2}(\hat{y} \odot \hat{y})}$ by conjugated pointwise squaring and nearest rounding $\mathcal{Q}_{p+2}$ to $\widetilde{\mathbb{C}}_{p+2}$, repeat Steps 2–4 on $\hat{w}$ to obtain $\hat{z} \approx F_R \hat{w}$, round $-2^{2b-1} R^2 \operatorname{Im} \hat{z}_j$ to integers, and propagate radix-2^b carries (Proposition 7.2).

Inner products: $O((\log n)^2)$-bit integers, delegated to any multiplier with $\mathsf{M}(N) = N(\log N)^{O(1)}$.

Theorem 1. *Algorithm 1, implemented on a deterministic multitape Turing machine with a fixed finite alphabet and a fixed finite number of tapes, multiplies two n-bit integers exactly in worst-case time*

$$O\big(n(\log n)^{1-2^{-46}}\big).$$

Structure. *The outer algorithm uses only power-of-two DFTs on one grid, one fixed convolution kernel per spread axis, and one two-term cyclic recurrence per spread axis; it uses no Chinese remainder map, prime transform length, coprime-length resampling, Fourier-grid expansion, Bluestein chirp, or polynomial-ring convolution. Ordinary products of $O((\log n)^2)$ bits (and inputs below a fixed threshold n_*) are delegated to any independent multiplier with $\mathsf{M}(N) = N(\log N)^{O(1)}$ (for instance that of Schönhage and Strassen [SS71]), with no recursive call to Algorithm 1 itself.*

Theorem 2 (Machine-checked bound)**.** *The $O\big(n(\log n)^{1-2^{-46}}\big)$ multiplication bound of Theorem 1 is formally proved in Lean 4 [dMU21, mC20] in Mathlib's finite-alphabet multi-stack Turing machine model over $\mathbb{N}$ (statement `SubNLogN.Challenge.IntMulSubNLogN` in [Lean26], available in our github repository; Section 8), depending only on Lean's three standard logical axioms (`propext`, `Classical.choice`, `Quot.sound`) and assuming no result of [OAI26].*

Three technical ingredients underlie Theorem 1; Section 2 gives a self-contained overview of each.

(1) **A d-dimensional fractional-shift bridge** (Section 4). For $R = t_1 \cdots t_d = 2^k$ with powers of two $t_a \geq 2$, spreading the axes in descending order $a = d, d-1, \ldots, 2$ yields an exact triangular

Poisson identity $F_{\mathbf{t}}U = \mathcal{A}_2\mathcal{A}_3\cdots\mathcal{A}_d F_R x$ with non-commuting alias operators (Lemma 4.2). Modulating the sinc-Gaussian kernel on axis a by the negative half-bin phase $e^{-\pi i s/t_a}$ shifts all frequency samples $(u_a+\frac{1}{2})/t_a$ away from the Nyquist crossing. For alias-width parameter ω, each two-alias truncation B_a then satisfies $\|\mathcal{A}_a - B_a\|_\infty = O(2^{-\omega})$ and $\|B_a^{-1}\|_\infty \le \frac{4}{7}t_a \le t_a$ *uniformly* in the cycle length $A_a = \prod_{i<a} t_i$ (Lemma 4.4). For target precision p, at working precision $P = O(p + \log R)$ and grid precision $q = P + 2$, the resulting fixed-point bridge approximates $F_R x$ to error $O(2^{-p}\max(1, \|x\|_\infty))$ using $O(d)$ direct axis exposures (Theorem 4.7, Lemma 4.8).

(2) **Exact positional permutations from butterfly layers** (Section 5). With $S = \mathrm{diag}(1, i)$, the 2×2 identities $H_0^3 = \frac{1}{2}H_0$ and $(H_0 S)^3 = \frac{1+i}{4}I$ allow three calls to the full simultaneous butterfly layer to apply H_0 selectively to any subset T of a slice while acting as a scalar multiple of the identity on the complement (Lemma 5.1). Conjugating a recordwise sign diagonal CZ by two selective layers realizes slice-batched controlled address flips $\mathrm{CNOT}^{\otimes s} = 2^s H_T\,\mathrm{CZ}\,H_T$ (Lemma 5.4). Factoring an arbitrary permutation of the $m_0 = \log_2(R/t_d)$ record-address bit positions into two involutions then executes every positional permutation *bit-exactly* in time $O(Vqd^{\lambda'-1})$, lifting precision temporarily to $q + 8d$ fractional bits so intermediate dyadic denominators never truncate (Theorem 5.8).

(3) **Direct complex tensor FFT and exposed network parameters** (Sections 6–7 and Appendix A). Executing the chunk-grouping layout L and natural-order restoration $(LB)^{-1}$ via exact positional permutations eliminates the chunk-exchange bottleneck. This yields a direct decimation-in-frequency complex tensor FFT on the q-grid with error below $12\ell\cdot 2^{-q}$ where $r = t_d = 2^\ell$ (Theorem 6.5), without Bluestein's chirp or ring convolutions. We also re-derive the butterfly layer with exposed leaf, network, and guard parameters (Proposition A.11), certifying $\tau = 1 - 2^{-40}$ and $\sigma = 1 - 2^{-42}$ from the exact $h = 100$ network integers via $\ln 10^6 < 14$ and the guard exponent $C_1 = 11 + 3 = 14$ from the complex-network guard bound $B_{\mathrm{net}} < m^{11}$ (Proposition A.2; cf. [OAI26, §5.4]). At dimension $d = \lfloor k^\varepsilon \rfloor$, this shifts the binding constraint to the linear butterfly-layer gap $\kappa < \varepsilon(1-\lambda') < (1-\sigma)/C_1$, achieving $\kappa = 2^{-46}$.

Remark 1.1 (Printed constants and the Bluestein ceiling). Keeping the structural constants $\beta = \frac{1}{2}$, $\tau = \sigma = 1 - 2^{-50}$, and $C_1 = 20$ printed in [OAI26, Prop. 14] fixed and varying only the outer schedule exponents, our direct-FFT cost analysis already gives $\kappa = 2^{-56}$, the largest dyadic exponent 2^{-j} that our analysis admits with those constants (Section 7). Conversely, if the direct tensor FFT is replaced by a Bluestein middle transform whose layout is executed by chunk interchanges, the saving is capped quadratically at $\kappa < (1-\tau)^2/(C_1\tau) \approx 2^{-83.8}$ even with our certified parameters (largest dyadic value $\kappa = 2^{-84}$). At the parameters of [OAI26], the same Bluestein analysis reproduces $\kappa = 2^{-182}$ (Appendix C).

Remark 1.2 (Inner integer products without CRT). Theorem 1 requires only $\mathsf{M}(N) = N(\log N)^{O(1)}$ for the independent inner multiplier on $O((\log n)^2)$-bit operands, so classical Schönhage–Strassen multiplication [SS71] suffices as a black box. Instantiating M instead with the two-dimensional fractional-shift multiplier of [MS] eliminates Chinese remainder maps and prime lengths from the inner scalar products as well (Section 7). Its written analysis uses faithful phase and filter tables as in Section 4; Theorem 2 uses our independent Lean 4 formalization `IntMulTM.intMul_TM2_nlogn` (Section 8), separate from the Lean development of [MS].

What the written proof depends on. No result of [HvdH21] is used. Our written proof of Theorem 1 uses only two unchanged ingredients of [OAI26] as black boxes. These are its linear-time elementary tape-stream operations ([OAI26, Lemmas 3–4], recorded as Fact 3.3) and its two fixed finite linear networks at $h = 100$ with their finite algebraic rank and endpoint certificates ([OAI26, Props. 7–8], recorded as Fact 3.4). Every other step is either proved in this paper or re-derived in Appendix A

Table 1: Components of the outer reduction surrounding the two $h = 100$ finite networks of [OAI26].

[HvdH21] / [OAI26]	This paper	Where
CRT map onto d distinct odd primes; prime search by trial division	None: one power-of-two length $R = t_1 \cdots t_d = 2^k$	Thm. 1, §4.1
Gaussian resampling prime $\to 2^\ell$ ([HvdH21]; [OAI26, Lemmas 19–20])	Same-grid sinc-Gaussian spread; negative half-bin phase gives one uniform 2-term solver ($\|B_a^{-1}\|_\infty \le \frac{4}{7} t_a$; cf. [MS])	Lemmas 4.2, 4.4; App. B
$O(d^2)$ adjacent axis-swap passes ([OAI26, §9])	$O(d)$ direct axis exposures	Lemma 4.8
$O(k/K)$ chunk-swap layout; quadratic cap $\kappa < (1-\tau)^2/(C_1\tau)$ ([OAI26, Lemma 16])	Exact Walsh-layer positional permutations; linear cap $\kappa < (1-\sigma)/C_1$	Thm. 5.8, §7.4
Bluestein chirp + twist + ring convolution in $\mathbb{C}[y]/(y^r+1)$ ([OAI26, Prop. 18, Lemma 21])	Direct complex tensor FFT; no polynomial-ring multiplication	Thm. 6.5, Cor. 6.6
Middle error amplified by $M = R/r$; $\log_2 M$-bit output widening ([OAI26, Prop. 18])	Error $< 12\ell \cdot 2^{-q}$; no output widening ($\delta_{\mathrm{mid}} = 0$)	Thm. 6.5, Cor. 6.6
$\Theta(rP)$-bit packed products ([OAI26, Lemma 17], invoking [HvdH21])	$O(P^2)$-bit integer products; any $\mathsf{M}(N) = N(\log N)^{O(1)}$ suffices (e.g. [SS71])	Thm. 1, §7.3
Separate precision budgets for resampling, chirp, and ring convolution	Single contract $P = O(p + \log R)$ with one master margin	Thm. 4.7
Main-route chain uses [HvdH21] and [OAI26] Lemmas 3–4, 9–10, 12–13, 15–21, Props. 7–8, 11, 14, 18	Zero results of [HvdH21]; only [OAI26] Lemmas 3–4, Props. 7–8 as black boxes (9–10, 12–13, Props. 11, 14 re-derived in App. A; 15–21, Prop. 18 unused; none in Lean 4)	§3.4, §8, App. A

by following the arguments of [OAI26, §§4–5] with all exponents and guard parameters exposed. This includes the parameter-exposed interchange, packed selected-bit addition, and simultaneous butterfly layer (Appendix A), the d-dimensional bridge (Section 4), the Walsh-layer permutations (Section 5), the direct tensor FFT (Section 6), and the cost assembly (Section 7). Lemmas 15–21 and Proposition 18 of [OAI26] (the synthetic transforms and layout, packed products, ring convolution, resampling, and chirp) are not needed on the main route and are analyzed only for comparison in Appendix C. The Lean 4 proof of Theorem 2 uses neither black box: it constructs and verifies both $h = 100$ networks, their certificates, and all Turing machine stream procedures from scratch.

1.3 Comparison of exponents

Table 2 compares the previous bound of [OAI26] with our algorithm under both the structural constants printed in [OAI26, Prop. 14] and our certified parameter-exposed constants. Throughout, $k = \log_2 R$, the dimension is $d = \lfloor k^\varepsilon \rfloor$, and the chunk width is $K = \lfloor d^c \rfloor$.

Row 1 is the bound of [OAI26] at its own schedule ($\varepsilon = 2^{-75}$, $c = 2^{-56}$). Retuning (ε, c) under the same structural constants, and using $O(d)$ direct axis exposures, already raises the Bluestein route to 2^{-108} at power-of-two schedule exponents (Row 2, Appendix C), while any chunk-exchange layout remains quadratically capped below $\beta(1-\tau)^2/(C_1\tau) \approx 2^{-105.3} < (1-\tau)^2/(C_1\tau) \approx 2^{-104.3}$ by (1). Comparing Rows 2 and 3 isolates the structural effect of our architecture with the structural constants $(\beta, \tau, \sigma, C_1)$ of [OAI26, Prop. 14] held fixed. Replacing Bluestein ring convolutions and chunk-exchange layouts by exact Walsh-layer positional permutations and the direct tensor FFT removes the chunk-layout bottleneck, the saving becomes linear rather than quadratic in the network deficit, and κ rises from 2^{-108} to 2^{-56}. Comparing Rows 3 and 4 isolates the effect of exposing parameters. Taking leaf threshold $\beta = 1/64$, guard exponent $C_1 = 14$ (cf. [OAI26, §5.4]), and the

Table 2: Asymptotic saving κ in $O(n(\log n)^{1-\kappa})$. Row 1 is the bound of [OAI26] (whose parameter choices [OAI26, §9.1] notes are deliberately conservative). Row 2 is proved in Appendix C, and Rows 3 and 4 in Section 7; Row 4 is formally verified in Lean 4 (Theorem 2). Additional comparison rows, including exposed-parameter Bluestein retunings (2^{-85}, 2^{-84}) and $C_1 = 20$ (2^{-47}), appear in Appendix C.

Algorithm	Front end	Middle transform	(β, C_1)	$(1-\tau, 1-\sigma)$	Binding gap	κ
[OAI26]	CRT + resampling	Bluestein + chunk swaps	$(\frac{1}{2}, 20)$	$(2^{-50}, 2^{-50})$	$\varepsilon c(1-\tau)$	2^{-182}
Bluestein, retuned (ε, c)	Frac.-shift bridge	Bluestein + chunk swaps	$(\frac{1}{2}, 20)$	$(2^{-50}, 2^{-50})$	$\varepsilon c(1-\tau)$	2^{-108}
Ours (printed)	Frac.-shift bridge	Direct FFT + Walsh perm.	$(\frac{1}{2}, 20)$	$(2^{-50}, 2^{-50})$	$\varepsilon(1-\lambda')$	2^{-56}
Ours (Thms. 1, 2)	Frac.-shift bridge	Direct FFT + Walsh perm.	$(\frac{1}{64}, 14)$	$(2^{-40}, 2^{-42})$	$\varepsilon(1-\lambda')$	2^{-46}

sharper $h = 100$ network exponents $1-\tau = 2^{-40}$ and $1-\sigma = 2^{-42}$ certified via $\ln 10^6 < 14$ raises κ to 2^{-46}. Even with $C_1 = 14$, $1-\tau = 2^{-40}$, and β retuned, a chunk-exchange Bluestein layout remains quadratically capped below $(1-\tau)^2/(C_1\tau) \approx 2^{-83.8}$ (largest dyadic value $\kappa = 2^{-84}$; Appendix C).

1.4 Formal verification, executability, and use of AI tools

Machine-checked verification as trust anchor. Asymptotic integer-multiplication algorithms combine delicate fixed-point rounding budgets, dyadic guard allocations, sign and conjugation conventions, and sequential tape-head invariants in which subtle mismatches easily escape informal inspection. To remove doubt about hidden hypotheses or bookkeeping gaps in the written proofs, we formalized the $\kappa = 2^{-46}$ complexity bound in Lean 4 (Theorem 2; Section 8). To our knowledge, this is the first machine-checked proof of an $o(n \log n)$ integer-multiplication bound, as well as the first independent machine verification of the $h = 100$ finite-network certificates of [OAI26]. Building on Mathlib, the development comprises roughly 299,000 lines across 1,062 files for the sub-$n \log n$ proof, plus an independent Lean 4 formalization of the two-dimensional multiplier of [MS] (949 files, about 185,000 lines; headline `IntMulTM.intMul_TM2_nlogn`) developed for this paper and kernel-checked as part of it. The companion manuscript [MS] distributes its own, separate Lean development (headline `IntMul.tmIntMul_evalsToInTime_n_log_n`). Our development [Lean26] exports a single closed theorem replayed through the Lean kernel with `leanchecker` and independently checked with `comparator` and `nanoda`, reducing the trust base to Lean's kernel, the Mathlib definitions named in the statement, the three standard axioms, and the 80-line statement file `SubNLogN/Challenge.lean` (its core definitions are quoted in Section 8).

Executability. Every component of Algorithm 1 (the d-dimensional bridge, the Walsh-layer gadgets, the direct tensor FFT, and the parameter-exposed interchange and butterfly-layer engines) is implemented in exact arithmetic in Python and tested bit-for-bit. The sub-$n \log n$ machinery runs end to end at test scale on small test-network fixtures, and the $h = 100$ certificates, including the 140-digit guard constant C_0, are recomputed in exact arithmetic ([Code26], available in our github repository; Appendix E).

Separately, a C/GMP program implements the two-dimensional [MS]-type $O(n \log n)$ multiplier in its windowed sinc-Gaussian variant (`algo=1`: real kernel with additive precision schedule and Nyquist-column alternating-sum identity). This is not Algorithm 1, not the negative-carrier variant, and not the sub-$n \log n$ algorithm. Single-threaded on one Zen 3 core (one vCPU of a lightly loaded KVM guest; measurement details in [CGMP26]), it multiplies two 10^8-bit integers in 92.5 s (excluding 5.5 s table setup), about 153× slower than GMP's `mpz_mul` (0.606 s) on the same operands (162× including tables; 10.4×, 6.31 s vs. 0.606 s, on 64 threads). It agrees bit-for-bit with GMP across all

tested sizes 10^7–10^9 bits. At 10^9 bits it takes 2613 s vs. 8.10 s single-threaded (323× excluding 26.0 s table setup) with 67.3 GiB (about 72 GB) peak RAM (at 10^9 bits the natural threshold selects a second recursion level, $2^{27} \to 2^{14} \to 128$, whose parameters lie outside the regime covered by the written analysis of [MS]; the output is nevertheless bit-exact). Source, script, and raw logs are in [CGMP26]. This shows only that the fractional-shift framework is implementable with modest constants; by contrast, [HvdH21] takes dimension 1729 in its proof and has, to our knowledge, never been executed. Since the sub-$n \log n$ regime begins at astronomical sizes (Section 9.2), Algorithm 1 runs only at test scale (Appendix E); no practical speedup is claimed.

Use of AI tools. This work was developed with substantial assistance from AI agents (Gemini 4 and other AI models), working under the direction of the authors. That assistance spanned the algorithm, the proofs in this paper, the exact-arithmetic reference implementation, the C/GMP implementation of the two-dimensional multiplier and its benchmark harness, and the Lean 4 formalization. The authors have checked the written proofs in this paper. No author has checked the formal Lean proof line by line. Rather, the correctness of the formal development rests on kernel verification by Lean 4 (replayed through the Lean kernel with `leanchecker` and independently checked with `comparator` and `nanoda`), and the authors have checked the formal statement in `SubNLogN/Challenge.lean` and its correspondence to Theorem 1. Agreement among AI systems was used only as a working heuristic during exploration, never as mathematical evidence: every claim in this paper is established by a complete written proof, an explicit citation to the literature, or a kernel-checked formal theorem.

1.5 Organization

Section 2 gives a self-contained technical overview of the algorithm and its analysis. Section 3 fixes notation, the Turing machine and fixed-point grid conventions, DFT identities, and the two black-box facts from [OAI26]. Section 4 develops the d-dimensional fractional-shift bridge, its negative half-bin kernel, its finite-precision contract, and its $O(d)$ axis exposures. Section 5 proves the three-layer selective Walsh identities, slice-batched controlled flips, and exact positional permutations, and Section 6 builds the direct complex tensor FFT. Section 7 analyzes the cost gaps and proves Theorem 1. Section 8 details the Lean 4 formalization (Theorem 2), and Section 9 discusses the exponent ceiling, activation sizes, what the reduction simplifies and at what cost, and open problems. In the appendices, Appendix A proves the parameter-exposed interchange and butterfly-layer bounds and certifies the $h = 100$ network deficits. Appendix B gives deferred bridge proofs (the two carrier impossibility propositions and series evaluation of phase tables), and Appendix C analyzes the Bluestein middle transform and additional parameter rows for comparison. Appendix D records the notation map and statements cited from [OAI26], and Appendix E summarizes the exact-arithmetic reference implementation.

2 Technical overview

This section explains the main ideas of Algorithm 1 and its analysis without tracking explicit precision constants, giving forward pointers to the formal statements in Sections 3–8.

2.1 The algorithm in five steps

Below the fixed cutoff n_* of (46), Algorithm 1 delegates directly to the independent inner multiplier M. For $n \geq n_*$, set $b = \lfloor \log_2 n \rfloor$, $N_{\mathrm{dig}} = \lceil n/b \rceil$, and let $R = 2^k$ be the least power of two with

$R \geq 2N_{\mathrm{dig}}$. Factor $R = t_1 \cdots t_d$ into balanced powers of two $\mathbf{t} = (t_1, \ldots, t_d)$ with $d = \lfloor k^{\varepsilon} \rfloor$, $t_d = r = 2^{\ell}$, $t_i \in \{r/2, r\}$, and $A_a = \prod_{i<a} t_i$ $(1 \leq a \leq d+1)$. Algorithm 1 multiplies two n-bit integers on this single power-of-two grid in five steps.

1. *Complex digit packing* (Proposition 7.2). Split the two inputs into radix-2^b digits $x_j^{(0)}, x_j^{(1)} \in [0, 2^b)$ $(0 \leq j < N_{\mathrm{dig}})$ and pack both into one complex vector $x_j = 2^{-b}(x_j^{(0)} + i x_j^{(1)})$ of length R (padded with zeros for $j \geq N_{\mathrm{dig}}$), reducing integer multiplication to two forward length-R DFTs at working precision $P = \Theta(k) = \Theta(\log n)$ $(q = P + 2)$.
2. *Descending fractional-shift spreading* (Section 4). Pre-scale into $\widetilde{\mathbb{C}}_q$ and spread axes $a = d, \ldots, 2$ in descending order on the same $t_1 \times \cdots \times t_d$ grid by the kernel $g_a(s) = e^{-\pi i s/t_a} \operatorname{sinc}(s) e^{-\pi s^2/\alpha^2}$ $(\alpha = 2\sqrt{\omega \ln 2/\pi}$, $\omega = \Theta(k))$ shifted by $\delta_a(\mathbf{j}) = (\sum_{i<a} j_i A_i)/A_a$ ($\mathbf{j} = (j_1, \ldots, j_d)$, $j = \sum_{i=1}^{d} j_i A_i$, $0 \leq j_i < t_i$) and folded modulo t_a (Definition 4.1, Lemma 4.2; exposures by Lemma 4.8), producing U and absorbing Cooley–Tukey twiddles without Nyquist singularities.
3. *Direct complex tensor FFT* (Sections 5–6). Evaluate $Y \approx F_{\mathbf{t}} U$ on $\widetilde{\mathbb{C}}_q$ directly over $\mathbb{C}$, executing the chunk and bit-reversal layouts $L, (LB)^{-1}$ bit-exactly via three-layer selective Walsh permutations (Theorem 5.8) and the record-axis butterfly rounds with combined twiddles (Theorem 6.5).
4. *Ascending two-term cyclic deconvolution* (Lemmas 4.4–4.6). Invert the non-commuting two-alias operators $B_2, \ldots, B_d$ in ascending order $a = 2, \ldots, d$ by solving $c_0 Z_f + c_1 Z_{(f - R/A_a) \bmod R} = Y_f$ on cycles of length A_a (with $c_\nu = \hat{g}_a(u_a/t_a - \nu)$ for $f = \sum_{i=1}^{d} u_i R/A_{i+1}$ and $\|B_a^{-1}\|_\infty \leq \frac{4}{7} t_a$ uniform in A_a), updating $Y \leftarrow Z$ after each axis, and rescale to obtain $\hat{y} \approx F_R x$.
5. *Conjugated squaring and carry propagation* (Proposition 7.2). Form $\hat{w} = \overline{\mathcal{Q}_{p+2}(\hat{y} \odot \hat{y})}$, repeat Steps 2–4 on $\hat{w}$ to obtain $\hat{z} \approx F_R \hat{w}$, recover the exact integer convolution $x^{(0)} \circledast x^{(1)}$ by rounding $-2^{2b-1} R^2 \operatorname{Im} \hat{z}_j$ via the two-call identity (8), and propagate radix-2^b carries.

2.2 Where the $n \log n$ barrier is broken, and what caps the saving

Because $R = \Theta(n/\log n)$ and working precision $P = \Theta(k) = \Theta(\log n)$ suffices to round the output coefficients to exact integers, the stored array has logical bit volume $V = \Theta(RP) = \Theta(n)$. On a multitape Turing machine, a standard one-dimensional FFT performs $k = \Theta(\log n)$ sequential passes over volume V, costing $\Theta(n \log n)$ steps from tape movement alone.

Harvey and van der Hoeven [HvdH21] reach $O(n \log n)$ by folding a one-dimensional convolution of odd-prime-product length $S = s_1 \cdots s_d$ into d dimensions. A Chinese remainder map splits the length onto the pairwise coprime odd primes $s_1, \ldots, s_d$, Gaussian resampling interpolates each prime axis onto a larger power-of-two length t_i, and multidimensional FFTs evaluate the product. In [HvdH21] the dimension d is a large constant. In [OAI26], by contrast, $d = \lfloor k^{\varepsilon} \rfloor$ grows with n, and two fixed finite linear networks at $h = 100$ ($m = h^3 = 10^6$; Fact 3.4) give a power saving across the d axes.

(i) A *bit-shear network* of W_b roles with $s_b < W_b m^{\tau}$ $(\tau < 1)$ interchanges two disjoint contiguous u-bit address fields in time $O(Vu^{\tau})$ rather than $O(Vu)$ (Lemma A.8).

(ii) A *complex phase network* of W_c roles with $s_c < W_c m^{\sigma}$ $(\sigma < 1)$ executes the normalized butterfly $H_0 = \frac{1}{2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ simultaneously on one selected bit ρ of each of $D \leq d$ consecutive K-bit address chunks $(K = \lfloor d^c \rfloor)$ in time $O(Vd^{\lambda'})$ with $\lambda' < 1$, rather than $O(Vd)$ (Proposition A.11).

Since $\ell = \lceil k/d \rceil$ bit-levels across $d - 1$ record axes then cost $O(V\ell d^{\lambda'}) = O(Vkd^{\lambda'-1})$, one would hope for an exponent saving $\kappa \approx \varepsilon(1 - \lambda')$ linear in the network deficit $1 - \sigma$.

The chunk-layout cap. Why does [OAI26] instead obtain $\kappa = 2^{-182}$? Before the simultaneous butterfly layer can act, the $m_0 = \log_2(R/r)$ record-address bits, initially stored axis by axis, must be rearranged by a layout permutation L that interleaves K-bit chunks across axes. Executing L via

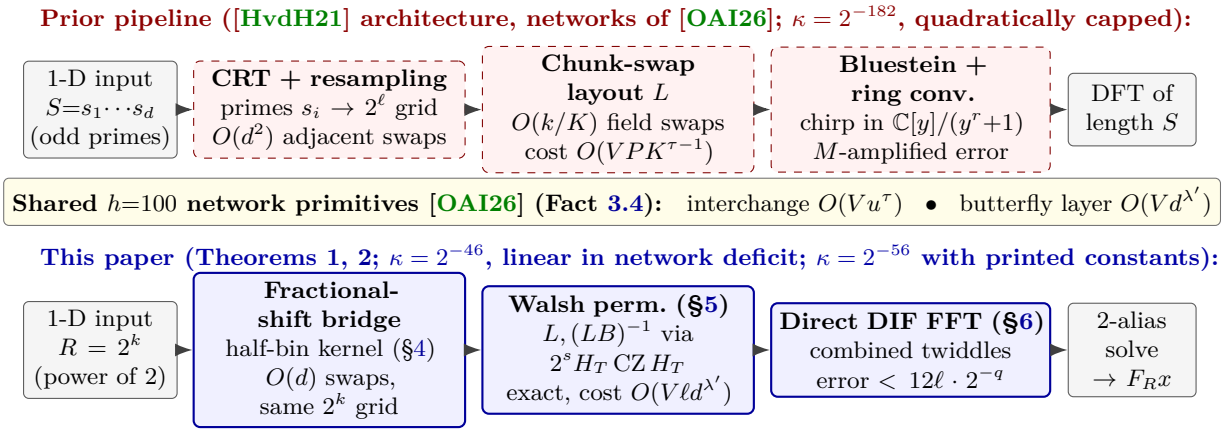


Figure 1: Comparison of sub-$n \log n$ multiplication pipelines. Both build on the same two finite-network primitives of [OAI26], but we replace the CRT/resampling front end, the chunk-swap address layout, and the Bluestein ring-convolution middle transform. Boxes removed by this paper: CRT map, prime lengths, Gaussian resampling, chunk-swap layout, Bluestein chirp, ring convolution; added: a fixed-kernel spread, an exact Walsh-layer permutation, and a two-term deconvolution.

$O(k/K)$ width-K chunk interchanges costs $O(V \cdot (k/K)K^\tau) = O(VPK^{\tau-1})$, whose saving factor $K^{1-\tau} = \Theta(P^{\varepsilon c(1-\tau)})$ requires c to be large. Yet inside the butterfly layer recursion (stopping below d^β chunks), moving K-bit chunks into aligned basis positions forces $\tau(1 + c/\beta) < \lambda < 1$, hence $c < \beta(1-\tau)/\tau$. Meanwhile, keeping the internal precision guard $\Delta = \lceil C_0 d^{C_1} \rceil$ sublinear in P forces $\varepsilon < 1/C_1$. Multiplying these bounds traps any chunk-exchange layout beneath a *quadratic* ceiling:

$$\kappa < \varepsilon c(1-\tau) < \frac{\beta(1-\tau)^2}{C_1 \tau} < \frac{(1-\tau)^2}{C_1 \tau}. \tag{1}$$

Even with our certified exponent $1 - \tau = 2^{-40}$ and $C_1 = 14$, (1) caps κ below $2^{-83.8}$; at the schedule $\varepsilon = 2^{-75}$, $c = 2^{-56}$ of [OAI26, (38)] the first term of (1) is 2^{-181}, giving 2^{-182}, while the cap itself is $\approx 2^{-104}$ at the printed structural constants. Figure 1 contrasts the architecture of [OAI26] with ours, highlighting the stages we remove and add.

2.3 A CRT-free front end: what the bridge removes and how

Cooley–Tukey twiddles as fractional shifts. Consider first dimension two [MS]: factor $R = t_1 r$ into powers of two, write input indices as $j = j_1 + t_1 j_2$ and reversed-digit frequencies as $f = u_1 r + u_2$ $(0 \le j_1, u_1 < t_1,\ 0 \le j_2, u_2 < r)$. Expanding the normalized transform $(F_R x)_f = \frac{1}{R} \sum_{j=0}^{R-1} x_j e^{2\pi i j f / R}$ yields

$$(F_R x)_{u_1 r + u_2} = \frac{1}{t_1} \sum_{j_1=0}^{t_1-1} e^{2\pi i j_1 u_1 / t_1} \left(\frac{1}{r} \sum_{j_2=0}^{r-1} x_{j_1 + t_1 j_2} \, e^{2\pi i (j_2 + j_1/t_1) u_2 / r} \right). \tag{2}$$

The inner sum is the length-r DFT of row j_1, except that every sample is evaluated at the *fractional shift* $j_2 + \delta_2$ with $\delta_2 = j_1/t_1 \in [0, 1)$. Rather than multiplying by twiddles in frequency (which couples the axes), we spread each row j_1 in the time domain by a localized kernel $g(s)$ folded modulo r at offset δ_2. By Poisson summation, the 2-D tensor DFT $F_{t_1, r} U$ of the spread array U is a weighted sum of aliases of $F_R x$ on the *exact same* $t_1 \times r$ grid:

$$(F_{t_1, r} U)_{u_1, u_2} = \sum_{\nu \in \mathbb{Z}} \hat{g}\Big(\frac{u_2}{r} - \nu \Big) \, (F_R x)_{((u_1 - \nu) \bmod t_1)\, r + u_2}.$$

For the sinc-Gaussian base kernel $g_0(s) = \mathrm{sinc}(s)e^{-\pi s^2/\alpha^2}$ ($\alpha \asymp \sqrt{\omega}$ for alias-width parameter ω), the Fourier transform $G(\xi) = \int_{\xi-1/2}^{\xi+1/2} \alpha e^{-\pi\alpha^2 v^2}\, dv$ is a smoothed indicator of $[-\frac{1}{2}, \frac{1}{2}]$. For $\xi \in (0,1)$, all aliases $\nu \notin \{0,1\}$ lie in Gaussian tails of total weight $O(2^{-\omega})$, leaving a two-term cyclic bidiagonal system $c_0 Z_{u_1} + c_1 Z_{u_1-1} = Y_{u_1}$ per column.

The Nyquist singularity and negative half-bin modulation. If the spreading kernel is real-valued, conjugate symmetry $\hat{g}(-\xi) = \overline{\hat{g}(\xi)}$ forces $|c_0| = |\hat{g}(\frac{1}{2})| = |\hat{g}(-\frac{1}{2})| = |c_1|$ at the Nyquist sample $u_2 = r/2$ ($\xi = \frac{1}{2}$). On a cycle of length L, the bidiagonal matrix $c_0 I + c_1 T$ is then singular whenever g is even and L is even (since $c_0 = c_1$), and for *any* real Schwartz kernel its inverse has ℓ^∞ norm $\Omega(L)$ whenever invertible (Proposition B.2). In dimension two, $L = t_1$ and [MS] handles the single Nyquist column via a separate alternating-sum identity. In d dimensions, however, cycle lengths reach $L = A_a = \prod_{i<a} t_i \le R/r$, so an $\Omega(L)$ conditioning loss on each of the $d-1$ spread axes would cost $\sum_{a=2}^{d} \log_2 A_a = \Theta(dk)$ extra bits of working precision. That overhead is superlinear in k once $d \to \infty$, destroying the $O(k)$-bit budget on which $V = \Theta(n)$ rests.

We eliminate the singularity uniformly across all dimensions and cycle lengths by modulating the kernel on each axis $a \in \{2, \dots, d\}$ with a *negative half-bin phase*:

$$g_a(s) = e^{-\pi i s/t_a}\ \mathrm{sinc}(s)\, e^{-\pi s^2/\alpha^2}, \qquad \hat{g}_a(\xi) = G\Big(\xi + \frac{1}{2t_a}\Big). \tag{3}$$

This shifts the frequency samples from integer bins u_a/t_a to half-bin midpoints $y = (u_a + \frac{1}{2})/t_a \in (0,1)$ (Figure 2). Because $t_a \ge 2$ is even, no integer $u_a \in [t_a]$ ever lands on the crossing point $y = \frac{1}{2}$ (which corresponds to $u_a = (t_a - 1)/2 \notin \mathbb{Z}$). Every sample satisfies $|y - \frac{1}{2}| \ge \frac{1}{2t_a}$, ensuring a separation $|c_0 - c_1| = |G(y) - G(y-1)| \ge \frac{7}{4t_a}$ and an inverse norm

$$\|B_a^{-1}\|_\infty \le \frac{1}{|c_0 - c_1|} \le \frac{4}{7}\, t_a \le t_a$$

completely independent of the cycle length $L = A_a$ (Lemma 4.4). The negative sign in (3) is essential: a positive carrier $e^{+\pi i s/t_a}$ shifts $u_a = 0$ to $y = -\frac{1}{2t_a} \notin (0,1)$, encroaching on the omitted $\nu = -1$ passband and degrading the alias error from $O(2^{-\omega})$ to $\Omega(\omega^{-1/2} 2^{-\omega/4})$ at $t_a = 2$ (Proposition B.1). Because $e^{-\pi i s/t_a}$ is anti-periodic of period t_a, periodizing g_a folds g_0 with alternating signs $(-1)^{(e-s)/t_a}$.

Triangular non-commuting factorization in d dimensions. For general $R = t_1 \cdots t_d = 2^k$, put $A_a = \prod_{i<a} t_i$, write time indices as $j = \sum_{i=1}^{d} j_i A_i$ (j_1 least significant) and frequencies in reversed digit order $f(\mathbf{u}) = \sum_{i=1}^{d} u_i R/A_{i+1}$ (u_d least significant). On axis $a \ge 2$, the fractional shift $\delta_a(\mathbf{j}) = (\sum_{i<a} j_i A_i)/A_a \in [0,1)$ depends on the *original* digits below a. We therefore spread the axes in *descending* order $a = d, d-1, \dots, 2$, forming $U = \mathcal{S}_2 \mathcal{S}_3 \cdots \mathcal{S}_d x$, so that whenever $\mathcal{S}_a$ acts, coordinates $1, \dots, a-1$ are still untouched. Dual to this in frequency space, shifting f by $-\nu R/A_a$ modifies only the more significant frequency digits $u_1, \dots, u_{a-1}$ while leaving $u_a, \dots, u_d$ fixed. Consequently, Poisson summation along all $d-1$ spread axes factors into an exact *triangular* operator product (Lemma 4.2):

$$F_{\mathbf{t}} U = \mathcal{A}_2 \mathcal{A}_3 \cdots \mathcal{A}_d (F_R x), \qquad (\mathcal{A}_a Z)_f = \sum_{\nu \in \mathbb{Z}} G\Big(\frac{u_a(f) + 1/2}{t_a} - \nu\Big)\, Z_{(f - \nu R/A_a) \bmod R}. \tag{4}$$

The operators $\mathcal{A}_a$ do not commute, so deconvolution applies the two-alias cyclic inverses $B_2^{-1}, B_3^{-1}, \dots, B_d^{-1}$ in ascending order. Because $\|(B_2 \cdots B_d)^{-1}\|_\infty \le (4/7)^{d-1}(R/t_1) \le R/t_1$,

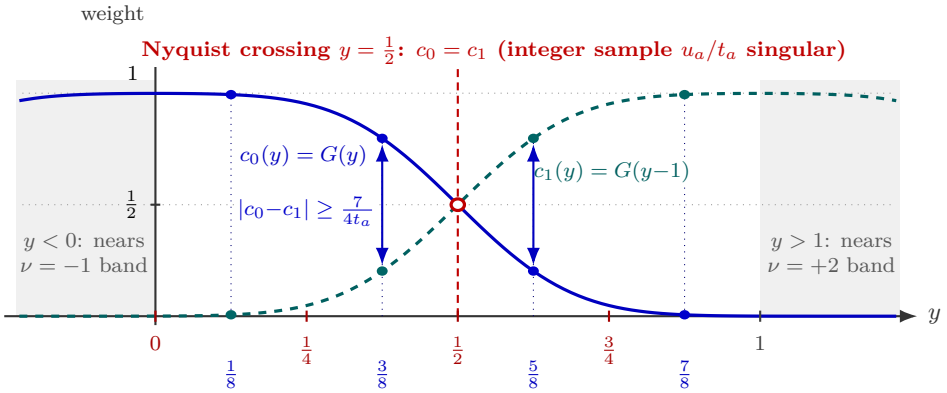


Figure 2: Retained alias weights $c_0(y) = G(y)$ and $c_1(y) = G(y-1)$ plotted at $\omega = 8$ for $t_a = 4$. Integer-bin samples $u_a/t_a \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$ (red ticks) include the Nyquist crossing $y = \frac{1}{2}$ (at $u_a = t_a/2$), where $c_0 = c_1$, making the two-alias operator singular. Negative half-bin samples $y = (u_a + \frac{1}{2})/t_a \in \{\frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}\}$ (solid dots) stay strictly inside $(0, 1)$ and at distance $\geq \frac{1}{2t_a}$ from $\frac{1}{2}$, giving $|c_0 - c_1| \geq \frac{7}{4t_a}$ and uniform conditioning $\|B_a^{-1}\|_\infty \leq \frac{4}{7}t_a$.

working precision $P = \Theta(p + \log R)$ and grid precision $q = P + 2$ control both alias and fixed-point errors to $O(2^{-p} \max(1, \|x\|_\infty))$ (Theorem 4.7). Finally, exposing each axis a by a single direct field interchange with the innermost axis d rather than a chain of adjacent swaps reduces the exposure overhead from $O(d^2)$ to $O(d)$ interchanges of cost $O(Vd(1 + \ell^\tau))$ (Lemma 4.8).

2.4 Butterfly layers as exact positional permutations

The central new idea of our middle transform is that the simultaneous butterfly layer of Proposition A.11 can itself execute *any* permutation of the bit positions of a binary record address, bit-exactly and at the cost of $O(1)$ butterfly layers per slice.

Selective Walsh layers via three-layer cancellation. Consider an array of 2^{m_0} disk-valued records on the q-grid ($m_0 = \log_2(R/r) = O(q)$), arranged so that a *slice* selects one bit position from each of $D \leq d$ consecutive K-bit address chunks. One black-box call to Proposition A.11 applies $H_0 = \frac{1}{2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ to *all* D slice positions simultaneously (H_{all}); we cannot disable individual chunks inside its recursive complex-network schedule. Observe, however, two 2×2 identities involving the phase gate $S = \text{diag}(1, i)$:

$$H_0^3 = \tfrac{1}{2} H_0, \qquad (H_0 S)^3 = \tfrac{1+i}{4} I. \tag{5}$$

Indeed, $H_0^2 = \frac{1}{2}I$ immediately gives $H_0^3 = \frac{1}{2}H_0$, while cubing $H_0 S = \frac{1}{2}\left(\begin{smallmatrix} 1 & i \\ 1 & -i \end{smallmatrix}\right)$ yields $(H_0 S)^2 = \frac{1}{4}\left(\begin{smallmatrix} 1+i & 1+i \\ 1-i & i-1 \end{smallmatrix}\right)$ and $(H_0 S)^3 = \frac{1+i}{4} I$. Suppose we want to apply H_0 only on an arbitrary subset T of the D slice positions ($s = |T|$) and act as the identity on the complement U ($q_{\text{sel}} = D - s$). Before each of *three* full layer calls H_{all}, we scan the array and multiply every record at address x by the scalar phase $(S_U z)(x) = i^{\sum_{j \in U} x_j} z(x)$. Because operators on distinct address bits tensor and commute, $(H_{\text{all}} S_U)^3$ acts as $\frac{1}{2}H_0$ on every bit in T and collapses to $\frac{1+i}{4} I$ on every bit in U. Rescaling by $2^s [2(1-i)]^{q_{\text{sel}}}$ recovers the selective operator H_T exactly (Lemma 5.1; Figure 3(a)).

Batched controlled flips without address movement. Next, conjugating $\text{diag}(1, -1)$ by H_0 gives $2H_0 \,\text{diag}(1, -1) H_0 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, whereas $2H_0 I H_0 = I$. Consider s pairwise disjoint control-target bit pairs (c_j, t_j) whose targets $T = \{t_1, \ldots, t_s\}$ lie in one slice, while the controls c_j may lie *anywhere*

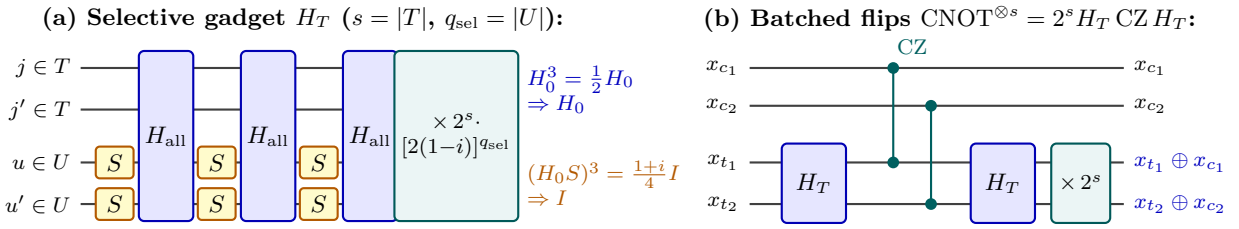


Figure 3: Turning simultaneous butterfly layers into exact controlled flips. (a) Inserting $S = \text{diag}(1, i)$ on U before three full layers cancels U to $\frac{1+i}{4} I$ while leaving $\frac{1}{2} H_0$ on T. (b) Sandwiching the recordwise sign diagonal CZ between two selective H_T gadgets flips all target address bits $t_j \in T$ by distant controls c_j in six layer calls.

in the m_0-bit record address. Define the recordwise sign diagonal $(\text{CZ}\, z)(x) = (-1)^{\sum_{j=1}^{s} x_{c_j} x_{t_j}} z(x)$. Then (Lemma 5.4; Figure 3(b))

$$\text{CNOT}^{\otimes s} = 2^s H_T \,\text{CZ}\, H_T \tag{6}$$

moves every record from address x to the address with $x_{t_j} \mapsto x_{t_j} \oplus x_{c_j}$ for all $j = 1, \ldots, s$, using six full butterfly-layer calls and linear scans (Proposition 5.5). Crucially, CZ never routes data between c_j and t_j: it computes the parity $\sum_j x_{c_j} x_{t_j}$ mod 2 from each record's address in $q^{O(1)}$ steps (absorbed by the record payload of $\Theta(rq)$ bits, Lemma A.5) and negates the record in place.

Every positional permutation is $O(1)$ CNOT stages, and is bit-exact. Every permutation π of the m_0 record-address bit positions factors on its cycles as $\pi = I_1 \circ I_0$, a product of two involutions, and each involution is a set of disjoint two-bit swaps implemented by three stages of disjoint CNOTs (Lemma 5.7). We partition the m_0 positions into $N_{\text{slice}} \le m_0/d + K$ slices of $D \le d$ chunks, plus fewer than K leftover positions handled by linear two-digit stream swaps. Grouping each of the six CNOT stages by target slice then executes any positional permutation π in at most $36 N_{\text{slice}}$ layer calls, costing $O(V[(m_0/d + K)d^{\lambda'} + K]) = O(V \ell d^{\lambda'})$ steps, the same order as the FFT butterfly rounds themselves (Theorem 5.8).

Why does fixed-point truncation introduce zero error across complex linear combinations? Before each CNOT batch on a q-grid disk array, we append $8d$ zero fractional bits to lift to precision $q_* = q + 8d$ (Lemma 5.3). Each normalized D-bit layer divides by 2^D ($D \le d$), while S_U and CZ multiply by units in $\{\pm 1, \pm i\}$, so throughout both H_T gadgets every intermediate entry lies on a dyadic grid of denominator exponent at most $q + s + 3D \le q + 4d < q_*$. Because Proposition A.11 evaluates each layer *exactly* internally (using its guard Δ) before a single final truncation Q_{q_*}, every truncation discards only trailing zeros (Lemma 5.2). Once $\text{CNOT}^{\otimes s}$ completes, the array is an exact permutation of the original q-grid records, so stripping the $8d$ zero bits returns to the q-grid bit-exactly and resets denominator growth before the next batch.

2.5 A direct tensor FFT and where 2^{-46} comes from

Direct decimation-in-frequency tensor FFT. With exact positional permutations available at layer cost, Bluestein's chirp reduction and polynomial-ring convolutions over $\mathbb{C}[y]/(y^r + 1)$ disappear completely. We evaluate $F_{\mathbf{t}} U$ on the q-grid ($t_d = r = 2^\ell$, $t_i \in \{r/2, r\}$; Theorem 6.5) in four steps.

(1) Run a standard length-r DIF FFT inside each contiguous record along axis d.

(2) Apply the exact positional permutation L (Lemma 6.3) to interleave K-bit chunks of the $d - 1$ record axes.

(3) Execute the ℓ DIF bit-levels across all $d - 1$ record axes in parallel: at most K prefix/leftover levels by exact individual butterflies in $O(VdK)$, and the remaining chunk levels by one call to

Proposition A.11 per level in $O(V\ell d^{\lambda'})$. At each level v, multiplying $d-1$ separately rounded twiddles would accumulate $O(d\ell \cdot 2^{-q})$ error. Instead, we combine all participating axis twiddles into a *single* exact index $J(x) = \sum_i (r/2^{v+1}) b_i j_i^{<v} \bmod r$, where 2^{v+1} is the DIF subproblem length, $b_i \in \{0,1\}$ is axis i's branch bit at level v, and $j_i^{<v}$ is its lower v-bit index. Looking up that index in one $2r$-entry disk-valued phase table (Lemma 6.1) incurs error $< 6 \cdot 2^{-q}$ per level regardless of d.

(4) Apply the exact positional permutation $(LB)^{-1} = B^{-1}L^{-1}$ (Theorem 5.8) to undo both the chunk layout L and the DIF bit-reversal B in one pass.

Because every exact DIF round is an ℓ^∞ contraction and the positional maps are exact, the ℓ last-axis levels and at most ℓ record-axis rounds accumulate total error strictly below $12\ell \cdot 2^{-q}$ (Corollary 6.6), with no R/r error amplification and no output bit-widening.

Cost balance and the exponent $\kappa = 2^{-46}$. Dividing the cost of one bridge call by $V = RP = \Theta(n)$ leaves the five normalized terms of Table 5, where $P = \Theta(\log n)$, $d = \lfloor k^\varepsilon \rfloor = \Theta(P^\varepsilon)$, $\ell = \Theta(P^{1-\varepsilon})$, $K = \lfloor d^c \rfloor = \Theta(P^{\varepsilon c})$, and $\mathsf{m}(N) = \mathsf{M}(N)/N = (\log N)^{O(1)}$. Individual prefix butterfly rounds cost $O(dK)$ (gap $1-\varepsilon(1+c)$ below P^1), simultaneous layers and the exact positional maps $L, (LB)^{-1}$ cost $O(\ell d^{\lambda'})$ (gap $\varepsilon(1-\lambda')$), and the last-axis FFT and combined twiddle products cost $O(\ell\,\mathsf{m}(P))$ (gap ε up to $\log P$). Direct axis exposures and restorations cost $O(d(1+\ell^\tau))$ (gap $(1-\varepsilon)(1-\tau)$), and spreading and two-alias cyclic solves cost $O\big(d(\mathsf{m}(P)+\mathsf{m}(P^2))\big)$ (gap $1-\varepsilon$ up to $\log P$). With the external chunk-swap term $PK^{\tau-1}$ gone, we are free to choose c tiny so internal layer basis changes never bind. This leaves the butterfly-layer gap $g_{\text{lay}} = \varepsilon(1-\lambda') < (1-\sigma)/C_1$ as the sole bottleneck, now *linear* in the complex network's deficit $1-\sigma$.

At $h = 100$ ($m = 10^6$), the complex network of [OAI26] has exact relative deficit $\eta_c = (W_c m - s_c)/(W_c m) = 73/19{,}906{,}842{,}167{,}500$. Using the elementary rational bound $\ln 10^6 < 14$ (via $e^7 > \sum_{j=0}^{11} 7^j/j! > 1000$, sharpening $\ln m < 20$ in [OAI26]), Proposition A.2 certifies $s_c/W_c < m^\sigma$ at $\sigma = 1-\xi$ with $\xi = 2^{-42}$ (since $73 \cdot 2^{42} > 14 \cdot 19{,}906{,}842{,}167{,}500$) and $\tau = 1-4\xi = 1-2^{-40}$ for the bit network. Moreover, the guard bound $B_{\text{net}} < 10^{66} = m^{11}$ admits guard exponent $C_1 = 11+3 = 14$ (Proposition A.11; cf. [OAI26, §5.4]). Setting $\varepsilon = 1/15$, $\beta = 1/64$, $c = \xi/256$, $\lambda = 1 - \frac{63}{64}\xi$, and $\lambda' = 1 - \frac{31}{32}\xi$ gives $g_{\text{lay}} = \varepsilon(1-\lambda') = \frac{31}{480}\xi > \frac{1}{16}\xi = 2^{-46}$, yielding $\kappa = 2^{-46}$ with positive slack (Section 7.5). In short, Bluestein's chunk layout is capped quadratically at $(1-\tau)^2/(C_1\tau) \approx 2^{-83.8}$, whereas Walsh-layer permutations reach $(1-\sigma)/C_1 \approx 2^{-45.8}$ linearly and already achieve $\kappa = 2^{-56}$ under the printed constants of [OAI26, Prop. 14].

2.6 Formal verification

Section 8 describes our Lean 4 formalization [Lean26] of Theorem 2. It constructs a single finite-alphabet multi-stack Turing machine in Mathlib's `Turing.TM2ComputableInTime` model and proves `SubNLogN.Challenge.IntMulSubNLogN` (exact multiplication of all $(a,b) \in \mathbb{N}^2$ within $C\,L\,(\log_2 L)^{1-2^{-46}}$ steps for input length $L \geq N_0$) from Lean's three standard axioms alone. That development verifies both $h = 100$ finite networks, their rank and endpoint certificates, all stream and parameter-transfer procedures, and the analytical error bounds from scratch. The formal proposition does not express syntactic CRT-freeness and formalizes only the $\kappa = 2^{-46}$ Walsh machine (deriving 2^{-47}, 2^{-56}, and $o(n \log n)$ by exponent monotonicity). It differs from the text primarily in two respects. First, it uses tie-upward nearest rounding (our bridge theorems are proved in Lean for arbitrary nearest rounding). Second, it instantiates the inner $O(N \log N)$ multiplier and its root offset via our independent Lean 4 formalization (`IntMulTM.intMul_TM2_nlogn`, kernel-checked within [Lean26]) of the two-dimensional multiplier of [MS], which distributes its own separate Lean development.

3 Preliminaries

This section fixes notation (Section 3.1) and specifies the multitape Turing machine model and fixed-point arithmetic conventions (Section 3.2). It then defines the discrete Fourier transforms used in the algorithm (Section 3.3) and states as black-box facts the two ingredients from [OAI26] used in our written proofs (Section 3.4).

3.1 Notation

Logarithms are natural (ln) unless written $\log_2$. Following [OAI26], we write $\lg n = \max\{\lceil \log_2 n \rceil, 1\}$, so $\lg R = \log_2 R$ whenever $R \geq 2$ is a power of two. For a positive integer N we write $[N] = \{0, \ldots, N-1\}$. For a vector x of complex numbers, $\|x\|_\infty = \max_j |x_j|$ denotes the largest complex modulus; for a linear map A, $\|A\|_\infty$ is the induced ℓ^∞ operator norm, i.e., the maximum absolute row sum. For an array of polynomial records in $\mathcal{R}_r = \mathbb{C}[y]/(y^r + 1)$, $\|\cdot\|_\infty$ denotes the largest modulus of any complex coefficient across the array. Table 3 collects the global notation of the paper.

3.2 Machine model and fixed-point arithmetic

Our complexity bounds refer to the standard multitape Turing machine model of [SS71, HvdH21, OAI26]: a deterministic machine with one fixed finite alphabet and a fixed finite number of one-dimensional tapes. In one step the machine reads and writes at most one cell on each tape and moves each head by at most one cell. The running time is the total number of steps, charging every sequential scan, head movement, rewind, counter update, and workspace cleanup. The input is $x\#y$ with $x, y \in \{0,1\}^n$, and the output is the $2n$-bit binary representation of the product [OAI26, §1].

Following [OAI26, §2], an array on an address set $[a_1] \times \cdots \times [a_t]$ (with $a_i \geq 1$) is stored in lexicographic order with the first coordinate most significant; addresses are implicit in the tape order and are not stored per entry. Adjacent spectator fields may be collapsed into a single field whose range is their product. An address permutation f moves the entry at address x to $f(x)$, so that $(fA)(f(x)) = A(x)$. The *logical bit volume* V of an array is the total number of stored data bits; temporary copies and parked streams on work tapes are charged in time without replacing V in recursive bounds. Integer parameters in procedure headers are encoded with the self-delimiting code $\Gamma(v) = 0^{b_v - 1}\mathrm{bin}_{b_v}(v)$ of length $2b_v - 1$, where $b_v = \lfloor \log_2 v \rfloor + 1$ for $v \geq 1$ [OAI26, (1)].

Independent integer multiplier. Ordinary products of N-bit integers are delegated to an independent deterministic multiplier of worst-case cost $\mathsf{M}(N) = N\,\mathsf{m}(N)$ (for instance that of Schönhage and Strassen [SS71], with $\mathsf{m}(N) = O(\lg N \lg\lg N)$). It is invoked as a black-box subroutine on dedicated work tapes that are clean on return, leaving the caller's other tapes untouched and making no recursive call to the outer algorithm.

Grids, rounding, and faithful tables. Stored complex numbers lie on dyadic fixed-point grids, and our error analysis uses two rounding operators.

Definition 3.1 (Fixed-point grids and rounding)**.** For an integer precision $P \geq 0$, define the *fixed-point grid* $\widetilde{\mathbb{C}}_P = \{2^{-P}(a + ib) : a, b \in \mathbb{Z}\}$ (also called the *P-grid*) and the *disk grid* $\mathbb{D}_P = \{z \in \widetilde{\mathbb{C}}_P : |z| \leq 1\}$. At precision $q \geq 0$, a *disk-grid record* is an element $\sum_{j<r} z_j y^j \in \mathcal{R}_r = \mathbb{C}[y]/(y^r + 1)$ with every $z_j \in \mathbb{D}_q$, stored in coefficient order $j = 0, \ldots, r-1$; a *disk array* is an array of records with $\|z\|_\infty \leq 1$. Each real or imaginary component is stored as 2^{-q} times a signed w-bit two's-complement integer, most significant bit first, with the real component before the imaginary one [OAI26, Prop. 14];

Table 3: Notation used throughout the paper. Bridge-specific quantities are defined in Section 4, and network parameters in Section 3.4.

Symbol	Meaning
$n,\ \kappa$	bit length of each input integer; log-saving exponent in $O(n(\lg n)^{1-\kappa})$
$R = t_1 \cdots t_d,\ k = \lg R$	bridge transform length ($t_i \geq 2$ powers of two, d dimension, $r = t_d = 2^\ell$) and its binary logarithm
$M = R/r,\ \mathsf{M}(N),\ \mathsf{m}(N)$	number of records; cost of the independent N-bit integer multiplier; $\mathsf{m}(N) = \mathsf{M}(N)/N$
$p,\ P$	target precision of a bridge call (errors in units of 2^{-p}); internal working precision of the bridge
$q,\ v = 2^{-q}$	dyadic grid precision of Sections 5–6, instantiated by the bridge at $q = P + 2$; unit of that grid
w	two's-complement component width; $w = q + 2 = P + 4$ in the bridge schedule (24)
$\omega,\ \alpha$	alias width parameter and Gaussian width parameter of the bridge kernel
$a \in \{2, \ldots, d\},\ \nu \in \mathbb{Z}$	spread-axis index of the bridge; alias index in Poisson summation
$g_a,\ G$	negative half-bin carrier on axis a; Fourier transform of $\operatorname{sinc}(s)\, e^{-\pi s^2/\alpha^2}$
$f(\mathbf{u}),\ A_a$	scalar frequency index with digits $u_1, \ldots, u_d$; $A_a = \prod_{i<a} t_i$, the cycle length of B_a
$c_0,\ c_1,\ c_{\max}$	retained alias weights $G((u+\frac{1}{2})/t)$ and $G((u+\frac{1}{2})/t - 1)$; $\max(c_0, c_1)$
$\mathcal{A}_a,\ B_a$	full alias operator on axis a; two-term cyclic operator (its two-alias truncation)
$\mathcal{R}_r = \mathbb{C}[y]/(y^r + 1)$	record ring specifying the r-coefficient record layout
$\widetilde{\mathbb{C}}_q,\ \mathbb{D}_q$	fixed-point grid $2^{-q}(\mathbb{Z} + i\mathbb{Z})$; disk grid $\{z \in \widetilde{\mathbb{C}}_q : \lvert z\rvert \leq 1\}$
$Q_q,\ \mathcal{Q}_q$	componentwise truncation toward zero to $\widetilde{\mathbb{C}}_q$; componentwise nearest rounding to $\widetilde{\mathbb{C}}_q$
V	logical bit volume of a stored array
$K = \lfloor d^c \rfloor,\ D$	chunk width and number of active K-bit chunks ($1 \leq D \leq d$) in a simultaneous butterfly layer
$A,\ q_{\mathrm{rad}},\ \Gamma$	layer and stream interface symbols: prefix range (written P in [OAI26, Prop. 14]), fixed radix (written q in [OAI26]), and self-delimiting header code [OAI26, (1)]
$H_0,\ S$	normalized butterfly $(u, v) \mapsto ((u+v)/2, (u-v)/2)$ and phase gate $\operatorname{diag}(1, i)$
$h = 100,\ m = h^3 = 10^6$	ground-set size parameters of the fixed triple networks of [OAI26]
$W_b, s_b;\ W_c, s_c$	role counts and recursive call counts of the bit and complex networks ($s_b < W_b m,\ s_c < W_c m$)
$\eta_b,\ \eta_c;\ \tau,\ \sigma$	relative rank deficits $(Wm - s)/(Wm)$ and network exponents ($s_b/W_b < m^\tau,\ s_c/W_c < m^\sigma$)
$\beta,\ \nu_{\mathrm{net}},\ C_1$	leaf-stopping exponent, guard degree $\min\{j \geq 1 : m^j \geq B_{\mathrm{net}}\} = 11$, and guard exponent of the layer
$\varepsilon,\ c,\ \lambda,\ \lambda'$	schedule and recurrence exponents; $d = \lfloor k^\varepsilon \rfloor$, $K = \lfloor d^c \rfloor$
$S,\ L,\ B,\ K$	also reused locally (defined at use): suffix range or pre-scale/series width S, layout map or cycle/input length L, bit-reversal or suffix range B, general chunk width K (and K_0, K_*)

any width $w \geq q + 2$ represents every disk-grid component, including ± 1. For $z \in \mathbb{C}$ and any integer $P \geq 0$:

- $Q_P(z)$ truncates the real and imaginary parts of z toward zero to $2^{-P}\mathbb{Z}$ [OAI26, (26)];
- $\mathcal{Q}_P(z)$ rounds the real and imaginary parts of z to a nearest element of $2^{-P}\mathbb{Z}$ (for concreteness

with ties to even, though every bound we prove holds for any nearest-rounding tie rule).

Both operators act coefficientwise on records and arrays.

For every $z \in \mathbb{C}$ and integer $P \geq 0$,

$$|Q_P(z) - z| < \sqrt{2} \cdot 2^{-P}, \qquad |Q_P(z)| \leq |z|, \qquad |\mathcal{Q}_P(z) - z| \leq \tfrac{\sqrt{2}}{2} \cdot 2^{-P}. \tag{7}$$

In particular, Q_q maps the closed unit disk into $\mathbb{D}_q$, so any exact disk-preserving operator followed by Q_q keeps every stored coefficient in $\mathbb{D}_q$ (whereas nearest rounding can leave the unit disk near ± 1). Conversely, inside the bridge's spreading and deconvolution stages, where norms are controlled explicitly, $\mathcal{Q}_q$ halves the per-step rounding error. Precomputed transcendental entries (filter taps, deconvolution parameters, and unit phases) are never assumed to be exact nearest roundings. They are *faithful* approximations computed by truncated rational series with certified error at most 2^{-q} for real entries and $2 \cdot 2^{-q}$ for complex taps and disk-valued unit phases.

3.3 Discrete Fourier transforms

We fix the sign and normalization of the complex transforms and recall the convolution identity used at the top level.

Definition 3.2 (Normalized transforms)**.** For an integer $R \geq 1$, the normalized positive one-dimensional DFT $F_R \colon \mathbb{C}^R \to \mathbb{C}^R$ is $(F_R x)_u = \frac{1}{R} \sum_{j \in [R]} x_j e^{2\pi i j u / R}$. For $\mathbf{t} = (t_1, \ldots, t_d)$ with $R = t_1 \cdots t_d$, the normalized positive tensor DFT $F_{\mathbf{t}} = F_{\mathbf{t}}^{+} = F_{t_1,\ldots,t_d}$ acts on arrays indexed by $[t_1] \times \cdots \times [t_d]$ via

$$(F_{\mathbf{t}} U)_{u_1,\ldots,u_d} = \frac{1}{R} \sum_{j_1,\ldots,j_d} U_{j_1,\ldots,j_d} \exp\Big(2\pi i \sum_{i=1}^{d} \frac{j_i u_i}{t_i}\Big).$$

By the triangle inequality, $\|F_R\|_\infty \leq 1$ and $\|F_{\mathbf{t}}\|_\infty \leq 1$. The cyclic convolution $(a \circledast b)_j = \sum_{l \in [R]} a_l b_{(j-l) \bmod R}$ satisfies $F_R(a \circledast b) = R\,(F_R a) \odot (F_R b)$, where $\odot$ denotes entrywise multiplication. For real vectors $a, b \in \mathbb{R}^R$ and $z = a + ib$, the *two-call identity* (cf. [MS]) recovers $a \circledast b$ from two forward transforms of length R:

$$a \circledast b = -\frac{R^2}{2} \operatorname{Im}\Big(F_R\big(\overline{(F_R z) \odot (F_R z)}\big)\Big). \tag{8}$$

Indeed, $\sum_{u \in [R]} e^{2\pi i (j-k) u / R}$ equals R for $j = k$ and 0 otherwise, so $F_R(\overline{F_R w}) = R^{-1} \overline{w}$ for every $w \in \mathbb{C}^R$; applying this to $w = z \circledast z = (a \circledast a - b \circledast b) + 2i(a \circledast b)$, for which $(F_R z) \odot (F_R z) = R^{-1} F_R w$, gives $F_R(\overline{(F_R z) \odot (F_R z)}) = R^{-2} \overline{w} = R^{-2}\big((a \circledast a - b \circledast b) - 2i(a \circledast b)\big)$, whose imaginary part is $-2R^{-2}(a \circledast b)$.

For completeness, and for the Bluestein comparison in Appendix C, we also record the synthetic transforms of [OAI26, §6] over the negacyclic ring $\mathcal{R}_r = \mathbb{C}[y]/(y^r + 1)$ ($r = 2^\ell \geq 2$). Multiplication by y in $\mathcal{R}_r$ is an ℓ^∞ isometry. For any power of two $t \mid 2r$, setting $\omega_t = y^{2r/t} \in \mathcal{R}_r$ defines contractions $F_t^{\mp} \colon \mathcal{R}_r^t \to \mathcal{R}_r^t$ by $(F_t^{\mp} z)_j = t^{-1} \sum_{k<t} \omega_t^{\mp jk} z_k$ satisfying $F_t^{+} F_t^{-} = t^{-1} I$. With B_t the permutation sending index $j \in [t]$ to its $(\lg t)$-bit binary reversal, in-place decimation in frequency and its reverse compute $B_t F_t^{-}$ and $F_t^{+} B_t^{-1}$; for $\mathbf{t}' = (t_1, \ldots, t_{d-1})$ with each $t_i \mid 2r$ we write $F_{\mathbf{t}'}^{\pm} = \bigotimes_{i=1}^{d-1} F_{t_i}^{\pm}$ and $B = \bigotimes_{i=1}^{d-1} B_{t_i}$. Our direct complex tensor FFT in Section 6 makes no use of ring products or synthetic transforms in $\mathcal{R}_r$, retaining $\mathcal{R}_r$ solely to specify the contiguous r-coefficient record format.

Motivation from two-dimensional fractional shifts. In dimension two ($R = t_1 r$ with $j = j_1 + t_1 j_2$ and $u = u_2 + r u_1$), the Cooley–Tukey decomposition expresses $(F_R x)_{u_2 + r u_1}$ through the inner sums $\sum_{j_2 \in [r]} x_{j_1 + t_1 j_2} e^{2\pi i (j_2 + j_1/t_1) u_2 / r}$, reading the twiddle factor $e^{2\pi i j_1 u_2 / (t_1 r)}$ as a uniform fractional shift of row j_1 by j_1/t_1 along the second axis [MS]. Spreading each row by a sinc-Gaussian filter shifted by j_1/t_1 and applying Poisson summation relates $F_{t_1, r}$ of the spread array on the *same* $t_1 \times r$ grid to a two-alias cyclic mixture of $F_R x$ along each column. With a real kernel, however, the two retained alias weights coincide at the Nyquist column $u_2 = r/2$, requiring a separate alternating-sum identity in [MS] and ruling out cycle-length-independent conditioning in higher dimensions. Section 4 resolves both issues for general $d \geq 2$ via a negative half-bin phase carrier and a descending multi-axis spreading schedule.

3.4 Ingredients from [OAI26]

Our written proofs use two results of [OAI26] as black boxes: its linear-time elementary tape stream primitives and its two fixed finite linear networks at $h = 100$ together with their algebraic certificates. Appendix D provides the symbol-renaming map from [OAI26] (in particular, its prefix range P becomes A, its fixed radix q becomes q_{rad}, and its box size T corresponds to R) along with full restatements.

Fact 3.3 (Elementary tape operations [OAI26, Lemmas 3–4])**.** *Fix an integer radix $q_{\mathrm{rad}} \geq 2$. On a stored array of logical bit volume V on a fixed number of finite-alphabet tapes, the following operations run in $O(V)$ steps (including counter initialization, head resets, and clearing visited workspace):*

(1) applying any fixed permutation to the pair of digit values at two specified radix-q_{rad} address positions, or moving one specified radix-q_{rad} digit position to another while preserving the order of intervening positions;

(2) splitting an array on a specified radix-q_{rad} digit, applying a fixed pointwise map to the aligned streams, and merging them back;

(3) within each prefix of an array on $[A] \times [R_{\mathrm{row}}] \times [B]$, distributing whole rows cyclically into a fixed number W of role streams and, when $W \mid R_{\mathrm{row}}$, merging them in any fixed order of row classes;

(4) when q_{rad} is prime and a target field $y \in [q_{\mathrm{rad}}^b]$ follows control fields of $O(b)$ digits each across arbitrary spectators, shifting $y \mapsto y + a(\mathrm{controls}) \pmod{q_{\mathrm{rad}}^b}$ for any offset computable in $b^{O(1)}$ time (in particular any fixed rational linear combination with denominators coprime to q_{rad}), or multiplying y modulo q_{rad}^b by a fixed rational unit prime to q_{rad};

(5) exact fixed-width signed binary addition, subtraction, sign change, bit-order reversal within a component, and multiplication or division by a power of two when guard bits prevent overflow or inexact truncation.

Fact 3.4 (Finite linear networks with a rank deficit [OAI26, Props. 7–8])**.** *At ground-set parameter $h = 100$ and dimension $m = h^3 = 10^6$, there exist two fixed finite linear circuits (over the three-element subsets of $[h]$) with the following certified role counts, recursive rank sums, and endpoint actions for arbitrary initial values on all auxiliary scratch roles:*

(a) Bit shear network*: a fixed pointwise XOR circuit on W_b roles equipped with rational $m \times m$ terminal and gate matrices whose completed scalar action is a role permutation ρ_b satisfying $M_{\mathrm{out}(\rho_b(w))} - M_{\mathrm{in}(w)} = I_m$ for every input role w (implementing the common address shear $(H, D) \mapsto (H + D, D)$ on m-tuples of address fields). Its role count W_b and total edge-difference rank $s_b = \sum_e \mathrm{rank}_{\mathbb{Q}}(M_{\mathrm{head}(e)} - M_{\mathrm{tail}(e)})$ are*

$$W_b = 177\,176\,569\,091\,445\,000\,000, \qquad s_b = 177\,176\,569\,088\,785\,861\,287\,000\,000 < W_b m,$$

giving relative rank deficit $\eta_b = (W_b m - s_b)/(W_b m) = 339/22\,587\,335\,000\,000$.

(b) Complex phase network*: a fixed scalar linear circuit on* W_c *roles with coefficients in* $\{0, \pm 1, \pm\frac{1}{2}\}$ *and nested binary subspace labels in* $\mathbb{F}_2^m$ *whose edge-residual orthonormal bases have total dimension* s_c*, implementing* $C^{\otimes m}$ *with* $C = \frac{1+i}{2} I + \frac{1-i}{2} X$ *on every role after prescribed source/sink diagonal phases and bank restoration, with*

$$W_c = 1\,873\,807\,244\,643\,542\,670\,000, \qquad s_c = 1\,873\,807\,244\,636\,671\,267\,308\,000\,000 < W_c m,$$

giving relative rank deficit $\eta_c = (W_c m - s_c)/(W_c m) = 73/19\,906\,842\,167\,500$.

Parameter-exposed tape procedures. As printed, [OAI26, Lemma 12 and Prop. 14] fix $\tau = \sigma = 1 - 2^{-50}$, $\beta = \frac{1}{2}$, and $C_1 = 20$. In Appendix A we follow [OAI26, §§4–5] with all structural parameters exposed, deriving from Facts 3.3 and 3.4 (and Lemma A.5 for long-record address scans):

- *Arbitrary-width address interchange* (Lemma A.8): for every fixed rational $\tau \in (0,1)$ with $s_b/W_b < m^\tau$, two disjoint u-bit address fields are interchanged across arbitrary spectator fields in $O(Vu^\tau)$ steps (and in $O(V(1+u^\tau))$ steps when their widths differ by one bit).
- *Simultaneous butterfly layer* (Proposition A.11, via packed selected-bit addition, Lemma A.9): put $B_{\mathrm{net}} = s_c + 64(W_c + m + 1)^3$, $C_0 = 128\, m\, B_{\mathrm{net}}^2$, and guard degree $\nu_{\mathrm{net}} = \min\{j \geq 1 : m^j \geq B_{\mathrm{net}}\} = 11$ (Appendix A, (47)). Let $\tau, \sigma \in (0,1)$ be any fixed rationals with $s_b/W_b < m^\tau$ and $s_c/W_c < m^\sigma$, let $\beta \in (0,1)$ be a leaf exponent and $C_1 \geq \nu_{\mathrm{net}} + 3 = 14$ a guard exponent with $\varepsilon C_1 < 1$, and let c, λ, λ' be exponents satisfying $\max\{\tau, \sigma\} < \lambda < 1$, $\tau(1 + c/\beta) < \lambda$, and $\max\{\lambda, \sigma + \beta(1-\sigma)\} < \lambda' < 1$. For any such fixed parameters, applying H_0 simultaneously to bit $\rho \in [K]$ ($K = \lfloor d^c \rfloor$) of $D \leq d$ consecutive K-bit chunks of disk-grid records in $\mathcal{R}_r$ at grid precision q evaluates the exact layer H using $\Delta = \lceil C_0 d^{C_1} \rceil = o(q)$ internal guard bits and truncates once. It returns $Q_q(Hz) \in \mathbb{D}_q$ with $\|Q_q(Hz) - Hz\|_\infty < \sqrt{2} \cdot 2^{-q}$ in $O(Vd^{\lambda'})$ steps.

Using the rational bound $\ln 10^6 < 14$ in place of $\ln 10^6 < 20$, Proposition A.2 certifies the sharper network exponents $\tau = 1 - 2^{-40}$ and $\sigma = 1 - 2^{-42}$ from the integers in Fact 3.4. Independently, the base-10 comparison $m^{10} < B_{\mathrm{net}} < m^{11}$ yields $\nu_{\mathrm{net}} = 11$, certifying the guard exponent $C_1 = \nu_{\mathrm{net}} + 3 = 14$ (cf. [OAI26, §5.4]).

4 The fractional-shift bridge

This section relates the one-dimensional transform F_R to the tensor transform $F_{\mathbf{t}}$ on the same power-of-two grid $R = t_1 \cdots t_d$, with no Chinese remainder map, prime axis length, or grid expansion. It then turns that identity into a fixed-point multitape algorithm with a complete error budget.

In Sections 4.1–4.3 we carry out the exact analysis. The input is spread along the axes $a = d, d-1, \ldots, 2$, in descending order, with a negative half-bin phase carrier. Lemma 4.2 expresses the tensor DFT of the spread array exactly as $\mathcal{A}_2 \cdots \mathcal{A}_d F_R x$, a product of non-commuting alias operators $\mathcal{A}_a$. By Lemma 4.4, each $\mathcal{A}_a$ lies within $0.120 \cdot 2^{-\omega}$ of a two-term cyclic operator B_a whose inverse has ℓ^∞ norm at most $4t_a/7 \leq t_a$, uniformly in its cycle length A_a. Applying $B_2^{-1}, \ldots, B_d^{-1}$ in chronological order recovers $F_R x$ up to (23). Two negative results in Appendix B (Propositions B.1 and B.2) show that within this two-alias framework both the sign of the half-bin carrier and the use of a complex kernel are needed for uniform bounds. Sections 4.4–4.5 then establish the fixed-point algorithm and precision contract (Lemma 4.6 and Theorem 4.7, with the series evaluation of the filter and phase tables proved in Lemma B.3 of Appendix B). Section 4.6 gives the $O(d)$ direct axis-exposure schedule on tapes (Lemma 4.8).

4.1 Index conventions and the carrier

Let $R = t_1 \cdots t_d = 2^k$ with every $t_i \geq 2$ a power of two, and put $A_a = \prod_{i<a} t_i$ for $1 \leq a \leq d+1$, so that $A_1 = 1$ and $A_{d+1} = R$. We write input indices as $j = \sum_{i=1}^{d} j_i A_i$ with $0 \leq j_i < t_i$, so j_1 is the least significant digit, and identify $x \in \mathbb{C}^R$ with the tensor $x_{j_1,\ldots,j_d}$. Frequencies use the reversed digit order:

$$f(\mathbf{u}) = \sum_{i=1}^{d} u_i \frac{R}{A_{i+1}}, \qquad 0 \leq u_i < t_i, \tag{9}$$

so u_d is the least significant digit. The map $\mathbf{u} \mapsto f(\mathbf{u})$ is a bijection onto $[R]$, and we write $u_a(f)$ for the a-th frequency digit of $f \in [R]$. (Throughout, k is reserved for $\lg R$.) For $2 \leq a \leq d$, the *fractional shift* of axis a is

$$\delta_a(\mathbf{j}) = \frac{\sum_{i<a} j_i A_i}{A_a} \in [0, 1), \tag{10}$$

which depends only on the input digits below a. In a multi-axis Cooley–Tukey decomposition of F_R, the twiddle factor coupling axis a to the lower axes is a fractional shift by $\delta_a(\mathbf{j})$ along axis a, generalizing the two-dimensional observation of [MS].

As in [MS], the base kernel is a sinc-Gaussian whose Fourier transform G is a smoothed indicator of $[-\frac{1}{2}, \frac{1}{2}]$. On each spread axis $a \geq 2$ we modulate it by a negative half-bin phase, shifting the frequency samples from u/t_a to the half-bin midpoints $(u + \frac{1}{2})/t_a$.

Definition 4.1 (Kernel and carrier)**.** Let $\omega \geq 8$ be the alias width, and fix α satisfying

$$2\sqrt{\omega \ln 2/\pi} \leq \alpha \leq 4\sqrt{\omega \ln 2/\pi}, \tag{11}$$

with default $\alpha = 2\sqrt{\omega \ln 2/\pi}$. Put $\alpha_\pi = \sqrt{\pi}\,\alpha$, so $\alpha_\pi \geq 2\sqrt{\omega \ln 2}$. With $\operatorname{sinc}(s) = \sin(\pi s)/(\pi s)$ for $s \neq 0$, $\operatorname{sinc}(0) = 1$, and Fourier transform $\hat{\phi}(\xi) = \int_{\mathbb{R}} \phi(s) e^{-2\pi i \xi s}\, ds$, the transform of $\operatorname{sinc}(s)\, e^{-\pi s^2/\alpha^2}$ is

$$G(\xi) = \int_{\xi-1/2}^{\xi+1/2} \alpha e^{-\pi \alpha^2 s^2}\, ds = \tfrac{1}{2}\big[\operatorname{erf}\big(\alpha_\pi(\xi + \tfrac{1}{2})\big) - \operatorname{erf}\big(\alpha_\pi(\xi - \tfrac{1}{2})\big)\big]. \tag{12}$$

On axis $a \in \{2, \ldots, d\}$, the *negative half-bin carrier* is

$$g_a(s) = e^{-\pi i s/t_a}\, \operatorname{sinc}(s)\, e^{-\pi s^2/\alpha^2}, \qquad \hat{g}_a(\xi) = G\Big(\xi + \frac{1}{2t_a}\Big), \tag{13}$$

and its *folded filter* is the t_a-periodic function of $e \in \mathbb{Z}$ given by

$$H_\delta^{(a)}(e) = \sum_{s \in e + t_a\mathbb{Z}} g_a(s + \delta). \tag{14}$$

Because the carrier phase $e^{-\pi i s/t_a}$ is antiperiodic of period t_a rather than periodic, the modulated kernel itself must be folded as in (14). Since $e^{-\pi i(s+\delta)/t_a} = (-1)^{(e-s)/t_a} e^{-\pi i(e+\delta)/t_a}$ for $s \in e + t_a\mathbb{Z}$,

$$H_\delta^{(a)}(e) = e^{-\pi i(e+\delta)/t_a} \sum_{s \in e + t_a\mathbb{Z}} (-1)^{(e-s)/t_a}\, g_0(s + \delta), \qquad g_0(s) = \operatorname{sinc}(s)\, e^{-\pi s^2/\alpha^2}. \tag{15}$$

Periodizing g_0 first and multiplying by the carrier afterwards would drop the alternating signs $(-1)^{(e-s)/t_a}$. Identity (15) also shows that $H_\delta^{(a)}$ can be applied as a real negacyclic convolution between two diagonal phase scans; the algorithm of Section 4.4 evaluates the complex filter directly.

4.2 Triangular alias identity

We spread one axis at a time. For $2 \le a \le d$, let $\mathcal{S}_a$ spread axis a while holding all other coordinates fixed:

$$(\mathcal{S}_a V)_{\dots,k_a,\dots} = \sum_{j_a \in [t_a]} V_{\dots,j_a,\dots}\, H^{(a)}_{\delta_a(\mathbf{j})}(j_a - k_a), \tag{16}$$

where $\delta_a(\mathbf{j})$ is read from the coordinates below a of the current array. We apply these operators in the descending order $a = d, d-1, \dots, 2$ and set $U = \mathcal{S}_2\mathcal{S}_3 \cdots \mathcal{S}_d x$. When axis a is spread, coordinates $1, \dots, a-1$ are still the original input digits $j_1, \dots, j_{a-1}$, because they are spread only at later stages. Hence

$$U_{j_1,k_2,\dots,k_d} = \sum_{j_2,\dots,j_d} x_{\mathbf{j}} \prod_{a=2}^{d} H^{(a)}_{\delta_a(\mathbf{j})}(j_a - k_a). \tag{17}$$

In the opposite order, δ_a would be evaluated after lower digits had already been replaced by spread indices, and (17) would fail.

For $2 \le a \le d$, the *alias operator* $\mathcal{A}_a$ on $\mathbb{C}^R$ collects the frequency aliases $f - \nu R/A_a$ ($\nu \in \mathbb{Z}$) generated by Poisson summation along axis a, weighted by half-bin samples of G:

$$(\mathcal{A}_a Z)_f = \sum_{\nu \in \mathbb{Z}} G\Big(\frac{u_a(f) + 1/2}{t_a} - \nu\Big)\, Z_{(f - \nu R/A_a) \bmod R}. \tag{18}$$

The weights of $\mathcal{A}_a$ depend only on the digit $u_a(f)$, whereas shifting f by a multiple of R/A_a alters only the more significant digits $u_1, \dots, u_{a-1}$. Thus in the product $\mathcal{A}_2\mathcal{A}_3 \cdots \mathcal{A}_d$ each factor shifts only digits that no factor to its right reads. This triangular dependency yields Lemma 4.2.

Lemma 4.2 (Triangular alias identity)**.** *With $U = \mathcal{S}_2\mathcal{S}_3 \cdots \mathcal{S}_d x$ and $Z = F_R x$,*

$$F_{\mathbf{t}} U = \mathcal{A}_2\mathcal{A}_3 \cdots \mathcal{A}_d\, Z, \tag{19}$$

where the operators compose from right to left. Explicitly, for every $\mathbf{u} \in [t_1] \times \cdots \times [t_d]$,

$$(F_{\mathbf{t}} U)_{\mathbf{u}} = \sum_{\boldsymbol{\nu} \in \mathbb{Z}^{d-1}} \Big[\prod_{a=2}^{d} G\Big(\frac{u_a + 1/2}{t_a} - \nu_a\Big)\Big]\, Z_{(f(\mathbf{u}) - \sum_{a=2}^{d} \nu_a R/A_a) \bmod R}, \tag{20}$$

where $\boldsymbol{\nu} = (\nu_2, \dots, \nu_d)$. The operators $\mathcal{A}_a$ do not commute in general, and (19) holds in the stated order.

Proof. We first derive (20) by Poisson summation along each spread axis, and then expand the operator product (19).

Explicit formula. For any Schwartz function ϕ and real z, ξ, applying Poisson summation to $s \mapsto \phi(z - s)e^{2\pi i \xi s}$ yields

$$\sum_{e \in \mathbb{Z}} \phi(z - e) e^{2\pi i e \xi} = \sum_{\nu \in \mathbb{Z}} \hat{\phi}(\xi - \nu)\, e^{2\pi i z(\xi - \nu)}. \tag{21}$$

No symmetry of ϕ is required, so (21) applies directly to the complex carriers g_a. Substitute (17) into $F_{\mathbf{t}} U$. For fixed $\mathbf{j}$ and $a \in \{2, \dots, d\}$, unfolding the t_a-periodic filter $H^{(a)}_{\delta_a}$ and applying (21) with $\phi = g_a$, $z = j_a + \delta_a(\mathbf{j})$, and $\xi = u_a/t_a$ gives

$$\sum_{k_a \in [t_a]} H^{(a)}_{\delta_a}(j_a - k_a)\, e^{2\pi i k_a u_a / t_a} = \sum_{\nu_a \in \mathbb{Z}} \hat{g}_a\Big(\frac{u_a}{t_a} - \nu_a\Big)\, e^{2\pi i (j_a + \delta_a)(u_a/t_a - \nu_a)}.$$

By the Gaussian decay of g_a and $\hat{g}_a$, all sums converge absolutely and may be interchanged:

$$(F_{\mathbf{t}}U)_{\mathbf{u}} = \sum_{\boldsymbol{\nu}\in\mathbb{Z}^{d-1}} \Big[\prod_{a=2}^{d} \hat{g}_a\Big(\frac{u_a}{t_a} - \nu_a\Big)\Big] \frac{1}{R}\sum_{\mathbf{j}} x_{\mathbf{j}} \exp\Big(2\pi i\Big[\frac{j_1u_1}{t_1} + \sum_{a=2}^{d}(j_a+\delta_a)\Big(\frac{u_a}{t_a}-\nu_a\Big)\Big]\Big).$$

Since $A_i/A_{a+1} \in \mathbb{Z}$ for $i > a$, we have $j/A_{a+1} \equiv (j_a+\delta_a)/t_a \pmod 1$ and $j/A_a \equiv \delta_a \pmod 1$; moreover $j/t_1 \equiv j_1/t_1 \pmod 1$ and $\nu_a j_a \in \mathbb{Z}$. The phase in brackets is therefore congruent modulo 1 to

$$\frac{j}{R}\Big(\sum_{i=1}^{d} u_i \frac{R}{A_{i+1}} - \sum_{a=2}^{d} \nu_a \frac{R}{A_a}\Big),$$

so the inner normalized sum over $\mathbf{j}$ equals $Z_{(f(\mathbf{u})-\sum_{a=2}^{d}\nu_a R/A_a) \bmod R}$. Substituting $\hat{g}_a(u_a/t_a-\nu_a) = G((u_a+\frac{1}{2})/t_a-\nu_a)$ yields (20).

Product form. Subtracting $\nu R/A_a$ from f modifies only the digits $u_1,\dots,u_{a-1}$, because R/A_a is the positional weight of u_{a-1} in (9) and borrows propagate toward u_1; the lower digits $u_a,\dots,u_d$ remain unchanged. When $(\mathcal{A}_2\mathcal{A}_3\cdots\mathcal{A}_d Z)_{f(\mathbf{u})}$ is expanded from left to right, every frequency shift accumulated before the weight of $\mathcal{A}_a$ is evaluated comes from factors $\mathcal{A}_{a'}$ with $a' < a$, which leave the digit u_a invariant. Every weight in the expansion is therefore evaluated at the original digit u_a of $f(\mathbf{u})$, matching the right-hand side of (20). In the reverse order a shift from $\mathcal{A}_{a+1}$ can alter u_a before $\mathcal{A}_a$ reads it, so the factors do not commute. □

Consequently, deconvolution must invert the factors from left to right: first B_2^{-1}, then B_3^{-1}, and so on through B_d^{-1}, where B_a is the two-alias truncation of $\mathcal{A}_a$ analyzed next.

Remark 4.3 (Non-commutativity in dimension three). Take $\mathbf{t} = (2,2,2)$, so $R = 8$, $A_2 = 2$, $A_3 = 4$, $j = j_1 + 2j_2 + 4j_3$, and $f = 4u_1 + 2u_2 + u_3$, with retained aliases $f - 4\nu_2 - 2\nu_3$ ($\nu_2,\nu_3 \in \{0,1\}$). For $x = e_3$ ($(j_1,j_2,j_3) = (1,1,0)$), we have $\delta_2 = \frac{1}{2}$, $\delta_3 = \frac{3}{4}$, and $Z_f = \frac{1}{8}e^{2\pi i\, 3f/8}$. At $\mathbf{u} = (1,0,1)$ ($f = 5$), the four retained aliases $(\nu_2,\nu_3) = (0,0),(1,0),(0,1),(1,1)$ land at frequencies $5,1,3,7$, with values $1,-1,i,-i$ times Z_5; with $c_0 = G(\frac{1}{4})$ and $c_1 = G(\frac{3}{4})$, axis 2 contributes weights (c_0,c_1) and axis 3 weights (c_1,c_0), so up to omitted aliases (20) gives $\frac{c_0-c_1}{8\sqrt{2}}\big((c_0+c_1)+i(c_0-c_1)\big)$. Similarly, writing $c_{a,\nu}(u) = G((u+\frac{1}{2})/t_a-\nu)$ for the weights of the two-term operator B_a in (22) below and taking Z to be the unit vector at frequency 6, evaluating B_2B_3Z and B_3B_2Z at frequency 0 gives $c_{2,0}(0)c_{3,1}(0)$ and $c_{3,1}(0)c_{2,0}(1)$, which differ since $c_{2,0}(0) = G(\frac{1}{4}) > G(\frac{3}{4}) = c_{2,0}(1)$ and $c_{3,1}(0) > 0$.

4.3 Two-term cyclic deconvolution

Because each $\mathcal{A}_a$ sums over all $\nu \in \mathbb{Z}$, the algorithm inverts its two-alias truncation. For $2 \le a \le d$, the *two-term cyclic operator* B_a retains the terms $\nu \in \{0,1\}$ of (18): with $y = (u_a(f)+\frac{1}{2})/t_a \in (0,1)$,

$$(B_aZ)_f = c_0 Z_f + c_1 Z_{(f-R/A_a) \bmod R}, \qquad c_0 = G(y), \quad c_1 = G(y-1). \tag{22}$$

Each orbit of $f \mapsto (f - R/A_a) \bmod R$ has length A_a, and the digit $u_a(f)$ is constant along every orbit. Thus B_a decomposes as a direct sum of cyclic bidiagonal operators $B\colon Z_j \mapsto c_0 Z_j + c_1 Z_{j-1}$ on cycles of length $L = A_a$, each with constant coefficients (c_0,c_1). In the balanced shape of Section 4.4, A_a grows up to R/r, far larger than t_a, so the conditioning of B^{-1} must be independent of the cycle length L.

The negative half-bin carrier makes both the omitted-alias error uniform in t_a and the inverse norm uniform in L. Every sample $y = (u+\frac{1}{2})/t_a$ lies strictly inside $(0,1)$, keeping all omitted aliases at distance more than $\frac{1}{2}$ from the Gaussian center, and since t_a is even no sample hits the Nyquist midpoint $y = \frac{1}{2}$ where $c_0 = c_1$.

Lemma 4.4 (Negative half-bin kernel and conditioning)**.** *Let $\omega \geq 8$, let α satisfy (11), and let $t = t_a \geq 2$ be a power of two ($2 \leq a \leq d$). For every frequency digit $0 \leq u < t$, with $y = (u + \frac{1}{2})/t$, $c_0 = G(y)$, and $c_1 = G(y-1)$:*

(i) $\|\mathcal{A}_a\|_\infty \leq 1$ *and* $\|B_a\|_\infty \leq 1$*;*

(ii) (omitted aliases) $\|\mathcal{A}_a - B_a\|_\infty < 0.120 \cdot 2^{-\omega}$*;*

(iii) (passband) $c_{\max} := \max(c_0, c_1) \geq \frac{1}{2} - 0.060 \cdot 2^{-4\omega} > 0.49999999$*;*

(iv) (separation) $|c_0 - c_1| \geq 7/(4t)$*;*

(v) (conditioning) *on a cycle of any length $L \geq 1$, the operator $B\colon Z_j \mapsto c_0 Z_j + c_1 Z_{j-1}$ is invertible with $\|B^{-1}\|_\infty \leq 1/|c_0 - c_1| \leq 4t/7$ (with equality in the first bound when L is even); consequently $\|B_a^{-1}\|_\infty \leq 4t_a/7 \leq t_a$;*

(vi) (product) $\|(B_2 \cdots B_d)^{-1}\|_\infty \leq (4/7)^{d-1} R/t_1 \leq R/t_1$*, and for $Y = F_{\mathbf{t}} U$,*

$$\|(B_2 \cdots B_d)^{-1} Y - F_R x\|_\infty \leq 0.120\,(d-1)\,\frac{R}{t_1}\,2^{-\omega}\,\|x\|_\infty. \tag{23}$$

Proof. Parts (i)–(iv) estimate the Gaussian-smoothed weights $G(y-\nu)$; parts (v)–(vi) bound the cyclic inverses.

(i) Norms. The intervals $[\xi-\nu-\frac{1}{2}, \xi-\nu+\frac{1}{2}]$ ($\nu \in \mathbb{Z}$) tile $\mathbb{R}$, so $\sum_{\nu\in\mathbb{Z}} G(\xi-\nu) = \int_{\mathbb{R}} \alpha e^{-\pi\alpha^2 s^2}\,ds = 1$ with all terms positive. The absolute row sums of $\mathcal{A}_a$ and B_a are at most $\sum_{\nu\in\mathbb{Z}} G(y-\nu) = 1$ and $c_0 + c_1 < 1$, respectively.

(ii) Omitted aliases. Removing the intervals for $\nu \in \{0, 1\}$ leaves $(-\infty, y - \frac{3}{2}] \cup [y + \frac{1}{2}, \infty)$, so the omitted alias weight at y is $\Psi(y) = \frac{1}{2}\operatorname{erfc}(\alpha_\pi(y + \frac{1}{2})) + \frac{1}{2}\operatorname{erfc}(\alpha_\pi(\frac{3}{2} - y))$, which bounds every absolute row sum of $\mathcal{A}_a - B_a$. Since $\operatorname{erfc}'' > 0$ on $(0, \infty)$, Ψ is convex on $[0, 1]$ and symmetric about $\frac{1}{2}$, whence $\Psi(y) \leq \Psi(0)$ for all $y \in (0, 1)$. Using $\operatorname{erfc} z \leq e^{-z^2}/(\sqrt{\pi} z)$ ($z > 0$) and $\alpha_\pi/2 \geq \sqrt{\omega \ln 2}$,

$$\Psi(0) \leq \frac{2^{-\omega}}{2\sqrt{\pi\omega\ln 2}} + \frac{2^{-9\omega}}{6\sqrt{\pi\omega\ln 2}} < 0.120 \cdot 2^{-\omega} \qquad (\omega \geq 8).$$

Because every half-bin sample $y = (u + \frac{1}{2})/t$ lies in $(0, 1)$, this bound is uniform in $t \geq 2$.

(iii) Passband. Since G is even and strictly decreasing on $[0, \infty)$ and $\min(y, 1-y) \leq \frac{1}{2}$, we have $c_{\max} \geq G(\frac{1}{2}) = \frac{1}{2}\operatorname{erf}(\alpha_\pi) = \frac{1}{2} - \frac{1}{2}\operatorname{erfc}(\alpha_\pi)$, and $\frac{1}{2}\operatorname{erfc}(\alpha_\pi) \leq e^{-\alpha_\pi^2}/(2\sqrt{\pi}\,\alpha_\pi) < 0.060 \cdot 2^{-4\omega}$ for $\omega \geq 8$.

(iv) Separation. For $0 \leq e \leq \frac{1}{2}$, put $\Delta_G(e) = G(\frac{1}{2} - e) - G(\frac{1}{2} + e)$. Differentiating (12) and using $\pi\alpha^2 = \alpha_\pi^2$ gives

$$\Delta_G'(e) = \alpha\big[2e^{-\alpha_\pi^2 e^2} - e^{-\alpha_\pi^2(1-e)^2} - e^{-\alpha_\pi^2(1+e)^2}\big] = 2\alpha e^{-\alpha_\pi^2 e^2}\big(1 - e^{-\alpha_\pi^2}\cosh(2\alpha_\pi^2 e)\big) > 0$$

on $[0, \frac{1}{2}]$, so Δ_G is strictly increasing on $[0, \frac{1}{2}]$. For $0 \leq e \leq \frac{1}{4}$, we have $e^{-\alpha_\pi^2}\cosh(2\alpha_\pi^2 e) \leq e^{-\alpha_\pi^2(1-2e)} \leq e^{-\alpha_\pi^2/2} \leq 2^{-2\omega} \leq 2^{-16}$, so $\Delta_G'(e) \geq 2\alpha(1 - 2^{-16})e^{-\alpha_\pi^2 e^2}$, and integrating from 0 yields $\Delta_G(e) \geq (1 - 2^{-16})\operatorname{erf}(\alpha_\pi e)$. Because $t \geq 2$ is even, the midpoint $(t-1)/2$ is a half-integer, so every $u \in [t]$ satisfies $e = |y - \frac{1}{2}| = |u - \frac{t-1}{2}|/t \geq \frac{1}{2t}$, with $\frac{1}{2t} \leq \frac{1}{4}$. By concavity of erf on $[0, \infty)$ and $2/t \leq 1$,

$$\operatorname{erf}\Big(\frac{\alpha_\pi}{2t}\Big) \geq \frac{2}{t}\operatorname{erf}\Big(\frac{\alpha_\pi}{4}\Big) \geq \frac{2}{t}\operatorname{erf}(\sqrt{2\ln 2}),$$

since $\alpha_\pi/4 \geq \frac{1}{2}\sqrt{\omega \ln 2} \geq \sqrt{2 \ln 2}$. The tail bound of (iii) gives $\operatorname{erf}(\sqrt{2 \ln 2}) > 1 - \frac{1}{4\sqrt{2\pi \ln 2}} > \frac{22}{25}$, and therefore

$$t\,|c_0 - c_1| \geq t\,\Delta_G\Big(\frac{1}{2t}\Big) > 2\,(1 - 2^{-16})\,\frac{22}{25} > \frac{7}{4}.$$

(v) Conditioning. Set $\rho = \min(c_0, c_1)/c_{\max} < 1$ and let T be the cyclic shift $(TZ)_j = Z_{(j-1) \bmod L}$. If $c_0 > c_1$, then $BZ = Y$ reads $(I + \rho T)Z = Y/c_0$; if $c_1 > c_0$, it reads $(I + \rho T^{-1})Z = T^{-1}Y/c_1$. For $S \in \{T, T^{-1}\}$, the Neumann identity gives $(I + \rho S)^{-1} = (1 - (-\rho)^L)^{-1} \sum_{i<L} (-\rho S)^i$. Since the L powers of S occupy distinct columns in each row, every absolute row sum of $(I + \rho S)^{-1}$ equals $(1 + \rho + \cdots + \rho^{L-1})/|1 - (-\rho)^L|$, which is $1/(1-\rho)$ for even L and $\frac{1-\rho^L}{1+\rho^L} \cdot \frac{1}{1-\rho} \leq \frac{1}{1-\rho}$ for odd L. Because $c_{\max}(1-\rho) = |c_0 - c_1|$, it follows that $\|B^{-1}\|_\infty \leq 1/|c_0 - c_1| \leq 4t/7$ by (iv). Since B_a is a direct sum of such operators on cycles of even length A_a, $\|B_a^{-1}\|_\infty \leq 4t_a/7 \leq t_a$.

(vi) Product. Submultiplicativity of $\|\cdot\|_\infty$ and (v) give $\|(B_2 \cdots B_d)^{-1}\|_\infty = \|B_d^{-1} \cdots B_2^{-1}\|_\infty \leq \prod_{a=2}^{d} (4t_a/7) = (4/7)^{d-1} R/t_1 \leq R/t_1$. Telescoping

$$\mathcal{A}_2 \cdots \mathcal{A}_d - B_2 \cdots B_d = \sum_{a=2}^{d} B_2 \cdots B_{a-1} (\mathcal{A}_a - B_a) \mathcal{A}_{a+1} \cdots \mathcal{A}_d$$

and applying the product bound, parts (i)–(ii), $\|F_R x\|_\infty \leq \|x\|_\infty$, and Lemma 4.2 yields (23). □

Notice that the bridge has unit grid expansion in every dimension. Appendix B proves two converse bounds within this two-alias framework. With the positive carrier $e^{+\pi i s/t_a} g_0(s)$ and retained pair $\nu \in \{0, 1\}$, the sample $u = 0$ lands at $y = -\frac{1}{2t_a} \notin (0, 1)$ and loses uniform $O(2^{-\omega})$ alias decay (Proposition B.1). For any real-valued kernel, $|c_0| = |c_1|$ at the Nyquist sample $\xi = \frac{1}{2}$, forcing $\|B^{-1}\|_\infty \geq \Omega(L)$ on cycles of length $L = A_a$ (Proposition B.2).

4.4 The fixed-point bridge algorithm and middle-transform contract

Throughout the remainder of this section, $R = t_1 \cdots t_d = 2^k$ has *balanced shape*: $d \geq 2$, $t_d = r = 2^\ell$ with $\ell \geq 2$, and $t_i \in \{r/2, r\}$ for $1 \leq i < d$. We write $X = \max(1, \|x\|_\infty)$ for the input x, and $v = 2^{-q}$ for the unit of the q-grid, where $q = P + 2$ is the dyadic grid precision and P is the internal working precision of the bridge.

The bridge interacts with the multidimensional transform at its core solely through the following interface.

Definition 4.5 (Middle-transform contract)**.** A *middle transform at precision q with error factor E_{mid} and widening δ_{mid}* is a procedure that accepts a complete tensor U in natural axis order $(j_1, \ldots, j_d)$, with the last axis contiguous, entries on $\widetilde{\mathbb{C}}_q$, and $\|U\|_\infty \leq 1/8$, and returns an array $\widehat{Y}$ on $\widetilde{\mathbb{C}}_q$ in natural frequency order $(u_1, \ldots, u_d)$, with component width at most $q + 2 + \delta_{\mathrm{mid}}$ $(0 \leq \delta_{\mathrm{mid}} \leq P)$ and

$$\|\widehat{Y} - F_{\mathbf{t}} U\|_\infty \leq E_{\mathrm{mid}}\, v, \qquad E_{\mathrm{mid}} \leq 16R(k+1).$$

In our main algorithm, Corollary 6.6 (Section 6) supplies a middle transform at precision q with $E_{\mathrm{mid}} = 12\ell$ and $\delta_{\mathrm{mid}} = 0$ for every parameter tuple (q, d, r, K) satisfying the hypotheses of Theorem 6.5 at precision q. Lemma 7.1 verifies those hypotheses along the root family of Section 7. (The Bluestein middle transform analyzed for comparison in Appendix C, Lemma C.4, likewise satisfies Definition 4.5, with $E_{\mathrm{mid}} = \sqrt{2}\,M(3\ell + 1) + 12d + 16$ and $\delta_{\mathrm{mid}} = \lg M$, where $M = R/r$.)

Given $x \in \widetilde{\mathbb{C}}_{p+2}^R$ with $\|x\|_\infty \leq 2^p$, the bridge computes $\widehat{F}_R x$ in five steps:

(1) *Pre-scaling.* Scan x once to find an integer H with $X \leq 2^H < 4X$, set $S = 2^{H + \lceil (\lg(R/t_1))/2 \rceil + 4}$, and form $x' = x/S$ exactly on the q-grid.

(2) *Spreading.* Reorder the input digits $(j_d, \dots, j_1)$ into natural tensor order $(j_1, \dots, j_d)$ and spread the axes $a = d, d-1, \dots, 2$ in descending order via (16), exposing each axis a as the innermost axis by Lemma 4.8. Along each exposed line of length t_a, spreading is a complex cyclic convolution with the reflected filter $e \mapsto H^{(a)}_{\delta_a(\mathbf{j})}(-e)$ truncated to unfolded offsets $|s| \le P$, whose taps are generated once per line. We evaluate each line by overlap-add with signed Kronecker packing (cf. [MS]). First, partition the line into blocks of $W_t = \min(2P+1, t_a)$ entries, and pack each block's signed real and imaginary numerators $(u_j)_{j<W_t}$ (and those of the W_t-tap reflected filter) into one signed integer $\sum_{j<W_t} u_j B_0^j$ in base $B_0 = 2^{4P+16}$ via a one-pass borrow scan across $(4P+16)$-bit slots. Second, compute four exact real integer products per block, and extract the exact convolution coefficients sequentially in the balanced range $[-B_0/2, B_0/2)$ (subtracting each balanced residue modulo B_0 and shifting right by $4P+16$ bits). Third, accumulate the overlaps at fractional scale $2q$ (wrapping modulo t_a) and round each completed line entry once to $\widetilde{\mathbb{C}}_q$ by $\mathcal{Q}_q$.

(3) *Middle transform.* Apply a middle transform at precision q (Definition 4.5) to the computed spread array $\widehat{U}$, producing $\widehat{Y}$.

(4) *Deconvolution.* Apply two-pass cyclic streaming solvers for $B_2^{-1}, B_3^{-1}, \dots, B_d^{-1}$ in chronological order on the array in natural one-dimensional frequency order (Lemma 4.6(iv)–(v) and Section 4.6).

(5) *Output.* Multiply the deconvolved array exactly by S and round once to $\widetilde{\mathbb{C}}_{p+2}$ by $\mathcal{Q}_{p+2}$.

All precomputed tables (folded taps, solver parameters, and unit phases) are faithful approximations certified to error at most v for real entries and $2v$ for complex taps and unit phases (Lemma 4.6(iii)).

4.5 Component error bounds and the precision contract

We first establish the per-stage numerical and bit-complexity bounds of the bridge components, and then combine them into the end-to-end precision contract.

Lemma 4.6 (Finite-precision spreading and deconvolution)**.** *Let $P \ge 20$, $q = P+2$, $v = 2^{-q}$, and $8 \le \omega \le P$. Let $\alpha^2 = 4\lambda_G \omega \ln 2/\pi$ with a fixed rational $\lambda_G \in [1,4]$ (so α satisfies (11)), and let $t = t_a$ $(2 \le a \le d)$.*

(i) (Filter norms and spatial tail) *For every $\delta \in [0,1)$, $\|H^{(a)}_\delta\|_2 \le 1$ and $\|H^{(a)}_\delta\|_1 \le \sqrt{t}$ over one period $[t]$. Truncating the unfolded kernel to $|e| \le P$ leaves a spatial tail $\sum_{|e|>P} |g_a(e+\delta)| < 0.10 \cdot 2^{-P} = 0.40\, v$.*

(ii) (Spreading) *With complex taps certified to error at most $2v$, step (2) of Section 4.4 maps any q-grid array z with $\|z\|_\infty \le 1$ to a q-grid array within $4(P+1)v$ of $\mathcal{S}_a z$. Each line of length t uses at most $2t/W_t$ blocks ($W_t = \min(2P+1, t)$), with four exact products of integers of at most $W_t(4P+16)+1$ bits per block and $O(tP)$ bit operations for packing, extraction, and exact accumulation.*

(iii) (Tables and phases) *For every integer $b \ge \max(8, P)$, each unfolded kernel sample, alias weight c_0, c_1, solver parameter $c_{\max}^{-1}$ and $\rho = \min(c_0, c_1)/c_{\max}$, and rational-angle unit phase used by the bridge can be computed deterministically with certified error at most 2^{-b} in $O((b+k)^3)$ bit operations, without any transcendental oracle. In particular, every truncated folded tap can be stored on $\widetilde{\mathbb{C}}_q$ with certified error below v (in $O(P(P+k)^3)$ bit operations per tap), the solver parameters $\hat{\mu}, \hat{\rho}$ on $2^{-q}\mathbb{Z}$ with $\hat{\rho} \ge 0$ and error below v, every unit phase e as a disk-grid entry $\hat{e} \in \mathbb{D}_q$ with $|\hat{e} - e| < 2v$, and a d-factor tensor chirp in $\mathbb{D}_q$ with error at most $4dv$.*

(iv) (Cyclic solver) *Let $L \geq 2$ be an even cycle length (every bridge cycle length A_a, $a \geq 2$, is even). Suppose $v \leq 1/(64t)$ and the stored parameters satisfy $|\hat{\mu} - c_{\max}^{-1}| \leq v$, $|\hat{\rho} - \rho| \leq v$, $\hat{\rho} \geq 0$, and $|\hat{\gamma} - (1-\hat{\rho}^L)^{-1}| \leq v$. Then the two-pass cyclic sweep, with every product rounded by $\mathcal{Q}_q$, returns an approximation $\widehat{Z}$ to $Z = B^{-1}Y$ satisfying*

$$\|\widehat{Z} - Z\|_\infty \leq 32t^2\, v \max(1, \|Y\|_\infty),$$

and every intermediate state has modulus strictly below $7t\max(1, \|Y\|_\infty)$.

(v) (Closure parameter) *Under $v \leq 1/(64t)$ and even $L \geq 2$, computing $\hat{\rho}^L$ from the stored $\hat{\rho}$ by binary powering with truncation at precision $Q = q + \lceil \lg L \rceil + 2\lceil \lg(4t) \rceil + 10$, followed by one guarded division and rounding to $2^{-q}\mathbb{Z}$, produces $\hat{\gamma}$ with $|\hat{\gamma} - (1-\hat{\rho}^L)^{-1}| \leq v$ using $O(P + \lg L + \lg t)$-bit arithmetic (which is $O(P+k)$ for $L = A_a \leq R$).*

(vi) (Product norm) $\|(B_2 \cdots B_d)^{-1}\|_\infty \leq R/t_1$.

Proof. (i) Filter norms and spatial tail. For $u \in [t]$, unfolding $H_\delta^{(a)}$ and applying (21) gives $\sum_{e\in[t]} H_\delta^{(a)}(e) e^{-2\pi i e u/t} = \sum_{\nu\in\mathbb{Z}} G(\frac{u+1/2}{t} - \nu) e^{2\pi i \delta(u/t - \nu)}$, whose modulus is at most $\sum_{\nu\in\mathbb{Z}} G(\frac{u+1/2}{t} - \nu) = 1$. Parseval's identity over $\mathbb{Z}_t$ yields $\|H_\delta^{(a)}\|_2 \leq 1$, and Cauchy–Schwarz gives $\|H_\delta^{(a)}\|_1 \leq \sqrt{t}$. For the spatial tail, $|g_a(s)| \leq e^{-\pi s^2/\alpha^2}/(\pi|s|)$. Since $\alpha \leq 4\sqrt{\omega \ln 2/\pi}$ and $\omega \leq P$, we have $\alpha\sqrt{P \ln 2/\pi} \leq (4\ln 2/\pi)\sqrt{\omega P} < P$. Thus on each side of $[-P, P]$, the first omitted sample contributes less than $2^{-P}/(\pi P)$, and the remaining samples contribute at most $\int_P^\infty e^{-\pi s^2/\alpha^2}(\pi s)^{-1}\, ds \leq 2^{-P}/(2\pi P \ln 2)$ (via $z = \pi s^2/\alpha^2$ and $\int_b^\infty e^{-z} z^{-1}\, dz \leq e^{-b}/b$ for $b \geq P\ln 2$). Summing both sides gives $\sum_{|e|>P} |g_a(e+\delta)| \leq (\frac{2}{\pi P} + \frac{1}{\pi P \ln 2}) 2^{-P} < 0.06 \cdot 2^{-P} < 0.10 \cdot 2^{-P} = 0.40\, v$ for $P \geq 20$.

(ii) Spreading. Truncating the unfolded kernel to $|s| \leq P$ leaves at most $2P+1$ effective taps (folded modulo t when $t < 2P+1$), each certified to error at most $2v$, while the omitted spatial tail adds at most $0.40\, v$ by (i). Hence the effective filter error has ℓ^1 norm at most $(4P + 2.4)v$; since $\|z\|_\infty \leq 1$ and the single final rounding $\mathcal{Q}_q$ adds at most $\frac{\sqrt{2}}{2} v < v$, each output entry differs from $\mathcal{S}_a z$ by at most $(4P+3.4)v < 4(P+1)v$. Because $\|z\|_\infty \leq 1$ and every certified tap has modulus below 2 (since $\|H_\delta^{(a)}\|_\infty \leq \|H_\delta^{(a)}\|_2 \leq 1$ by (i)), each numerator satisfies $|u_j| < 2 \cdot 2^q < B_0/2$ ($B_0 = 2^{4P+16}$), so $|\sum_{j<W_t} u_j B_0^j| < B_0^{W_t}/2 < 2^{W_t(4P+16)}$ has at most $W_t(4P+16)+1$ signed bits and is formed by a linear borrow scan in $O(W_t P)$ bit operations. Likewise, every real product coefficient c_m and overlap sum at fractional scale $2q$ has magnitude at most $16(2P+1)2^{2q} < 2^{4P+15} = B_0/2$, so $c_m \in (-B_0/2, B_0/2)$ is the unique balanced residue of $\sum_{k\geq m} c_k B_0^{k-m}$ modulo B_0. Reading $(4P+16)$-bit slots from low to high, subtracting c_m, and shifting right by $4P+16$ bits recovers all c_m and accumulates overlaps at scale $2q$ with no wrap or intermediate rounding. Partitioning the line into $\lceil t/W_t \rceil \leq 2t/W_t$ blocks gives four exact integer products of width at most $W_t(4P+16)+1$ bits per block, plus $O(tP)$ bit operations for packing, extraction, and accumulation.

(iii) Tables and phases. This is Lemma B.3, proved in Appendix B.

(iv) Cyclic solver. Assume $c_0 > c_1$; the opposite case $c_1 > c_0$ reduces to the same forward recurrence via the reorientation $z_i = Z_{(-i) \bmod L}$ and $y'_i = Y_{(1-i) \bmod L}$, encoding the one-step shift of the proof of Lemma 4.4(v). Let T be the cyclic shift $(TZ)_j = Z_{(j-1) \bmod L}$. Starting from $z = 0$, the first pass computes $b_j = \mathcal{Q}_q(\hat{\mu} Y_j)$ and updates $z \leftarrow b_j + \mathcal{Q}_q(-\hat{\rho} z)$ for $j = 0, \ldots, L-1$; it then closes the cycle via $\widehat{Z}_{L-1} = \mathcal{Q}_q(\hat{\gamma} z)$, and a second pass from $z = \widehat{Z}_{L-1}$ computes $\widehat{Z}_0, \ldots, \widehat{Z}_{L-2}$ by the same recurrence.

By Lemma 4.4(iii)–(iv) and $v \leq 1/(64t)$, we have $\rho \leq 1 - 7/(4t)$, so $0 \leq \hat{\rho} \leq 1 - 1/(2t)$, $\|(I+\rho T)^{-1}\|_\infty \leq t$, and $\|(I+\hat{\rho}T)^{-1}\|_\infty \leq 2t$; moreover $c_{\max}^{-1} < 2.0000001$, so $|\hat{\mu}| < 2.01$. By the resolvent identity, the exact solution $\tilde{Z} = \hat{\mu}(I+\hat{\rho}T)^{-1} Y$ for the stored parameters $(\hat{\mu}, \hat{\rho})$ satisfies

$\|\tilde{Z} - Z\|_\infty \le (2t + 2.01 \cdot 2t^2)v\|Y\|_\infty \le 6t^2 v\|Y\|_\infty$. Each step of the first pass introduces at most two nearest roundings of $\frac{\sqrt{2}}{2}v$ each (at most $2v$ total), damped by powers of $\hat{\rho}$, so at the end of the pass the accumulated prefix error is at most $2vS_L$ with $S_L = \sum_{j<L} \hat{\rho}^j$. Because L is even, $(I + \hat{\rho}T)^{-1}$ closes with denominator $1 - (-\hat{\rho})^L = 1 - \hat{\rho}^L$, giving the exact cancellation

$$\frac{S_L}{1 - \hat{\rho}^L} = \frac{1}{1 - \hat{\rho}} \le 2t.$$

Relative to $\tilde{Z}_{L-1}$, the prefix rounding error therefore contributes at most $2vS_L/(1 - \hat{\rho}^L) \le 4tv$ to $\widehat{Z}_{L-1}$. Since the prefix states satisfy $|z| \le S_L(\max_j |b_j| + 2v) < 4.1t \max(1, \|Y\|_\infty)$, the closure error $|\hat{\gamma} - (1 - \hat{\rho}^L)^{-1}| \le v$ adds at most $v|z| < 4.1tv \max(1, \|Y\|_\infty)$, and the final $\mathcal{Q}_q$ adds at most v, so $|\widehat{Z}_{L-1} - \tilde{Z}_{L-1}| \le 12tv \max(1, \|Y\|_\infty)$ independently of L. The second pass adds at most $2v/(1 - \hat{\rho}) \le 4tv$, giving

$$\|\widehat{Z} - Z\|_\infty \le (6t^2 + 16t)v \max(1, \|Y\|_\infty) \le 32t^2 v \max(1, \|Y\|_\infty).$$

Finally, $\|\tilde{Z}\|_\infty \le |\hat{\mu}|\,\|(I + \hat{\rho}T)^{-1}\|_\infty \|Y\|_\infty < 4.02t\|Y\|_\infty$, and every second-pass state lies within $16tv \max(1, \|Y\|_\infty) \le \frac{1}{4}\max(1, \|Y\|_\infty)$ of $\tilde{Z}$, so all states have modulus strictly below $7t \max(1, \|Y\|_\infty)$.

(v) Closure parameter. Binary powering of $\hat{\rho} \in [0, 1)$ with truncation toward zero at precision Q incurs error at most $2L \cdot 2^{-Q}$ and never exceeds $\hat{\rho}^L$. Since $1 - \hat{\rho}^L \ge 1 - \hat{\rho} \ge 1/(2t)$, both exact and truncated denominators are at least $1/(2t)$, where the derivative of $x \mapsto (1 - x)^{-1}$ is at most $4t^2$. The powering error therefore perturbs $(1 - \hat{\rho}^L)^{-1}$ by at most $4t^2 \cdot 2L \cdot 2^{-Q} \le v/2048$, and one guarded division followed by nearest rounding to $2^{-q}\mathbb{Z}$ stays within error v. Because $\hat{\gamma}$ is computed from the stored dyadic $\hat{\rho}$ (without any exact power oracle), the cancellation $S_L/(1 - \hat{\rho}^L) = 1/(1 - \hat{\rho})$ in (iv) holds with no condition relating L to 2^P.

(vi) Product norm. This is Lemma 4.4(vi). □

Theorem 4.7 (Bridge precision contract)**.** *Let $p \ge \max(8, k)$ and let $x \in \widetilde{\mathbb{C}}^R_{p+2}$ satisfy $\|x\|_\infty \le 2^p$, with $X = \max(1, \|x\|_\infty)$. Set*

$$\omega = p + k + \lceil \lg d \rceil + 20, \qquad P = 2p + 12k + 32\lceil \lg(p + k + 2) \rceil + 256, \qquad q = P + 2, \qquad w = q + 2, \tag{24}$$

take $\alpha = 2\sqrt{\omega \ln 2/\pi}$, certify real table entries to error $v = 2^{-q}$ and complex taps and unit phases to $2v$, and compute closure parameters at precision $Q = q + \lceil \lg L \rceil + 2\lceil \lg(4t) \rceil + 10$. Then for every middle transform at precision q satisfying Definition 4.5, the algorithm of Section 4.4 returns $\widehat{F}_R x \in \widetilde{\mathbb{C}}^R_{p+2}$ satisfying

$$\|\widehat{F}_R x - F_R x\|_\infty < \Big(\frac{\sqrt{2}}{8} + 2^{-12} + 2^{-23}\Big) 2^{-p} X < 0.177021 \cdot 2^{-p} X. \tag{25}$$

Specifically, before the final rounding the arithmetic error is at most $2^{-p-12}X$ and the omitted-alias error is at most $2^{-p-23}X$, while the final rounding adds at most $\frac{\sqrt{2}}{8}\,2^{-p}$. Every stored and intermediate component has width $O(P)$ bits (Table 4), with $P = O(p + \lg R)$.

Proof. Write $U = \mathcal{S}_2 \cdots \mathcal{S}_d x'$ for the exact spread array of $x' = x/S$ and $\widehat{U}$ for the computed spread array passed to the middle transform.

Margins. Put $z = p + k + 2 \ge 12$ and $b = \lceil \lg z \rceil$. From (24), $P + 1 \le 44z + 233 \le 64z$, so $\lg(P + 1) \le b + 6$; also $d \le k$. Consequently $\log_2(2^{14} d(P + 1)R^3 v) \le 18 + 4k + b - P \le -p - 12$, since $P - (p + 4k + b + 30) = p + 8k + 31b + 226 > 0$. This master margin

$$2^{14} d(P + 1)R^3 v \le 2^{-p-12} \tag{26}$$

implies $16d(P+1)\sqrt{R/t_1}\,v < \frac{1}{16}$, $E_{\mathrm{mid}}v \le 16R(k+1)v < \frac{1}{100}$, $v \le 1/(64\max_a t_a)$, and $64v\sum_{a=2}^{d} t_a < 1$, while $8 \le \omega \le P$, verifying all hypotheses of Lemma 4.6.

Pre-scaling. Let $C = \max(1, \max_j |\mathrm{Re}\, x_j|, \max_j |\mathrm{Im}\, x_j|)$ and $H = 1 + \lceil \log_2 C \rceil$; then $X \le 2^H < 4X$ and $H \le p+1$. With $S = 2^{H+\lceil(\lg(R/t_1))/2\rceil+4}$ we obtain $16\sqrt{R/t_1}\,X \le S < 128\sqrt{R/t_1}\,X$, whence $\|x'\|_\infty \le 1/(16\sqrt{R/t_1})$. Moreover x' has at most $(p+2) + H + \lceil(\lg(R/t_1))/2\rceil + 4 \le 2p + \lceil(\lg(R/t_1))/2\rceil + 7 \le q$ fractional bits, so $x' \in \widetilde{\mathbb{C}}_q^R$ is exact.

Spreading. By Lemma 4.6(i), $\|\mathcal{S}_a\|_\infty \le \sqrt{t_a}$, so every exact partial spread has ℓ^∞ norm at most $\|x'\|_\infty \prod_{a=2}^{d}\sqrt{t_a} = \|x'\|_\infty\sqrt{R/t_1} \le 1/16$. By Lemma 4.6(ii) and induction over the $d-1$ stages $a = d, \ldots, 2$, the spreading error $E_{\mathrm{spread}} = \|\widehat{U} - U\|_\infty$ satisfies

$$E_{\mathrm{spread}} \le 4(P+1)v \sum_{a=2}^{d}\prod_{j=2}^{a-1}\sqrt{t_j} \le 16d(P+1)\sqrt{R/t_1}\,v < \frac{1}{16}$$

by (26). Every computed partial spread therefore has norm at most $\frac{1}{16} + \frac{1}{16} = \frac{1}{8} \le 1$, maintaining the input bound of Lemma 4.6(ii) at each stage and ensuring $\|\widehat{U}\|_\infty \le 1/8$ for Definition 4.5.

Middle transform. Applying Definition 4.5 to $\widehat{U}$ and using $\|F_{\mathbf{t}}\|_\infty \le 1$, $E_{\mathrm{mid}} \le 16R(k+1)$, and $k+1 \le P+1$, the middle-transform output $\widehat{Y}$ satisfies $\|\widehat{Y} - F_{\mathbf{t}}U\|_\infty \le E_Y := E_{\mathrm{spread}} + E_{\mathrm{mid}}v \le 32d(P+1)Rv$ and $\|\widehat{Y}\|_\infty < \frac{1}{8} + \frac{1}{100} < 1$.

Deconvolution. Put $R_a = \prod_{j=2}^{a} t_j$ for $2 \le a \le d$ and $R_1 = 1$, so $R_d = R/t_1$. Because $32v\sum_{a=2}^{d} t_a < \ln 2$, induction with Lemma 4.6(iv) and $\|B_a^{-1}\|_\infty \le t_a$ shows that the input to solver stage a has norm at most $2R_{a-1}$. Stage a therefore introduces local error at most $32t_a^2 v \cdot 2R_{a-1}$, which the subsequent exact inverses $B_d^{-1}\cdots B_{a+1}^{-1}$ amplify by at most R_d/R_a. Since $t_a^2 R_{a-1}(R_d/R_a) = t_a(R/t_1)$, which does not depend on the cycle lengths A_a, the total solver error satisfies

$$E_{\mathrm{solver}} \le 64\,\frac{R}{t_1}\,v\sum_{a=2}^{d} t_a \le 64dR^2 v.$$

Meanwhile E_Y is amplified by $\|(B_2\cdots B_d)^{-1}\|_\infty \le R/t_1 \le R$ (Lemma 4.6(vi)). The normalized arithmetic error is thus at most $(R/t_1)E_Y + E_{\mathrm{solver}} \le 96d(P+1)R^2 v$, and after exact scaling by $S < 128\sqrt{R/t_1}\,X \le 128RX$,

$$E_{\mathrm{arith}} \le 2^{14}d(P+1)R^3 vX \le 2^{-p-12}X$$

by (26).

Aliases and output. Applying (23) to x' and rescaling by S bounds the omitted-alias error by $0.120(d-1)(R/t_1)2^{-\omega}X < 0.120\cdot 2^{-p-20}X < 2^{-p-23}X$, since $(d-1)2^{-\lceil \lg d\rceil} < 1$ and $R/t_1 < 2^k$. Finally, nearest rounding $\mathcal{Q}_{p+2}$ adds at most $\frac{\sqrt{2}}{2}2^{-p-2} = \frac{\sqrt{2}}{8}2^{-p} \le \frac{\sqrt{2}}{8}2^{-p}X$ (as $X \ge 1$). Using $\sqrt{2} < 1.4142136$ (since $1.4142136^2 > 2$), rational arithmetic gives $\frac{\sqrt{2}}{8} + 2^{-12} + 2^{-23} < 0.17702096 < 0.177021$, proving (25). All component widths match Table 4; any middle-transform widening δ_{mid} is stripped only after verifying $\|\widehat{Y}\|_\infty < 1$, which alters no value. □

4.6 Axis exposures and streaming layout

Spreading operates on the lines of one axis a at a time, which are contiguous on the tape only when that axis is innermost. Bubbling axis a inward through $O(d)$ adjacent swaps would incur cost $O(Vd^2\ell^\tau)$ and exceed linear-logarithmic time at our dimension exponents ε. Lemma 4.8 instead exchanges axis a directly with the innermost axis in a single call to the arbitrary-width interchange of Lemma A.8. (Note that the Walsh positional permutations of Section 5 are not used here, because bridge line operations carry $O(P)$-bit scalar entries rather than length-r polynomial payloads.)

Table 4: Component bit widths in the bridge precision contract (Theorem 4.7, $q = P + 2$). On the direct tensor FFT route of Section 6, $\delta_{\mathrm{mid}} = 0$.

Object	Width or scale
normalized input x', completed spread arrays	$q + 2 = P + 4$ bits per real/imaginary component
table evaluation (Lemma B.3)	$S = 2b + 8\lceil \lg b \rceil + 64$ fractional bits ($b \le q + O(\lg P)$) plus $O(b + k)$ integer bits; temporaries $\le 3S + O(k)$ fractional bits
exact overlap-add accumulators	fractional scale $2q$, plus $O(\lg P)$ integer bits
packed spread operands	$O(P)$ slots of $4P + 16$ bits
middle-transform output $\widehat{Y}$	$q + 2 + \delta_{\mathrm{mid}}$ bits ($0 \le \delta_{\mathrm{mid}} \le P$)
deconvolution states and prefixes	$q + k + 6 = P + k + 8$ bits
closure powering (Lemma 4.6(v))	$Q = q + \lceil \lg L \rceil + 2\lceil \lg(4t) \rceil + 10$ fractional, $O(\lg t)$ integer bits
final exact scaling by S and output	add $\log_2 S$ to exponent; one rounding $\mathcal{Q}_{p+2}$ to $\widetilde{\mathbb{C}}_{p+2}$

Lemma 4.8 (Axis exposures via chunk interchanges)**.** *Let $\tau \in (0, 1)$ be any rational interchange exponent satisfying $s_b/W_b < m^\tau$ in Lemma A.8 (for instance $\tau = 1 - 2^{-40}$ by Proposition A.2, or $\tau = 1 - 2^{-50}$ as in [OAI26, Lemma 12]). Let a complete tensor of logical bit volume V have axis bit-widths in $\{\ell - 1, \ell\}$ ($\ell \ge 2$) and a nonempty record payload. Given the selected axis and the lengths of the collapsed prefix, gap, and suffix fields as a fixed number of binary integers (their computation from the axis list is charged to the caller), any axis can be exposed as the innermost axis, and the original axis order restored afterwards, in $O(V(1 + \ell^\tau))$ steps. A sweep of line operations over all axes uses at most $2(d - 1)$ such exchanges, and reversing the axis order uses at most $\lfloor d/2 \rfloor$ more, so a constant number of sweeps costs $O(Vd(1 + \ell^\tau))$ steps.*

Proof. Collapse all axes preceding the chosen axis a into the prefix field A, all axes strictly between a and the innermost axis d into the gap field G, and the scalar payload into the suffix B of Lemma A.8. If axes a and d have equal bit-width (ℓ or $\ell - 1$), one call to Lemma A.8 exchanges them in $O(V\ell^\tau)$ steps, independently of the size of G. If their widths differ by one bit and the earlier field is longer, write it as (x_0, X) with $|X| = |Y|$ (Y the later field) and execute $(x_0, X, G, Y) \mapsto (x_0, Y, G, X) \mapsto (Y, G, x_0, X)$: one equal-width interchange of cost $O(V\ell^\tau)$ followed by one single-bit move (Fact 3.3(1) / Lemma A.4(2)) of cost $O(V)$, totaling $O(V(1 + \ell^\tau))$. If the later field is longer, execute $(X, G, y_0, Y) \mapsto (y_0, X, G, Y) \mapsto (y_0, Y, G, X)$. Running the inverse sequence after the line operation restores the layout. Axis d requires no exchange, so a full sweep uses at most $2(d - 1)$ exchanges, and reversing the d axes factors into $\lfloor d/2 \rfloor$ disjoint exchanges of the same form. □

Two streaming observations complete the tape implementation of Section 4.4:

- *Deconvolution without transposition.* In natural one-dimensional frequency order, write any frequency $f \in [R]$ as $f = j\,(R/A_a) + s$ with $0 \le j < A_a$ and $0 \le s < R/A_a$. For each fixed s, the entries $j = 0, 1, \ldots, A_a - 1$ form a single cycle of B_a (along which $u_a(f)$ is constant), while the tape stores, for $j = 0, 1, \ldots, A_a - 1$ in order, the contiguous row of all $s \in [R/A_a]$. On a sequential state tape, we maintain one $O(P)$-bit solver state record for each $s \in [R/A_a]$, holding the current state z and the column parameters $(\hat{\mu}, \hat{\rho}, \hat{\gamma})$ (which depend only on (a, u_a) and are replayed rather than recomputed). We then scan each data row j in lockstep with the state tape and rewind only the state tape after each row. This costs $O(V)$ tape movement per pass and per axis a, independently of A_a (with the $O(1)$ multiplications per entry charged in Section 7.3). Columns with $c_0 > c_1$ and $c_1 > c_0$ are processed in two masked passes; for $c_1 > c_0$, linear-time

row-block reversals implement the index reflections $j \mapsto (-j) \bmod A_a$ and $j \mapsto (1-j) \bmod A_a$ of the proof of Lemma 4.6(iv) and undo them afterwards. This lockstep scan relies on Definition 4.5 returning $\widehat{Y}$ in natural frequency order.

- *Amortized tap and table generation.* Each exposed axis a has $R/t_a \le 2R/r$ lines, and generating the $O(P)$ folded taps of a line takes $\mathrm{poly}(P,k)$ bit operations by Lemma 4.6(iii). Even regenerating taps for every line therefore costs $O(dR(P+k)^{O(1)}/r)$ steps, which is $o(RP)$ whenever r grows faster than every fixed polynomial in P (as holds along the root family of Lemma 7.1); the same applies to the solver and phase tables.

5 Walsh-layer permutations

This section and the next replace the Bluestein middle transform and chunk-exchange layout of [OAI26] by a direct complex tensor FFT. The key new tool is an exact procedure that permutes the *positions* of a binary record address using simultaneous butterfly layers, per-record diagonal phases, and dyadic scalings, together with elementary two-digit permutations at fewer than K leftover positions outside the chunks (Section 5.3). Three full layers apply $H_0 = \frac{1}{2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ exactly to any subset of the selected bits (Section 5.1), two such selective layers around a sign diagonal execute a batch of disjoint controlled flips (Section 5.2), and six stages of batched flips execute every positional permutation (Section 5.3).

Throughout this section and Section 6, stored arrays lie on the q-grid $\widetilde{\mathbb{C}}_q = 2^{-q}(\mathbb{Z}+i\mathbb{Z})$, which the bridge of Section 4 instantiates at $q = P+2$. For arrays of records, $\|\cdot\|_\infty$ denotes the largest complex modulus of a coefficient, and a *disk array* is one with $\|z\|_\infty \le 1$. By (7), componentwise truncation Q_q toward zero to $\widetilde{\mathbb{C}}_q$ satisfies

$$\|Q_q z\|_\infty \le \|z\|_\infty, \qquad \|Q_q z - z\|_\infty < \sqrt{2}\cdot 2^{-q}. \tag{27}$$

We invoke the simultaneous butterfly layer through Proposition A.11, proved in Appendix A from the elementary stream primitives of Fact 3.3 and the fixed triple networks at $h = 100$, $m = h^3 = 10^6$ of Fact 3.4. Let us recall its signature at grid precision q. Fix rational network exponents $0 < \tau < 1$ and $0 \le \sigma < 1$ satisfying $s_b/W_b < m^\tau$ and $s_c/W_c < m^\sigma$, rational parameters $0 < \beta < 1$, $c > 0$, $0 < \varepsilon < 1$, λ, λ' with $\max\{\tau, \sigma, \tau(1+c/\beta)\} < \lambda < 1$ and $\max\{\lambda, \sigma + \beta(1-\sigma)\} < \lambda' < 1$, a guard exponent $C_1 \ge 14$ with $\varepsilon C_1 < 1$, and band constants $a_d \le b_d$, $a_r \le b_r$, C_M, C_{desc} and $C_w > 1$. For all q above a fixed cutoff with

$$a_d q^\varepsilon \le d \le b_d q^\varepsilon, \qquad 2^{a_r q/d} \le r \le 2^{b_r q/d}, \qquad K = \lfloor d^c \rfloor,$$

and integers $1 \le D \le d$, $A, S \ge 1$ and $0 \le \rho < K$ satisfying $M = AS\,2^{KD}$, $\lg(2M) \le C_M q$ and $q+2 \le w \le C_w q$, the procedure accepts a disk-grid array in lexicographic order on $[A] \times [2^K]^D \times [S]$ together with the header $\Gamma(q)\Gamma(d)\Gamma(r)\Gamma(D)\Gamma(A)\Gamma(S)\Gamma(K)\Gamma(\rho+1)\Gamma(w)$ of length at most $C_{\mathrm{desc}} q$. Each address carries a length-r record in $\mathcal{R}_r$ stored in coefficient order, each real or imaginary component being 2^{-q} times a signed w-bit two's-complement integer written most significant bit first. Let H apply H_0 simultaneously to bit ρ (counted from the least significant end) of each of the D active chunks. Proposition A.11 evaluates Hz *exactly* over $\mathbb{Z}[i, 1/2]$ using $\Delta = \lceil C_0 d^{C_1} \rceil = o(q)$ internal guard bits ($q+\Delta$ fractional and $\Delta+3$ integer and sign bits per real component) before applying one final truncation Q_q. It returns $Q_q(Hz)$ in the original record and component format, as a disk array satisfying $\|Q_q(Hz) - Hz\|_\infty < \sqrt{2}\cdot 2^{-q}$, in $O(Vd^{\lambda'})$ steps, where $V = 2Mrw$.

5.1 Selective Walsh operations

Controlled flips require H_0 on an arbitrary subset of a *slice*, the set of D positions selected by one call of the butterfly layer (Definition 5.6 fixes the slices used for positional maps); three full layers provide this selective operator exactly. Positions in a binary record address are numbered from the most significant end, while within each K-bit chunk the offset ρ is counted from the least significant end. Let $S = \mathrm{diag}(1, i)$, and for any set U of record-address positions define the recordwise diagonal

$$(S_U z)(x) = i^{\sum_{j \in U} x_j}\, z(x), \tag{28}$$

which multiplies every coefficient of the record at address x by the same fourth-root phase.

Lemma 5.1 (Three-layer selection)**.** *Let a slice consist of D selected positions ($1 \le D \le d$) and let T be any subset of them, with $s = |T|$, U the selected positions of the slice not in T, and $q_{\mathrm{sel}} = |U| = D - s$. Let H_{all} apply H_0 on all D positions and H_T only on those in T. Then*

$$H_0^3 = \tfrac{1}{2} H_0, \qquad (H_0 S)^3 = \tfrac{1+i}{4} I, \qquad H_T = 2^s\, [2(1-i)]^{q_{\mathrm{sel}}}\, (H_{\mathrm{all}} S_U)^3. \tag{29}$$

If $q_{\mathrm{sel}} = 2f + e$ with $e \in \{0, 1\}$, then $2^s [2(1-i)]^{q_{\mathrm{sel}}} = 2^{s+3f+e} (-i)^f (1-i)^e$.

Proof. The first identity follows from $H_0^2 = \frac{1}{2} I$, and cubing $H_0 S = \frac{1}{2} \left(\begin{smallmatrix} 1 & i \\ 1 & -i \end{smallmatrix}\right)$ gives the second. Operators on different positions commute and tensor independently, so $(H_{\mathrm{all}} S_U)^3$ acts position by position: as $\frac{1}{2} H_0$ on each position in T and as $\frac{1+i}{4} I$ on each position in U. Hence $(H_{\mathrm{all}} S_U)^3 = 2^{-s} (\frac{1+i}{4})^{q_{\mathrm{sel}}} H_T$, and $(\frac{1+i}{4})^{-1} = 2(1-i)$. Finally, $[2(1-i)]^2 = -8i$ gives the last formula. □

In chronological order, the *gadget* of Lemma 5.1 applies $S_U, H_{\mathrm{all}}, S_U, H_{\mathrm{all}}, S_U, H_{\mathrm{all}}$ and then the scalar $2^s [2(1-i)]^{q_{\mathrm{sel}}}$; there are three S_U diagonals, including the initial one. The scalar is applied in one scan: a fourth-root phase, at most one multiplication by $1-i$ via $(\Re z, \Im z) \mapsto (\Re z + \Im z, \Im z - \Re z)$, and one binary shift. Every external diagonal and scalar correction is applied only at completed layer boundaries in the ordinary record layout, after the layer call has restored its role routing and removed its padding, with position masks kept as caller data on separate tapes.

Lemma 5.2 (Exact boundaries)**.** *Let the input be a disk array on the q-grid and let a slice have $1 \le D \le d$ selected positions. Put $q_* = q + 8d$, lift the representation exactly to precision q_*, and execute the gadget of Lemma 5.1 by completed calls of Proposition A.11 at precision q_*, each with its internal guard Δ. Then the gadget is exact: every final truncation Q_{q_*} of a layer call discards only zeros, and the output is $H_T z$, on a grid of denominator exponent at most $q + s$. After any diagonal with entries in $\{1, -1, i, -i\}$ (which preserves both the grid and the disk), a second gadget with the same T is again exact, with output on a grid of denominator exponent at most $q + 2s$. All completed outputs are disk arrays.*

Proof. A full normalized D-bit layer has Gaussian dyadic entries with denominator dividing 2^D, and S_U adds no denominator bits. The three layer outputs of the first gadget therefore lie on grids of denominator exponent at most $q + D$, $q + 2D$ and $q + 3D$, all strictly below $q_* = q + 8d$, so each Q_{q_*} is the identity; the internal computation of each layer is exact by Proposition A.11, whose internal denominators are controlled by Δ. After the exact scalar scan, the value equals $H_T z$, which has denominator exponent at most $q + s$. A diagonal with entries in $\{1, -1, i, -i\}$ multiplies every coefficient by a unit of $\mathbb{Z}[i]$, preserving both this denominator exponent and the modulus of every coefficient. The second gadget therefore again starts from a disk array, now on denominator exponent at most $q + s$; its three layer outputs have denominator exponents at most $q + s + 3D \le q + 4D < q_*$, and its completed output $H_T(\cdot)$ has exponent at most $q + 2s$. Each full

layer and each S_U is a contraction in $\|\cdot\|_\infty$, and the completed gadget output equals H_T applied to a disk array, hence is itself a disk array: this operator identity, rather than a bound on the intermediate scalar, verifies the disk-input condition of the next layer call. The scalar scan requires only a constant number of temporary integer bits. □

Lemma 5.3 (Lifting)**.** *Lifting a q-grid array to precision $q_* = q+8d$ costs $O(V)$: append $8d$ fractional zero bits to every numerator and use component width $w_* = w+8d$, so that $q_*+2 \le w_* \le C_w q_*$, and rebuild the header with q_*, w_* and the same d, r, K. Since $d = O(q^\varepsilon)$ with $\varepsilon < 1$, eventually $q_* \le 2q$, and the comparison bands $a_d q^\varepsilon \le d \le b_d q^\varepsilon$ and $2^{a_r q/d} \le r \le 2^{b_r q/d}$ at q imply $(a_d/2)q_*^\varepsilon \le d \le b_d q_*^\varepsilon$ and $2^{(a_r/2)q_*/d} \le r \le 2^{b_r q_*/d}$. Hence one fixed set of widened comparison constants serves both q and q_*.*

Proof. Because $C_w > 1$ and $w \le C_w q$, we have $w_* = w + 8d \le C_w(q + 8d) = C_w q_*$. Appending $8d$ low zero bits represents the exact same dyadic values at scale 2^{-q_*} (whereas changing only the precision field in the header would rescale the array), and the internal guarded width $q_* + 2\Delta + 3$ remains $O(q)$. All re-encoding scans are linear in V. □

5.2 Exact controlled flips

Two selective layers around a sign diagonal perform a batch of controlled flips exactly. For disjoint pairs (c_j, t_j), $1 \le j \le s$, of distinct record-address positions, let $T = \{t_1, \dots, t_s\}$ and define

$$(\mathrm{CZ}\, z)(x) = (-1)^{\sum_{j=1}^{s} x_{c_j} x_{t_j}}\, z(x). \tag{30}$$

Lemma 5.4 (Controlled flips)**.** *As array operators, $\mathrm{CNOT}^{\otimes s} = 2^s H_T\, \mathrm{CZ}\, H_T$, where the left-hand side moves the record at address x to the address obtained by $x_{t_j} \mapsto x_{t_j} \oplus x_{c_j}$ for all $1 \le j \le s$ and changes no coefficient.*

Proof. For a single pair (c_1, t_1), fix the control bit x_{c_1}. If $x_{c_1} = 0$, the target diagonal is I and $2H_0^2 = I$. If $x_{c_1} = 1$, the target diagonal is $\mathrm{diag}(1,-1)$ and $2H_0\,\mathrm{diag}(1,-1)H_0 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Tensoring over the s disjoint pairs gives the identity. Because the address map is an involution, the move and pull conventions coincide. □

Proposition 5.5 (Slice-batched controlled flips)**.** *Assume the hypotheses of Proposition A.11 at grid precision q. Suppose all targets $t_1, \dots, t_s$ of a batch of disjoint controlled flips lie among the selected positions of one slice, while the controls $c_1, \dots, c_s$ lie anywhere else in the record address (including selected positions of the same slice outside T). On disk arrays on the q-grid, the batch is performed exactly in $O(Vd^{\lambda'})$ steps by six completed layer calls at precision $q_* = q + 8d$ and a constant number of linear scans, returning the output in the original q-grid format.*

Proof. Lift the input from q to $q_* = q + 8d$ (Lemma 5.3), apply the first selective gadget for H_T, the sign diagonal CZ, and the second selective gadget for H_T, multiply exactly by 2^s, and lower from q_* back to q. Since CZ has entries in $\{\pm 1\}$, Lemma 5.2 applies: all six layer truncations discard only zeros. If a control c_j lies in the same slice outside T, the internal layers of a gadget mix that bit temporarily, but the completed gadget acts on it as the identity before CZ is applied. By Lemma 5.4, the value after multiplying by 2^s is an exact permutation of the input q-grid records, so lowering from q_* to q discards only zero low bits; this relies on the permutation identity rather than denominator counting alone. There is no control-before-target ordering requirement because CZ is a recordwise value diagonal rather than a controlled address rotation. For each record, the phases of (28) and the parity of (30) are computed once from its address and the caller's mask or pair

list in time polynomial in q, which is absorbed by the record length r (Lemma A.5), while lifting and lowering are elementary streams (Lemma A.4) and the lifted volume remains $O(V)$. Thus Proposition A.11 gives total time $O(Vd^{\lambda'})$. □

Because a completed batch is an exact record permutation on $\widetilde{\mathbb{C}}_q$, it neither amplifies inherited numerical error nor accumulates denominator bits across batches.

5.3 Positional permutations

Definition 5.6 (Positional maps and slices)**.** A *positional map* on m_0 record-address positions is a permutation π of $\{0, \dots, m_0 - 1\}$ under the move convention: the bit at position j is placed at position $\pi(j)$. It induces an array operator that permutes whole records while leaving each record's coefficient list unchanged; $f \circ g$ applies g first. Put $J = \lfloor m_0/K \rfloor$, partition the first JK positions into consecutive K-bit chunks, and group those chunks into consecutive groups of at most d chunks. For a group starting at position j_0 with D chunks ($1 \le D \le d$) and an offset $0 \le \rho < K$, the corresponding *slice* consists of the positions $j_0 + jK + (K - 1 - \rho)$ for $0 \le j < D$, with complete rectangular layer shape $[2^{j_0}] \times [2^K]^D \times [2^{m_0 - j_0 - DK}]$. The fewer than K remaining address positions are *leftover* positions.

The number of slices is

$$N_{\text{slice}} = K\lceil J/d \rceil \le \frac{m_0}{d} + K. \tag{31}$$

Lemma 5.7 (Two involutions)**.** *Every positional map is a composition $I_1 \circ I_0$ of two involutions. Each involution is a set of disjoint positional swaps, and each set of disjoint swaps is implemented by three stages of disjoint controlled flips.*

Proof. On a cycle $(i_0, \dots, i_{n_c - 1})$ with $\pi(i_j) = i_{j+1}$ (indices modulo n_c), define $I_0(i_j) = i_{-j}$ and $I_1(i_j) = i_{1-j}$. Both are involutions and $I_1(I_0(i_j)) = i_{j+1}$; defining them independently on every cycle of π gives $\pi = I_1 \circ I_0$. A swap of two positions b_1, b_2 factors into three controlled flips, $x_{b_1} \leftarrow x_{b_1} \oplus x_{b_2}$, $x_{b_2} \leftarrow x_{b_2} \oplus x_{b_1}$, $x_{b_1} \leftarrow x_{b_1} \oplus x_{b_2}$; within each of the three stages, distinct swaps act on disjoint pairs of positions and therefore commute. □

Theorem 5.8 (Positional permutations)**.** *Assume the parameter family of Proposition A.11 at grid precision q. Let a complete m_0-bit record-address cube, with $m_0 = O(q)$, carry disk-valued q-grid records of r complex coefficients, so that $V = 2^{m_0} \cdot 2rw$. For every positional map π on $\{0, \dots, m_0 - 1\}$, a fixed multitape procedure performs the induced record permutation exactly in*

$$O\Big(V\Big[\Big(\frac{m_0}{d} + K\Big)d^{\lambda'} + K\Big]\Big) \tag{32}$$

steps, which is $O(Vqd^{\lambda'-1})$ when $dK = o(q)$. The output is returned in the original q-grid format.

Proof. Schedule. Execute the three stages of I_0 followed by the three stages of I_1 from Lemma 5.7. Within each stage, partition the controlled-flip pairs according to the slice that contains their target bit; reordering batches within a stage is valid because the pairs of a stage have pairwise disjoint supports. Each slice batch satisfies Proposition 5.5 (including pairs whose control bit lies in a leftover position or in another slice). Any pair whose target bit is a leftover position is executed directly by a two-digit stream permutation (Lemma A.4(1)); since there are fewer than K leftover positions, each of the six stages performs fewer than K such permutations, costing $O(VK)$ in total.

Bookkeeping. Cycle decompositions, involution tables, slice masks and nine-field headers have size polynomial in q and are constructed by polynomially many scans of polynomial-size tables,

absorbed into $O(V)$ by the record length (Lemma A.5). Every slice acts on the complete cube of 2^{m_0} records, its two spectator ranges contribute at most $2m_0 + 2$ header bits, and the record length contributes $2 \lg r + 1 = O(q)$ bits, so a single pair of constants C_M, C_{desc} serves every slice call.

Cost. Six stages with at most N_{slice} slice batches per stage and six layer calls per nonempty batch use at most $36N_{\mathrm{slice}}$ completed layer calls. Combining (31) with Proposition 5.5 yields (32). Each layer completes its own padding and cleanup, and each batch returns to the q-grid before the next begins; internal packed basis changes (Lemmas A.8–A.9) are already accounted for in $O(Vd^{\lambda'})$. When $m_0 = O(q)$ and $dK = o(q)$, we have $(m_0/d + K)d^{\lambda'} + K = O(qd^{\lambda'-1})$. □

Note that Theorem 5.8 permutes the bit positions of a complete binary record address carrying a long contiguous payload via intermediate linear combinations on $\widetilde{\mathbb{C}}_{q_*}$. It does not execute arbitrary permutations of the 2^{m_0} records or apply to short payloads without a record factor r.

6 Direct tensor FFT

We now compute the middle transform of the bridge directly as a complex tensor FFT, using the simultaneous butterfly layers of Proposition A.11 for the record axes and the exact positional permutations of Theorem 5.8 for the layout and natural-order restoration. At grid precision q, its error is less than $12\ell \cdot 2^{-q}$, where $r = 2^\ell$ is the length of the last axis (Theorem 6.5), requiring no Bluestein chirp, no ring convolution in $\mathcal{R}_r$, no M-amplified error factor and no output widening (Corollary 6.6).

Throughout this section, let

$$t_d = r = 2^\ell, \qquad t_i \in \{2^{\ell-1}, 2^\ell\}\ (1 \le i < d), \qquad R = t_1 \cdots t_d, \qquad M = R/r, \qquad m_0 = \lg M.$$

The last coordinate is stored as the contiguous length-r coefficient list of each record; here $\mathcal{R}_r$ specifies only the record format, and no multiplication in that ring is performed. The exact transform is the normalized positive tensor DFT $F_{\mathbf{t}}^{+} = F_{\mathbf{t}}$ of Definition 3.2. Stored arrays lie on the q-grid $\widetilde{\mathbb{C}}_q$ with component width $q + 2 \le w \le C_w q$. Scalar integer products are delegated to an independent multiplier of cost $\mathsf{M}(N) = N\,\mathsf{m}(N)$ on N-bit integers (for instance Schönhage–Strassen [SS71]), never to a recursive call of the algorithm of this paper.

6.1 Phases and scalar products

The error analysis of Theorem 6.5 requires every round to start from a disk array, so both scalar primitives preserve the closed unit disk: phases (Lemma 6.1) and complex products (Lemma 6.2).

Lemma 6.1 (Disk-valued phases)**.** *For every precision $q \ge 8$ and $r = 2^\ell$ with $\ell \le q$, the $2r$ phases $\theta_j = e^{\pi i j/r}$, $0 \le j < 2r$, have deterministically computable approximations $\tilde{\theta}_j \in \widetilde{\mathbb{C}}_q$ satisfying*

$$|\tilde{\theta}_j| \le 1, \qquad |\tilde{\theta}_j - \theta_j| < 2 \cdot 2^{-q}.$$

The fourth roots of unity are returned exactly. Each entry costs $O(q^3)$ bit operations on dyadic numbers of $O(q)$ bits, on a fixed number of tapes.

Proof. Put $u_0 = 2^{-q}$ and $b = q + 5$. We compute with dyadic numbers of $S = 2b + 8\lceil \lg b \rceil + 64$ fractional bits, rounding each product and quotient once (error at most 2^{-S}), as in Lemma 4.6(iii) and Appendix B; exact products before a rounding use at most $2S$ fractional bits. We evaluate θ_j by rational series to within $u_0/16$, then shrink the result slightly and truncate it into the disk.

Reduction. Reduce the dyadic rational $\rho = j/r$ exactly modulo 2 and then, by the symmetries of sine and cosine, to $|\rho| \le \frac{1}{4}$. The symmetries contribute only a sign, a conjugation or a factor $\pm i$, which act exactly; the four exact fourth roots of unity are recognized directly from the index.

The angle. Machin's identity $\pi = 16 \arctan \frac{1}{5} - 4 \arctan \frac{1}{239}$, summed to $N = 4S$ terms with one rounded division per term, gives $\tilde{\pi}$ with $|\tilde{\pi} - \pi| \le 21N2^{-S}$ (its tails are below $20 \cdot 5^{-2N-1}$). Since ρ is dyadic with denominator dividing $r = 2^\ell$, the product $x = \tilde{\pi}\rho$ is formed exactly on the grid of $S + \ell$ fractional bits and rounded once to S fractional bits, adding at most 2^{-S}. Thus $|x| < 1$ and $|x - \pi\rho| \le (21N + 1)2^{-S}$.

Series. Sum $\cos x$ and $\sin x$ as the alternating Taylor polynomials $\sum_{n<N}(-1)^n T_n$, with $T_0 = 1$ and $T_n = T_{n-1}x^2/((2n)(2n-1))$ for the cosine, and $T_0 = x$ and $T_n = T_{n-1}x^2/((2n)(2n+1))$ for the sine. Because the step multipliers satisfy $x^2/2 < \frac{1}{3}$, every rounding residue is damped, so each sum has rounding error below $4N2^{-S}$ and tail below $2(1.6\,\mathrm{e}/N)^N$. Since cos and sin are 1-Lipschitz, the resulting complex approximation a satisfies

$$|a - \theta_j| \le 64N2^{-S} + 4(1.6\,\mathrm{e}/N)^N \le 2^{-b-4} < \frac{u_0}{16},$$

as $S - b - 4 = b + 8\lceil \lg b \rceil + 60$.

Into the disk. Set $\tilde{\theta}_j = Q_q((1 - u_0/8)a)$. Before truncation, the modulus is at most $(1 - u_0/8)(1 + u_0/16) < 1$ and the error from θ_j is at most $u_0/16 + (u_0/8)(1 + u_0/16) < u_0/4$, so by (27) we obtain $|\tilde{\theta}_j| \le 1$ and $|\tilde{\theta}_j - \theta_j| < (\sqrt{2} + \frac{1}{4})u_0 < 2u_0$.

Cost. All operands have $O(S) = O(q)$ bits and there are $O(N) = O(q)$ arithmetic steps, so schoolbook arithmetic takes $O(q^3)$ bit operations per entry without any transcendental oracle. □

Lemma 6.2 (Disk-preserving products)**.** *If $a, b \in \widetilde{\mathbb{C}}_q$ satisfy $|a|, |b| \le 1$, then $Q_q(ab)$ is computed by four exact products of $O(q)$-bit integers and $O(q)$ further work, lies in the disk $\mathbb{D}_q$, and satisfies $|Q_q(ab) - ab| < \sqrt{2} \cdot 2^{-q}$.*

Proof. Write $a = a_0 + ia_1$ and $b = b_0 + ib_1$, form $(a_0b_0 - a_1b_1) + i(a_0b_1 + a_1b_0)$ exactly on the $2q$-bit fractional grid (without rounding real products separately), and truncate both components toward zero by Q_q. Since $|ab| \le 1$, (27) gives both claims. □

6.2 The decimation-in-frequency schedule and its positional maps

The decimation-in-frequency (DIF) recursion runs on all $d - 1$ record axes in parallel, one bit level per round. A positional map L groups their address slots into K-bit chunks so that each chunk round is a single call of Proposition A.11, and a second positional map U restores natural frequency order (Lemma 6.3).

For a transform of even length u, one positive normalized DIF step applies H_0 followed, on the branch $b = 1$, by multiplication by $e^{2\pi ij/u}$:

$$(F_u^+ z)_{2v+b} = \frac{1}{u/2} \sum_{j<u/2} e^{2\pi ivj/(u/2)}\, e^{2\pi ibj/u}\, \frac{z_j + (-1)^b z_{j+u/2}}{2}, \qquad b \in \{0, 1\}. \tag{33}$$

Successive address slots from high to low thus receive frequency bits from low to high, so the completed DIF recursion produces bit-reversed frequency order.

Following the named-slot arrangement of [OAI26, Lemma 16], put $a_i = \lg t_i$ for $1 \le i \le d - 1$, and name the address slots of record axis i from most to least significant by $(i, a_i - 1), \ldots, (i, 0)$. Let $b_0 = (\ell - 1) \bmod K$ and $J_{\mathrm{blk}} = (\ell - 1 - b_0)/K$. The original slot list is $\mathcal{O} = ((i, v) : i = 1, \ldots, d-1,\ v = a_i - 1, \ldots, 0)$. The reordered list begins with the prefix

$$\mathcal{P} = ((i, \ell - 1) : a_i = \ell,\ i \text{ increasing}) \parallel ((i, v) : v = b_0 - 1, \ldots, 0 \text{ (outer)},\ i \text{ increasing (inner)})$$

and continues with the chunked list

$$\mathcal{C} = \bigparallel_{j=0}^{J_{\mathrm{blk}}-1} \bigparallel_{i=1}^{d-1} \big((i,v) : v = \ell - 2 - jK, \ldots, \ell - 1 - (j+1)K\big),$$

empty sublists being omitted. Let L be the positional map moving each named slot from its position in $\mathcal{O}$ to its position in $\mathcal{P}\|\mathcal{C}$, and let B be the per-axis bit reversal $(i,v) \mapsto (i, a_i - 1 - v)$ on $\mathcal{O}$.

Lemma 6.3 (Layout and natural-order restoration)**.** *In a round at named slot v with $b_0 \le v \le \ell - 2$, the selected positions in $\mathcal{P}\|\mathcal{C}$ are bit $\rho = (v - b_0) \bmod K$, counted from the least significant end, of $D = d - 1$ consecutive K-bit chunks. With n_P and n_S denoting the numbers of record-address bits before and after these $d - 1$ chunks, the round has layer shape $[2^{n_P}] \times [2^K]^{d-1} \times [2^{n_S}]$ for Proposition A.11 (unlike a slice of Definition 5.6, whose chunks start at position 0, here they start after $\mathcal{P}$). The DIF rounds on the $d - 1$ record axes, executed in layout L on input in the original order, have exact counterpart $L\,B\,F^{+}_{\mathbf{t}_{<d}}$, where $F^{+}_{\mathbf{t}_{<d}}$ is the tensor DFT over the first $d - 1$ axes. Natural frequency order is restored by the positional map*

$$U = (LB)^{-1} = B^{-1}L^{-1}. \tag{34}$$

Both L and U are compiled in time polynomial in q and executed exactly by Theorem 5.8.

Proof. The layout. The prefix $\mathcal{P}$ removes the extra top slot of each axis of length 2^ℓ and the lowest b_0 slots of all $d - 1$ record axes. The remaining $\ell - 1 - b_0 = J_{\mathrm{blk}}K$ slots of each record axis split into J_{blk} chunks of width K, which $\mathcal{C}$ orders first by chunk block j and then by axis $i \in \{1, \ldots, d-1\}$.

The rounds. In a round at named slot v with $b_0 \le v \le \ell - 2$, every record axis uses chunk block $j = \lfloor (\ell - 2 - v)/K \rfloor$, whose least significant slot is $\ell - 1 - (j+1)K = (J_{\mathrm{blk}} - j - 1)K + b_0$. Its offset from the least significant end of that chunk is therefore $(v - b_0) \bmod K$, while the preceding and following address bits act as spectators of lengths n_P and n_S.

The output order. Applying (33) inductively along each record axis yields the bit-reversed output order B. Relocating the named slots by L conjugates each physical round by L, so executing the rounds in layout L after applying L to the original-order input computes $LBF^{+}_{\mathbf{t}_{<d}}$, proving (34).

Compilation. The slot lists have $m_0 = O(q)$ entries, so computing ranks, compositions and inverses (forming U as $(L \circ B)^{-1}$) takes time polynomial in q. □

Remark 6.4. In general $B^{-1}L^{-1} \neq L^{-1}B^{-1}$, and (34) requires the former: for two record axes of bit widths $(4, 3)$ and $K = 2$ ($\ell = 4$, $b_0 = 1$), in zero-based destination-list notation $L = (0, 3, 4, 1, 5, 6, 2)$ and $B = (3, 2, 1, 0, 6, 5, 4)$, so $L \circ B$ sends position 0 to 1 whereas $B \circ L$ sends it to 3. Executing L via width-K chunk exchanges as in [OAI26, Lemma 16] (Appendix C) would reintroduce the normalized cost term $qK^{\tau-1}$; here L and U are executed instead by Theorem 5.8.

6.3 One combined twiddle per record

In a record-axis round each participating axis contributes a twiddle factor. Combining them into a single phase per record ensures that a round incurs only one phase approximation and one complex multiplication error per coefficient, regardless of d.

At named slot v, each participating record axis i has current subproblem length $u_i = 2^{v+1}$. For a record address x, reading the named-slot table gives its current branch bit $b_i = x_{(i,v)}$ and its unprocessed lower index $j_i^{<v} = \sum_{0 \le s < v} 2^s x_{(i,s)}$. Because the twiddle of axis i depends only on the bits of axis i, it commutes with the butterflies on all other axes, so one parallel round has exact counterpart

$$z \mapsto D_\zeta H z, \qquad \zeta(x) = \exp\Big(2\pi i \sum_{i \text{ participating}} \frac{b_i j_i^{<v}}{u_i}\Big), \tag{35}$$

where H applies H_0 to slot v of each participating axis and D_ζ multiplies record x by $\zeta(x)$. Since every u_i divides $r = 2^\ell$, the integer

$$J(x) = \sum_{i \text{ participating}} \frac{r}{u_i} b_i j_i^{<v} \bmod r$$

is computed exactly, and the round multiplies every coefficient of record x by the single phase entry $\tilde{\theta}_{2J(x)}$ from Lemma 6.1.

6.4 The direct tensor FFT

Theorem 6.5 combines the last-axis FFT, the record-axis rounds, and the exact layout permutations. Each of the ℓ last-axis levels and at most ℓ record-axis rounds adds error below $6 \cdot 2^{-q}$, while L and U are exact.

Theorem 6.5 (Direct complex tensor FFT)**.** *Assume the parameter family of Proposition A.11 at grid precision q. Suppose $q \geq 8$, $d \geq 2$, $r = 2^\ell$ with $2 \leq \ell < q$, $d\ell \leq C_\ell q$, $1 \leq K \leq \ell - 1$, $t_i \in \{r/2, r\}$ for $1 \leq i < d$, and $q + 2 \leq w \leq C_w q$. On a disk array on the q-grid in natural tensor order, a fixed multitape procedure returns a disk array on the q-grid in natural frequency order on every axis, in component width w (with no $\lg M$ widening), satisfying*

$$\|\widehat{F}_{\mathbf{t}}^{+} z - F_{\mathbf{t}}^{+} z\|_\infty < 12\ell \cdot 2^{-q}, \tag{36}$$

in

$$O(V[dK + \ell d^{\lambda'} + \ell\, \mathsf{M}(q+2)/q]) = O(V[dK + \ell d^{\lambda'} + \ell\, \mathsf{m}(q+2)]) \tag{37}$$

steps, where $V = 2Rw = \Theta(Rq)$.

Proof. The procedure executes four stages sequentially:

(i) Apply a normalized positive DIF FFT on every length-r contiguous record, followed by bit-reversal restoration inside each record.

(ii) Apply the positional map L on the record addresses (Theorem 5.8).

(iii) Run record-axis rounds at named slots $v = \ell - 1, \ldots, 0$ (at $v = \ell - 1$ only axes of length r participate). Chunk rounds ($b_0 \leq v \leq \ell - 2$) call Proposition A.11 with $D = d - 1$, prefix rounds ($v = \ell - 1$ and $v < b_0$) apply exact individual butterflies before a single truncation, and each completed butterfly round is followed by the combined twiddle multiplication of Section 6.3.

(iv) Apply the restoring positional map $U = (LB)^{-1}$ on the record addresses (Theorem 5.8).

Because the last-axis transform acts within records and commutes with record-axis operations and record permutations, (33) and Lemma 6.3 show that the exact counterpart of the four stages is $F_{\mathbf{t}}^{+}$.

Phase table. One master $2r$-entry phase table (Lemma 6.1) serves all rounds. Building it once costs $rq^{O(1)} = o(V)$, since $M \geq r/2$ and r exceeds every fixed polynomial in q under the band $r \geq 2^{a_r q/d}$ with $d \leq b_d q^\varepsilon$. Every subsequent table lookup is a charged sequential scan, so no random access is assumed.

Last axis. At a DIF level of block length u, stream the two halves of each block onto aligned tapes, form each half-sum and half-difference exactly and truncate to $\widetilde{\mathbb{C}}_q$ by Q_q, and multiply the $b = 1$ branch by its stage phase via Lemma 6.2. One scan of the master table extracts the stage table

$(\tilde\theta_{(2r/u)j})_{j<u/2}$, which is replayed with a local rewind across the blocks of each record. Splits, merges and the final bit reversal on the ℓ intra-record bit positions are elementary streams (Lemma A.4); this intra-record reversal is separate from U. Each level costs $O(rq + r\,\mathsf{M}(q+2))$ per record, and the intra-record bit reversal costs $O(rq\ell)$ per record.

Record axes. In each chunk round, Lemma 6.3 gives the layer shape, and setting $C_M = C_\ell + 1$ and $C_{\mathrm{desc}} = 32(C_\ell + C_w + 1)$ satisfies the header hypotheses of Proposition A.11: there are 2^{m_0} records with $m_0 \le d\ell \le C_\ell q$, so $\lg(2 \cdot 2^{m_0}) \le (C_\ell + 1)q$, and the nine-field header has length at most $2(C_\ell + 1)q + O(\lg q) \le C_{\mathrm{desc}} q$ for large q. In each prefix round, the at most $d-1$ participating H_0 factors are applied sequentially and exactly before one truncation Q_q: intermediate values lie in the disk on denominator exponent at most $q + d - 1$, and temporary sums have modulus at most 2, requiring $q + d + O(1) = O(q)$ bits. In both kinds of round, for each record x, the index $J(x)$ is computed in time polynomial in q, $\tilde\theta_{2J(x)}$ is fetched by one scan of the master table, and every coefficient of the record is multiplied by $\tilde\theta_{2J(x)}$ via Lemma 6.2. The per-record address and scan overhead is $O(V)$ by Lemma A.5. Every record-axis round thus implements

$$z \mapsto Q_q(\tilde D\, Q_q(Hz)), \tag{38}$$

where $\tilde D$ is the recordwise diagonal of stored phases.[1]

Error. Put $u_0 = 2^{-q}$, let z be a disk array, and set $z' = Q_q(Hz)$. By (27), Lemma 6.1 and Lemma 6.2,

$$\begin{aligned}\|Q_q(\tilde D z') - D_\zeta H z\|_\infty &\le \|Q_q(\tilde D z') - \tilde D z'\|_\infty + \|(\tilde D - D_\zeta)z'\|_\infty + \|D_\zeta(z' - Hz)\|_\infty \\ &< (\sqrt{2} + 2 + \sqrt{2})u_0 < 6u_0.\end{aligned}$$

The same $6u_0$ bound holds for each last-axis level (its $b = 0$ branch incurs one truncation, its $b = 1$ branch two truncations and one phase error). Every computed intermediate array entering a round is a disk array, because H_0 layers preserve the disk, Q_q does not increase modulus, $|\tilde\theta_j| \le 1$, and positional maps permute values. Since every exact round A is a contraction in $\|\cdot\|_\infty$, for computed input x and ideal input x_0 we have $\|\widehat{A}x - Ax_0\|_\infty \le \|\widehat{A}x - Ax\|_\infty + \|x - x_0\|_\infty$. Summing the local bound $6u_0$ over ℓ last-axis levels and at most ℓ record-axis rounds (with L and U exact by Theorem 5.8) gives (36).

Time. The at most $b_0 + 1 \le K$ prefix rounds cost $O(VdK)$, and the at most ℓ chunk rounds cost $O(V\ell d^{\lambda'})$. Scalar complex products across the last-axis FFT and the record-axis twiddles cost $O(V\ell + R\ell\,\mathsf{M}(q+2)) = O(V\ell\,\mathsf{M}(q+2)/q)$. By Theorem 5.8, the positional maps L and U cost $O(V[(m_0/d + K)d^{\lambda'} + K]) = O(V\ell d^{\lambda'})$, since $m_0/d \le \ell$ and $K \le \ell - 1$. All subroutines run sequentially on a fixed number of tapes and clean their workspaces on return. □

6.5 Interface with the bridge

Corollary 6.6 instantiates Theorem 6.5 as the middle transform of the fractional-shift bridge (Theorem 4.7), where p is the target precision, P is the bridge's internal working precision, and $q = P + 2$ is its grid precision, with component width $w = q + 2$.

Corollary 6.6 (Walsh middle transform)**.** *Assume the hypotheses of Theorem 6.5 at the bridge's grid precision $q = P + 2$ with component width $w = q + 2$ (verified for the root family in Lemma 7.1, with Lemma 5.3 covering the lifted precision $q + 8d$ of the internal controlled-flip batches). Applied*

[1] Unlike monomial twists in $\mathcal{R}_r$, general complex twiddles do not commute with componentwise truncation Q_q, and the rounded map need not be a contraction between arbitrary inputs. The error analysis below therefore compares each computed round directly against its exact contraction $D_\zeta H$ on disk inputs.

to a complete tensor U in natural axis order $(j_1, \dots, j_d)$ with the last axis contiguous and entries in $\mathbb{D}_q$ (in particular the bridge's prepared spread array, which satisfies $\|U\|_\infty \le 1/8$), Theorem 6.5 returns a disk array on the q-grid in natural frequency order $(u_1, \dots, u_d)$, in component width $q+2$, satisfying

$$\|\widehat{F}_{\mathbf{t}}^{+}U - F_{\mathbf{t}}^{+}U\|_\infty < 12\ell \cdot 2^{-q}. \tag{39}$$

In particular, it satisfies the middle-transform contract (Definition 4.5) of Theorem 4.7 with $E_{\mathrm{mid}} = 12\ell$ and output widening $\delta_{\mathrm{mid}} = 0$.

Proof. Immediate from Theorem 6.5 and Proposition 5.5 at grid precision q and width $w = q+2$; internal guard allocations and temporary lifts to $q + 8d$ return to width $q+2$ at completed boundaries. □

The middle transform uses Theorem 5.8 for L and U, where every record carries r coefficients. By contrast, the bridge's line-spreading axis exposures use the chunk interchanges of Lemma A.8 at cost $O(Vd(1+\ell^\tau))$ (Lemma 4.8), since a single-line operation has no long retained record payload.

7 Multiplier, cost analysis, and proof of the main theorem

This section assembles the integer multiplier along the Walsh route, verifies the size hypotheses of all components on the root parameter family, and proves exact recovery of the product from two bridge transforms. It then analyzes the normalized cost gaps, fixes the primary parameter tuple yielding $\kappa = 2^{-46}$ with explicit positive slack, and proves Theorem 1.

7.1 The algorithm and its root parameter family

Given two n-bit integers (for n above the cutoff n_* of Section 7.6; below n_* the machine invokes the independent inner multiplier directly), let $b = \lfloor \log_2 n \rfloor$ and $N_{\mathrm{dig}} = \lceil n/b \rceil$, and let $R = 2^k$ be the least power of two with $R \ge 2N_{\mathrm{dig}}$. Given fixed rational exponents ε, c (chosen in Section 7.5), define

$$p = 2b + 2k + 6, \qquad d = \lfloor k^\varepsilon \rfloor, \qquad \ell = \lceil k/d \rceil, \qquad r = 2^\ell, \qquad K = \lfloor d^c \rfloor. \tag{40}$$

Because $0 \le d\ell - k < d$, we set the first $d\ell - k$ axis lengths to $r/2$ and the remaining axis lengths, including t_d, to r; then $\prod_{i=1}^{d} t_i = R$. The alias width ω, internal working precision P, grid precision $q = P + 2$, and component width $w = q + 2$ are determined by the bridge schedule (24); we call these sizes the *root family*. Fixed rational powers are computed by exact integer comparisons: for $\varepsilon = \varepsilon_1/\varepsilon_2$, the integer $d = \lfloor k^\varepsilon \rfloor$ is the largest $y \in \mathbb{N}$ satisfying $y^{\varepsilon_2} \le k^{\varepsilon_1}$. Every Fourier length is a power of two; no prime length, Chinese remainder map, or coprime-length resampling occurs. As in Section 6, $\mathsf{M}(N) = N\,\mathsf{m}(N)$ denotes the cost of the independent inner multiplier on N-bit integers.

The multiplier packs the radix-2^b digits of the two inputs into one complex vector of length R and recovers the product polynomial from two one-dimensional transforms F_R (Proposition 7.2). Each F_R is evaluated by the fractional-shift bridge of Theorem 4.7, using direct axis exposures (Lemma 4.8) and spreading along axes $a = d, \dots, 2$ with the negative half-bin carrier. It then applies the direct complex tensor FFT middle transform of Corollary 6.6 (whose layout L and restoring map $U = (LB)^{-1}$ are executed by Theorem 5.8), followed by chronological deconvolutions $B_2^{-1}, \dots, B_d^{-1}$.

The next lemma verifies the size and band hypotheses of all components along the root family, uniformly in n, at both the bridge grid precision q and the lifted precision $q + 8d$ of the Walsh gadgets (Lemma 5.3).

Lemma 7.1 (Uniform root family)**.** *Fix any rational exponents $0 < \varepsilon < \frac{1}{14}$ and $c > 0$ with $\varepsilon(1+c) < 1$ (in particular the primary parameter tuple of Table 6, as well as the comparison rows of Appendix C). For all sufficiently large n: $RP = \Theta(n)$, $P = \Theta(q) = \Theta(k) = \Theta(\log n)$, $d = \Theta(P^{\varepsilon})$, $K = \Theta(P^{\varepsilon c})$, and $\ell = \Theta(P^{1-\varepsilon})$. At both precisions $q' \in \{q,\ q+8d\}$, the comparison bands of Proposition A.11 hold with the fixed constants*

$$a_d = \tfrac{1}{4}, \quad b_d = 1, \quad a_r = \tfrac{1}{128}, \quad b_r = \tfrac{1}{8}, \quad C_\ell = 1, \quad C_M = 2, \quad C_w = 2, \quad C_{\mathrm{desc}} = 128, \tag{41}$$

that is, $a_d(q')^{\varepsilon} \le d \le b_d(q')^{\varepsilon}$ and $2^{a_r q'/d} \le r \le 2^{b_r q'/d}$. Moreover $dK = o(P)$, $K = o(\ell)$, $K/\log P \to \infty$, and $r/P^j \to \infty$ for every fixed j.

Proof. Sizes. From $2N_{\mathrm{dig}} \le R < 4N_{\mathrm{dig}}$ we have $k \le b \le 2k$ for large n and $b/k \to 1$, so the bridge schedule (24) gives $P/k \to 20$ and $q/k \to 20$. In particular, $16k \le q \le 32k$ for all $k \ge 4096$, and $q + 8d \le 64k$. Since $R = \Theta(n/\log n)$, we also have $RP = \Theta(n)$.

Bands. Because $0 < \varepsilon < \frac{1}{14}$, once $k^{\varepsilon} \ge 2$ we have $\frac{1}{2}k^{\varepsilon} \le d \le k^{\varepsilon}$. Hence, for $q' \in \{q, q+8d\}$,

$$d \ge \tfrac{1}{2}\Big(\frac{q'}{64}\Big)^{\varepsilon} \ge \tfrac{1}{2}\,64^{-1/14}(q')^{\varepsilon} > \tfrac{1}{4}(q')^{\varepsilon}$$

(since $64^{1/14} = 2^{3/7} < 2$), while $d \le k^{\varepsilon} \le (q')^{\varepsilon}$. Because $k/d \le \ell \le 2k/d$ and $16k \le q' \le 64k$, we obtain $q'/(128d) \le \ell \le q'/(8d)$. The remaining constants of (41) hold as well: $d\ell \le k + d \le 2k < q'$ and $\lg(2 \cdot R/r) \le k+1 < 2q'$. The nine-field headers have length $O(k) \le C_{\mathrm{desc}} q'$, and the component widths $w = q + 2$ and $w_* = q + 2 + 8d$ satisfy $q' + 2 \le w_{q'} \le 2q'$.

Growth. Finally, $dK/k = O(k^{\varepsilon(1+c)-1}) \to 0$, which implies $K = o(\ell)$ since $\ell = \Theta(k/d)$; the limits $K/\log P \to \infty$ and $r/P^j \to \infty$ follow from $\varepsilon c > 0$ and $\ell \ge k^{1-\varepsilon}$. □

7.2 Exact recovery

By the two-call convolution identity (8), two bridge transforms and a pointwise squaring approximate the product polynomial to error well below $\frac{1}{2}$, so nearest-integer rounding recovers its exact coefficients.

Proposition 7.2 (Two-call root packing)**.** *Let $x_j^{(0)}, x_j^{(1)} \in [0, 2^b)$ $(0 \le j < N_{\mathrm{dig}})$ be the radix-2^b digits of the two n-bit inputs (with $x_j^{(0)} = x_j^{(1)} = 0$ for $N_{\mathrm{dig}} \le j < R$), and set $x_j = 2^{-b}(x_j^{(0)} + i x_j^{(1)})$ for $0 \le j < R$. Compute $\hat{y} \approx F_R x$ by the bridge, $\hat{w} = \overline{\mathcal{Q}_{p+2}(\hat{y} \odot \hat{y})}$, and $\hat{z} \approx F_R \hat{w}$ by the bridge. Then*

$$c_j = \mathrm{round}(-2^{2b-1} R^2 \operatorname{Im} \hat{z}_j)$$

equals the exact convolution coefficient $\sum_{u+v=j} x_u^{(0)} x_v^{(1)}$, and radix-$2^b$ carry propagation returns the exact $2n$-bit product. Outside the two bridge calls, the cost is $O(R\,\mathsf{M}(P) + n) = O(RP\,\mathsf{m}(P) + n)$.

Proof. The identity. Applying (8) to $2^{-b}x^{(0)}$ and $2^{-b}x^{(1)}$ gives $x^{(0)} \circledast x^{(1)} = -2^{2b-1}R^2 \operatorname{Im} F_R(\overline{(F_R x) \odot (F_R x)})$, which has no cyclic wrap because the product polynomial has degree at most $2N_{\mathrm{dig}} - 2 < R$.

The error. Put $u_0 = 2^{-p}$. Each x_j lies on the $(p+2)$-grid with $\|x\|_\infty < \sqrt{2}$, so Theorem 4.7 gives $\|\hat{y} - y\|_\infty < 0.177021\sqrt{2}\,u_0 < 0.25035\,u_0$, where $y = F_R x$ and $\|y\|_\infty \le \|x\|_\infty < \sqrt{2}$. Since $p \ge 14$, we have $\|\hat{y} \odot \hat{y} - y \odot y\|_\infty \le (\|\hat{y}\|_\infty + \|y\|_\infty)\|\hat{y} - y\|_\infty < 0.7082\,u_0$, and rounding to $\widetilde{\mathbb{C}}_{p+2}$ adds at most $\frac{\sqrt{2}}{8}u_0 < 0.17678\,u_0$. Hence $\|\hat{w} - \overline{y \odot y}\|_\infty < 0.8850\,u_0$ and $\|\hat{w}\|_\infty < 2.0001$. The second bridge call adds less than $0.177021 \cdot 2.0001\,u_0 < 0.35406\,u_0$, and $\|F_R\|_\infty \le 1$, so $\|\hat{z} - F_R(\overline{y \odot y})\|_\infty < 1.2391\,u_0$. Multiplying by $2^{2b-1}R^2 = 2^{2b+2k-1} = 2^{p-7}$ yields an error below $1.2391 \cdot 2^{-7} < \frac{1}{2}$, so rounding to

the nearest integer recovers every c_j exactly. On $\widetilde{\mathbb{C}}_{p+2}$, if $n_j \in \mathbb{Z}$ is the numerator of $\operatorname{Im} \hat{z}_j$, then $-2^{2b-1}R^2 \operatorname{Im} \hat{z}_j = -n_j/512$.

Carries and cost. Each coefficient satisfies $0 \le c_j \le N_{\mathrm{dig}}(2^b-1)^2$. Setting $s_0 = 0$, $\gamma_j = (c_j + s_j) \bmod 2^b$ and $s_{j+1} = \lfloor (c_j + s_j)/2^b \rfloor$ produces the radix-2^b digits γ_j of the product, which is strictly below 2^{2n}, emitting $2n$ bits. Pointwise squaring uses R complex multiplications on $O(P)$-bit integers, and all remaining steps are linear scans. □

By Proposition 7.2, the total running time $\mathsf{M}_{\mathrm{W}}(n)$ of the multiplier satisfies

$$\mathsf{M}_{\mathrm{W}}(n) \le 2\,\mathsf{T}_{\mathrm{W}}(R,P) + O(R\,\mathsf{M}(P) + n), \tag{42}$$

where $\mathsf{T}_{\mathrm{W}}(R,P)$ is the cost of one bridge call on the Walsh route, with no recursive call to M_{W} itself.

7.3 Cost analysis of a bridge call

Put $V = RP$. Because $\varepsilon C_1 < 1$, all internal and external component widths (including the layer guard $\Delta = \lceil C_0 d^{C_1} \rceil = o(P)$ and the $8d$-bit Walsh lift) are $O(P)$, so V is the stored bit volume up to constant factors. One-time and per-line or per-record setup work that is not repeated per coefficient (cycle decompositions, masks, headers, named-slot lists, filter taps, solver parameters, phase tables, and recursion descriptors) takes time polynomial in P per object across at most $(R/r + r)P^{O(1)}$ objects. Hence, for fixed constants $C_{\mathrm{set}}, D_{\mathrm{set}}$, the setup cost satisfies

$$\mathcal{S}(R,P) \le C_{\mathrm{set}}\big((R/r)P^{D_{\mathrm{set}}} + rP^{D_{\mathrm{set}}} + P^{D_{\mathrm{set}}}\big) = o(RP), \tag{43}$$

by Lemma 7.1 and $R/r \ge r/2$.

Operand widths of all inner integer products. Per-coefficient multiplications are delegated to an independent integer multiplier of cost $\mathsf{M}(N) = N\,\mathsf{m}(N)$ on N-bit operands, where we assume only $\mathsf{m}(N) = (\lg N)^{O(1)}$ (for instance Schönhage–Strassen [SS71], with $\mathsf{m}(N) = O(\lg N \lg\lg N)$). Its operands and output occupy dedicated work tapes, leaving the caller's other tapes untouched and cleaning its workspace on return; fixing a regular majorant $\overline{\mathsf{M}}(N) = A\,N(\lg(2N))^J$ of M gives $\overline{\mathsf{M}}(c'N) = O(\overline{\mathsf{M}}(N))$ for every constant $c' > 0$. Every ordinary integer product on the Walsh route occurs at one of four sites and has at most $O(P^2) = O((\log n)^2)$ bits:

(1) *Root packing* (Proposition 7.2): R complex squarings on $\widetilde{\mathbb{C}}_{p+2}$, using $O(P)$-bit integer products of total cost $O(R\,\mathsf{M}(P)) = O(V\,\mathsf{m}(P))$.

(2) *Direct tensor FFT* (Theorem 6.5 at $q = P+2$): ℓ last-axis levels and at most ℓ record-axis combined twiddles, each performing four exact $(q+2)$-bit integer products per complex multiplication (Lemma 6.2), of total cost $O(V\ell\,\mathsf{m}(P))$. (The butterfly layers of Proposition A.11 and the positional permutations of Theorem 5.8 perform no general multiplications.)

(3) *Cyclic deconvolution* (Lemma 4.6(iv)): $O(1)$ multiplications per entry on operands of at most $P + k + 8 = O(P)$ bits across $d-1$ axes, of total cost $O(Vd\,\mathsf{m}(P))$. Truncated binary powering on $O(P+k)$-bit integers for the closures in Lemma 4.6(v) and rational series evaluation on $O(P)$-bit integers in Lemma 4.6(iii) and Lemma 6.1 belong to the setup term (43).

(4) *Line spreading* (Lemma 4.6(ii)): on each of $d-1$ axes, overlap-add blocks of $W_t = \min(2P+1, t_a) = \Theta(P)$ entries packed via signed Kronecker substitution into $(4P+16)$-bit slots perform four exact products of integers of at most $W_t(4P+16)+1 = O(P^2)$ bits per block across $O(R/P)$ blocks per axis, costing $O(Vd\,\mathsf{m}(P^2))$.

In particular, because no polynomial-ring multiplication in $\mathcal{R}_r$ is performed, no $\Theta(rP)$-bit packed product arises, and both $\mathsf{m}(P)$ and $\mathsf{m}(P^2)$ are $(\lg P)^{O(1)}$.

Combining Theorem 5.8 (under which both L and $U = (LB)^{-1}$ cost $O(V[(m_0/d+K)d^{\lambda'}+K]) = O(V\ell d^{\lambda'})$), Theorem 6.5, Lemma 4.8 and Lemma 4.6 yields the normalized cost breakdown in Table 5.

Table 5: Cost of one bridge call on the Walsh route, divided by $V = RP$, with $d = \Theta(P^{\varepsilon})$, $\ell = \Theta(P^{1-\varepsilon})$ and $K = \Theta(P^{\varepsilon c})$. A row with gap g has normalized cost $O(P^{1-g})$, up to polylogarithmic factors in P where indicated (Section 7.4).

operation	normalized cost	gap below P^1
individual prefix butterfly rounds	dK	$1-\varepsilon(1+c)$
butterfly layers and positional layouts L, U	$\ell d^{\lambda'}$	$\varepsilon(1-\lambda')$
complex twiddles and last-axis FFT	$\ell\,\mathsf{m}(P)$	ε, up to logarithms
axis exposures and restorations	$d(1+\ell^{\tau})$	$(1-\varepsilon)(1-\tau)$
line spreading and cyclic solves	$d(\mathsf{m}(P)+\mathsf{m}(P^2))$	$1-\varepsilon$, up to logarithms

Summing the rows of Table 5 and absorbing (43) gives

$$\mathsf{T}_{\mathrm{W}}(R,P) = O\big(RP\,[dK + \ell d^{\lambda'} + \ell\,\mathsf{m}(P) + d(1+\ell^{\tau}) + d(\mathsf{m}(P)+\mathsf{m}(P^2))]\big). \tag{44}$$

Here the direct axis exposures of Lemma 4.8 contribute $d(1+\ell^{\tau})$ in place of the $d^2(1+\ell^{\tau})$ adjacent-axis schedule of [OAI26, §9]. Executing L and $U = (LB)^{-1}$ via Theorem 5.8 eliminates the external chunk-exchange term $PK^{\tau-1}$, while the internal packed basis changes of Proposition A.11 are already included in $d^{\lambda'}$.

7.4 Gaps and slack

By Table 5, a row with gap g contributes $O(V\,P^{1-g}(\lg P)^{O(1)})$ steps. An admissible parameter tuple therefore yields the saving exponent κ whenever $0 < \kappa < g_{\min}$, where $g_{\min}$ is the minimum of the five gaps

$$g_{\mathrm{pre}} = 1-\varepsilon(1+c), \quad g_{\mathrm{lay}} = \varepsilon(1-\lambda'), \quad g_{\mathrm{last}} = \varepsilon, \quad g_{\mathrm{axis}} = (1-\varepsilon)(1-\tau), \quad g_{\mathrm{short}} = 1-\varepsilon. \tag{45}$$

Any strict positive slack $s = g_{\min} - \kappa > 0$ absorbs fixed powers of $\lg P$: for every integer $j \geq 0$ and $P \geq 2$,

$$(\lg(2P))^j \leq 4^j (\ln P)^j \leq j!\,(4/s)^j\,P^s,$$

by the j-th term of the exponential series for $e^{s\ln P}$.

Linear vs. quadratic bottleneck. On the Walsh route, the binding gap is the butterfly-layer gap $g_{\mathrm{lay}} = \varepsilon(1-\lambda')$. Since the layer hypotheses require $\lambda' > \sigma$ and $\varepsilon < 1/C_1$, this gap is at most $(1-\sigma)/C_1$, which is *linear* in the complex network's exponent deficit $1-\sigma$. By contrast, if the layout L is executed by width-K chunk exchanges as on a Bluestein middle transform (Appendix C, (59)), the normalized cost term $PK^{\tau-1}$ introduces the additional gap $g_{\mathrm{chunk}} = \varepsilon c(1-\tau)$; because Proposition A.11 requires $c < \beta(1-\tau)/\tau$ and $\varepsilon < 1/C_1$, this forces

$$g_{\mathrm{chunk}} = \varepsilon c(1-\tau) < \frac{\beta(1-\tau)^2}{C_1\tau} < \frac{(1-\tau)^2}{C_1\tau},$$

capping κ quadratically in $1-\tau$.

7.5 Parameter choice

By Proposition A.2 (Appendix A), at $h = 100$ ($m = 10^6$) with $\xi = 2^{-42}$, the exact network counts of Fact 3.4 together with the rational bound $\ln 10^6 < 14$ certify that $\tau = 1 - 4\xi = 1 - 2^{-40}$ and $\sigma = 1 - \xi = 1 - 2^{-42}$ satisfy $s_b/W_b < m^\tau$ and $s_c/W_c < m^\sigma$. Likewise, $B_{\text{net}} < m^{11}$ ($\nu_{\text{net}} = 11$ in (47)) certifies that the guard exponent $C_1 = \nu_{\text{net}} + 3 = 14$ is admissible in Proposition A.11. Table 6 records the parameter tuple used to prove Theorem 1.

Table 6: Primary parameter tuple for the Walsh route ($\xi = 2^{-42}$), yielding $\kappa = 2^{-46}$ in Theorem 1.

κ	C_1	ε	β	c	τ	σ	λ	λ'
2^{-46}	14	1/15	1/64	$\xi/256$	$1-4\xi$	$1-\xi$	$1-63\xi/64$	$1-31\xi/32$

We verify the admissibility inequalities of Proposition A.11 and the gaps of (45) for Table 6 in exact rational arithmetic:

- *Layer and guard hypotheses.* We have $\max\{\tau, \sigma\} = 1 - \xi < \lambda = 1 - \frac{63}{64}\xi < 1$ with margin $\lambda - \sigma = \frac{\xi}{64} > 0$. Since $c/\beta = \xi/4$, we have $\tau(1 + c/\beta) = (1 - 4\xi)(1 + \xi/4) = 1 - \frac{15}{4}\xi - \xi^2$, giving the strict margin $\lambda - \tau(1 + c/\beta) = \frac{177}{64}\xi + \xi^2 > 0$. Moreover $\sigma + \beta(1 - \sigma) = 1 - \frac{63}{64}\xi = \lambda$, so $\max\{\lambda, \sigma + \beta(1 - \sigma)\} = \lambda < \lambda' = 1 - \frac{31}{32}\xi < 1$ with margin $\lambda' - \lambda = \frac{\xi}{64} > 0$. Finally, $C_1 = 14$ is admissible and $1 - \varepsilon C_1 = \frac{1}{15} > 0$.
- *Gaps and slack.* Evaluating (45) gives

$$g_{\text{lay}} = \frac{31}{480}\xi, \qquad g_{\text{axis}} = \frac{56}{15}\xi, \qquad g_{\text{last}} = \frac{1}{15}, \qquad g_{\text{short}} = \frac{14}{15}, \qquad g_{\text{pre}} = \frac{14}{15} - \frac{\xi}{3840} > \frac{13}{15}.$$

  The unique minimum is $g_{\min} = g_{\text{lay}} = \frac{31}{480}\xi$. Comparing against $\kappa = 2^{-46} = \frac{\xi}{16} = \frac{30}{480}\xi$ gives strict positive slack $s = g_{\min} - \kappa = \frac{\xi}{480} > 0$ (with ratio $g_{\min}/\kappa = 31/30$).

Remark 7.3 (Exponent with the printed constants of [OAI26], and comparison rows). To isolate the gain of the Walsh middle transform from parameter retuning, fix the structural constants $\beta = \frac{1}{2}$, $\tau = \sigma = 1 - 2^{-50} = 1 - 2u$ ($u = 2^{-51}$) and $C_1 = 20$ printed in [OAI26, Prop. 14]. With these constants, the printed statements [OAI26, Lemma 12, Prop. 14] apply directly without Appendix A. Taking $\varepsilon = 1/21$, $c = u/4$, $\lambda = 1 - 129u/128$ and $\lambda' = 1 - 63u/64$ satisfies all hypotheses of [OAI26, Prop. 14] (with $\tau(1 + c/\beta) = 1 - \frac{3}{2}u - u^2 < \lambda$, $\sigma + \beta(1 - \sigma) = 1 - u < \lambda'$, and $1 - \varepsilon C_1 = 1/21$) and gives $g_{\min} = g_{\text{lay}} = 3u/64 = 3 \cdot 2^{-57}$, yielding $\kappa = u/32 = 2^{-56}$ with slack 2^{-57}. Because any choice of the free parameters with these fixed structural constants must satisfy $\lambda' > \sigma + \beta(1 - \sigma) = 1 - u$ and $\varepsilon < 1/C_1 = 1/20$, one has $\kappa < g_{\text{lay}} = \varepsilon(1 - \lambda') < u/20 = \frac{8}{5}\,2^{-56} < 2^{-55}$; thus 2^{-56} is the largest exponent of the form 2^{-j} ($j \in \mathbb{N}$) that our analysis admits with those constants. Appendix C gives the full comparison with a Bluestein middle transform (reproducing 2^{-182} at the parameters of [OAI26, (38)] and reaching 2^{-84} with retuned parameters) together with the $C_1 = 20$ Walsh row ($\kappa = 2^{-47}$).

7.6 The cutoff and proof of Theorem 1

Because the tape procedures derived from Facts 3.3–3.4 have unspecified implied constants and eventual thresholds, we define the fallback threshold symbolically (see Section 9 on the resulting astronomical activation sizes).

For the tuple of Table 6, put $s = g_{\min} - \kappa = \xi/480 > 0$. Fix integers $C \geq 2$, $j \geq 1$ and $D_{\mathrm{set}} \geq 1$ such that every normalized cost term in (44) and (42) is at most $CP^{1-g_{\min}}(\lg(2P))^j$ and (43) holds with coefficient at most C and degree at most D_{set}. Let P_{tape} be a common precision threshold, under the bands (41), for the butterfly layer (Proposition A.11) and the arbitrary-width interchange (Lemma A.8), and let C_0 be the guard constant (47) of Proposition A.11 (Appendix A). For $\alpha_0 > 0$ and $b_* \geq 1$ put $\Phi(\alpha_0, b_*) = 2^{\lceil (64b_*/\alpha_0)^2 \rceil}$, and put $K_{\log} = 2^{\lceil \max\{2,(16j/s)^2,(8/s)\lg(2C)\} \rceil}$. Define the constant

$$K_* = \Big\lceil \max\Big\{4096,\ P_{\mathrm{tape}},\ (4C_0)^{1/(1-\varepsilon C_1)},\ \Phi(\varepsilon c, 1),\ 2^{2/\varepsilon},\ 2^{2/(1-\varepsilon(1+c))},\ \Phi\big(1-\varepsilon, \lceil D_{\mathrm{set}} + 2 + \lg(8C)\rceil\big),\ K_{\log}\Big\}\Big\rceil \tag{46}$$

(where K_* is a fixed threshold, not the chunk width K) and set $n_* = 2^{4K_*}$.

For $n \geq n_*$ we have $b = \lfloor \log_2 n \rfloor \geq 4K_*$, so $2N_{\mathrm{dig}} \geq 2n/b > 2^{2K_*}$ and hence $k \geq K_*$. Let us verify that $k \geq K_*$ activates all asymptotic bounds simultaneously. If $k \geq \Phi(\alpha_0, b_*)$ and $z = \log_2 k$, then $z \geq (64b_*/\alpha_0)^2$ implies $(\ln 2)\alpha_0 z \geq 32 b_* \sqrt{z} \geq \ln(b_*(z+5)^2)$, so $k^{\alpha_0} \geq b_*(z+5)^2$. Taking $\alpha_0 = \varepsilon c$ and $b_* = 1$ makes $K \geq 6$ and $K/\log P$ large (since $q \leq 32k$ gives $\log_2 q \leq z + 5$); taking $\alpha_0 = 1 - \varepsilon$ and $b_* = \lceil D_{\mathrm{set}} + 2 + \lg(8C) \rceil$ gives $r \geq 8CP^{D_{\mathrm{set}}+1}$ and $R/r \geq r/2 \geq 4CP^{D_{\mathrm{set}}+1}$, so the setup (43) is at most RP. The term $(4C_0)^{1/(1-\varepsilon C_1)}$ ensures $\Delta \leq C_0 k^{\varepsilon C_1} + 1 \leq k/2$, and the terms $2^{2/\varepsilon}$ and $2^{2/(1-\varepsilon(1+c))}$ ensure $d \geq 2$ and $dK \leq k/4$. Finally, $K_{\log}$ ensures $\ln C + j \ln\lg(2P) \leq \frac{s}{2} \ln P$, so half of the slack s survives all implied constants and polylogarithmic factors.

Proof of Theorem 1. Fix the parameter tuple of Table 6 ($\kappa = 2^{-46}$) and the cutoff $n_* = 2^{4K_*}$ from (46). Let the independent inner multiplier have cost $\mathsf{M}(N) = N(\log N)^{O(1)}$ (for instance Schönhage–Strassen [SS71]), and for $n < n_*$ invoke this inner multiplier directly on the two n-bit inputs. We verify parameter admissibility, correctness, outer CRT-freeness, running time, and machine uniformity for $n \geq n_*$.

Admissibility. Proposition A.2 certifies the network exponents $\tau = 1 - 2^{-40}$ and $\sigma = 1 - 2^{-42}$ and the admissibility of $C_1 = 14$ at $h = 100$, and the strict margins of Proposition A.11 hold as checked after Table 6. Since $\varepsilon = 1/15 < 1/14$ and $\varepsilon(1+c) < 1$, Lemma 7.1 guarantees the comparison bands (41), component widths, complete rectangular shapes and long-record conditions at both precisions $q = P + 2$ and $q_* = q + 8d$.

Correctness and outer structure. All transform lengths $R = 2^k$ and $t_1, \ldots, t_d \in \{r/2, r\}$ are powers of two; the outer reduction uses no Chinese remainder map, no prime transform length, no coprime-length resampling, and no Fourier-grid expansion. Inside each bridge call, Theorem 5.8 executes the layout map L and the restoring map $U = (LB)^{-1}$ exactly on $\widetilde{\mathbb{C}}_q$, returning from the lifted precision $q + 8d$ to $\widetilde{\mathbb{C}}_q$ at proved exact permutation boundaries. Theorem 6.5 and Corollary 6.6 therefore supply a middle transform at grid precision q with $E_{\mathrm{mid}} = 12\ell$ and $\delta_{\mathrm{mid}} = 0$. By Theorem 4.7 and Lemma 4.8, the two bridge transforms satisfy the precision bound (25), and Proposition 7.2 recovers the exact $2n$-bit product.

Inner products and running time. As verified in Section 7.3, every ordinary integer product in the algorithm has at most $O(P^2) = O((\log n)^2)$ bits and is executed by the independent multiplier M, with no recursive call to the algorithm of Theorem 1. By (42), (44), the logarithmic absorption of Section 7.4, and the positive slack $s = g_{\min} - \kappa = \xi/480 > 0$, the running time for $n \geq n_*$ satisfies $\mathsf{M}_{\mathrm{W}}(n) = O(RP \cdot P^{1-\kappa})$. Since $RP = \Theta(n)$ and $P = \Theta(\log n)$ by Lemma 7.1, this is $O(n(\log n)^{1-2^{-46}})$ uniformly for $n \geq n_*$. On the finitely many input lengths $n < n_*$, the direct call to the inner multiplier takes bounded time, absorbed into the implied constant.

Uniformity. All lists of axis lengths, masks, phase and filter tables, and positional-map decompositions are generated deterministically by the machine as tape data, and all subroutines

are sequential compositions of fixed-tape procedures with a fixed finite alphabet and tape count independent of n. □

Remark 7.4 (CRT-free inner multiplication). Theorem 1 treats the inner multiplier M as a black box on $O((\log n)^2)$-bit integers and does not depend on [MS]. If one instantiates M instead with the two-dimensional fractional-shift $O(N \log N)$ multiplier of [MS] in place of [SS71], then the inner integer products likewise use only power-of-two Fourier transforms without a Chinese remainder map or prime transform lengths. Its written analysis uses faithful phase and filter tables as in Section 4. In the formal proof of Theorem 2, the inner multiplier is supplied by an independent Lean 4 formalization of the two-dimensional multiplier of [MS] (headline `IntMulTM.intMul_TM2_nlogn`), developed for this paper and kernel-checked as part of [Lean26] (Section 8). The companion manuscript [MS] distributes its own, separate Lean development (headline `IntMul.tmIntMul_evalsToInTime_n_log_n`).

8 Formal verification in Lean 4

What is proved. Theorem 2 establishes the $O(n(\log n)^{1-2^{-46}})$ complexity bound of Theorem 1 in the Lean 4 proof assistant [dMU21] against its mathematical library Mathlib [mC20]; the formal development is [Lean26]. Our written proofs in Sections 4–7 take the elementary stream primitives (Fact 3.3) and the $h = 100$ finite-network certificates (Fact 3.4) of [OAI26] as black boxes. By contrast, the formal proof assumes no result of [OAI26], of [MS], or of any other paper as a hypothesis or axiom. Both triple networks of [OAI26, §3] at $h = 100$ and their rank and endpoint certificates, all multi-stack Turing machine procedures, the parameter-exposed reductions of Appendix A, and the $O(N \log N)$ inner integer multiplier are defined and proved inside Lean. The formalization uses only the three standard axioms of Lean's logic (`propext`, `Classical.choice`, and `Quot.sound`).

The formal statement. The top-level theorem of the development is

```
theorem SubNLogN.Assembly.intMul_challenge : SubNLogN.Challenge.IntMulSubNLogN
```

The proposition `IntMulSubNLogN` and its input encoding are defined in the self-contained statement file `SubNLogN/Challenge.lean`, inside `namespace SubNLogN.Challenge` with `Computability` open. We quote them verbatim, omitting only docstrings, the decoder with its helper functions and correctness lemmas, and two sanity checks:

```
inductive PairSym
  | bit (b : Bool)
  | sep
  deriving DecidableEq, Fintype

def encodeNatPair (p : ℕ × ℕ) : List PairSym :=
  (encodeNat p.1).map PairSym.bit ++ PairSym.sep :: (encodeNat p.2).map PairSym.bit

def encNatPair : Encoding (ℕ × ℕ) PairSym where
  encode := encodeNatPair
  decode := decodeNatPair
  decode_encode := decodeNatPair_encodeNatPair

def IntMulSubNLogN : Prop :=
```

```
∃ M : Turing.TM2ComputableInTime encNatPair.encode encodingNatBool.encode
    (fun p : ℕ × ℕ => p.1 * p.2),
  (∀ k, Finite (M.tm.Γ k)) ∧
  ∃ C N₀ : ℕ, 0 < C ∧ ∀ L ≥ N₀,
    (M.time L) ^ (2 ^ 46) ≤ (C * L) ^ (2 ^ 46) * (Nat.log2 L) ^ (2 ^ 46 - 1)
```

In words, `IntMulSubNLogN` asserts that there exists a single multi-stack Turing machine M in Mathlib's model, with finite stack alphabets on all stacks, that transforms bits(a) `sep` bits(b) into bits(ab) for all $a, b \in \mathbb{N}$ and halts within $C\,L\,(\log_2 L)^{1-2^{-46}}$ steps on all inputs of length $L \geq N_0$. Here bits(x) is least-significant-bit-first binary with no zero at the most-significant end, empty for $x = 0$.

Correspondence to Theorem 1.

(a) *Machine model.* `Turing.TM2ComputableInTime` packages a machine of Mathlib's `Turing.FinTM2` model with input/output alphabet identifications and a time bound `M.time`; the conjunct `∀ k, Finite (M.tm.Γ k)` enforces finiteness of every stack alphabet (Mathlib itself requires this only of the input alphabet). Each stack is a tape whose head remains at its top cell, and one `FinTM2` step executes a bounded number of push, pop, peek, and state-branch operations, costing $O(1)$ standard multitape steps. Translating between fixed-width n-bit operand encodings and `encNatPair`/`encodingNatBool` takes $O(n)$ steps (these routine model translations are not formalized).

(b) *Input, output, and worst-case time.* Execution starts with `encodeNatPair (a, b)` on the input stack and all other stacks empty, and must reach Mathlib's halting configuration holding bits(ab) on the output stack with all other stacks empty within $T(L) =$ `M.time L` steps, for every $(a, b) \in \mathbb{N}^2$. For $L = |\text{bits}(a)| + |\text{bits}(b)| + 1 \geq N_0$, taking 2^{46}-th roots of $T(L)^{2^{46}} \leq (CL)^{2^{46}} \lfloor \log_2 L \rfloor^{2^{46}-1}$ gives $T(L) \leq C\,L\,(\log_2 L)^{1-2^{-46}}$. For n-bit operands $0 \leq a, b < 2^n$ ($n \geq 2$), stripping leading zeros gives $1 \leq L \leq 2n+1$: inputs with $L < \max(N_0, 2)$ take at most $\max_{1 \leq L' < \max(N_0,2)} T(L') = O(1)$ steps, while inputs with $\max(N_0, 2) \leq L \leq 2n+1$ satisfy $T(L) \leq C(2n+1)(\log_2(2n+1))^{1-2^{-46}}$ by monotonicity, so together with the $O(n)$ fixed-width wrapper this yields $O\big(n(\log n)^{1-2^{-46}}\big)$.

(c) *Exponents.* Alongside `intMul_challenge` (and `intMul_TM2_sub_nlogn_46`), the development derives by exponent monotonicity with the same machine the bounds with exponents 2^{-47} (`intMul_TM2_sub_nlogn_47`) and 2^{-56} (`intMul_TM2_sub_nlogn_56`) as well as $o(n \log n)$ (`intMul_TM2_little0_nlogn`). Only the $\kappa = 2^{-46}$ Walsh parameterization is constructed as a formal machine; the printed-constants parameterization and the Bluestein comparison route (Appendix C) are not formalized separately.

(d) *Natural numbers.* The formal statement is over $\mathbb{N}$; signed n-bit integers reduce to $\mathbb{N}$ by a linear-time sign-and-magnitude wrapper (not formalized).

The formalized algorithm. Although `IntMulSubNLogN` is existential, inspection of the definitions confirms that the constructed machine implements the Walsh route at the $\kappa = 2^{-46}$ parameter tuple of Table 6 (`SubNLogN/Math/Cost/Row46.lean`: $C_1 = 14$, $\varepsilon = 1/15$, $\beta = 1/64$, $c = \xi/256$, $\tau = 1 - 4\xi$, $\sigma = 1 - \xi$, $\lambda = 1 - 63\xi/64$, $\lambda' = 1 - 31\xi/32$ with $\xi = 2^{-42}$). It combines the $h = 100$ bit and complex networks, the fractional-shift bridge (Section 4), the direct tensor FFT (Theorem 6.5, Corollary 6.6), and two-call root packing (Proposition 7.2). For the inner $O(N \log N)$ multiplier (and the fallback

below the cutoff K_*), it uses an independent Lean 4 formalization of the two-dimensional multiplier of [MS] (949 files, about 185,000 lines; headline `IntMulTM.intMul_TM2_nlogn`), developed for this paper and kernel-checked as part of it. The companion manuscript [MS] distributes its own, separate Lean development (headline `IntMul.tmIntMul_evalsToInTime_n_log_n`). Syntactic CRT-freeness is not expressed in the formal proposition. Table 7 maps the paper's components to their Lean theorems.

Table 7: Correspondence between the components of the $\kappa = 2^{-46}$ proof and Lean 4 theorems, given without their namespace prefixes (`SubNLogN` and its sub-namespaces under `Math`, `Machine`, `Assembly`, and `S5`; top-level and gate-checked names are listed in `SubNLogN/GateCheck.lean` and the archive `README.md`), except `IntMulTM.intMul_TM2_nlogn`.

Paper component	Lean 4 theorems
Fact 3.3; Lemmas A.4 and A.5	`FinTM2` stream kernels with proved step bounds (`k1PairPerm_holds`, `k2Distribute_holds`, `k3Add_holds`, `k4RotateCtl_holds`, `c3LongRecordScan_holds`) on a framed, re-entrant procedure calculus.
Fact 3.4; Definition A.1, Proposition A.2	`bitNetCert_holds`, `complexNetCert_holds`, `deficitsH100_holds`.
Lemma A.6, Proposition A.7, Lemmas A.8–A.10	`shearFactorization_holds`, `existsValidFac_holds`, `packedAdd_correct`, `cCoreProc_correct`; machine level: `c6_holds`, `c7Extract_holds`.
Proposition A.11 ($C_1 = 14$, $\beta = 1/64$)	`prop71_final`, `prop71Guard_holds` (layer machine inside `walshMiddle_holds`).
Section 5 (Lemmas 5.1, 5.4; Theorem 5.8)	`walsh_exact`, `cnot_batch`, `positional_perm`; machine level: `cnot1Machine_holds`, `positionalPermMachine_of_leaves` (parametric in the layer contracts; hypotheses discharged by `cnotScan_holds` and `posPermSched_holds`).
Lemmas 4.2–4.6; Theorem 4.7	`lemmaA`; `lemmaB_i`, …, `lemmaB_vi`; `lemmaC_ii` and companions; `bridge_precision` (for arbitrary nearest rounding).
Theorem 6.5, Corollary 6.6	`direct_fft`, `walsh_middle`; machine level: `walshMiddle_holds`.
Proposition 7.2	`root_packing`, `bridgeFixed_contract`, `rootW8_Fnew`.
Cost table, gaps, and $\kappa = 2^{-46}$ row (Sections 7.3–7.5)	`headlineCost_walshNat46_W8`, `kappa_lt_gaps`, `wrec_le_rpow`; step budget: `walshCostClause_walshMidData`.
Inner multiplier, $\mathsf{M}(N) = O(N \log N)$	`mulBig_holds` (assembled from the `WP` stage programs that also yield `IntMulTM.intMul_TM2_nlogn`); below-cutoff pipeline `w9_pipeline_headline_runs`; standalone $O(n \log n)$ machine `IntMulTM.intMul_TM2_nlogn` (transported to `encNatPair` by `S5.intMul_TM2_nlogn_subNLogNSpec`).
Theorems 1 and 2 ($\kappa = 2^{-46}$)	`intMul_TM2_sub_nlogn_46`, `intMul_challenge`.

Departures from the written text. The formal construction departs from the written text in the following minor ways:

(1) *Root parameters.* The formal proof uses the root schedule of [MS], $R = 2^{K_0}$ with $p_0 = 2b+2K_0+6$ and $K_0 \le k+2$ (so R is at most four times the value in Lemma 7.1), the same schedule as our independent formalization of the two-dimensional multiplier, and proves the cost chain directly for that family.

(2) *Rounding.* The machine rounds half-way ties upward rather than to even; Theorem 4.7,

Proposition 7.2, and the table bounds are proved in Lean for every nearest rounding.

(3) *Cutoff.* The threshold K_* is defined as the least admissible integer rather than the closed form (46).

(4) *Costs.* Machine steps are counted directly, unrolling the recurrences of Appendix A in the exact form $WF_k \le sF_{k-1} + Wc$.

Conversely, the formal development supplies three items left black-box or informal in the written text. These are a machine-checked proof of the $h = 100$ network certificates (Fact 3.4), a fully typed formulation of Definition A.1(b), and a re-entrant calling contract for the recursive exact complex core of Lemma A.10.

Trust base and kernel checks. Accepting the formal proof requires trusting only four components. First is the Lean 4 kernel (version 4.33.1). Second are the Mathlib definitions referenced in the statement at Mathlib commit `0df444a3` (`Turing.FinTM2`, `Turing.TM2ComputableInTime`, `Computability.encodeNat`, `Computability.decodeNat`, `encodingNatBool`, `Encoding`, and `Finite`), together with Lean core's `Nat.log2`. Third is the 80-line statement file `SubNLogN/Challenge.lean`, whose core specification is quoted above and which imports only two Mathlib modules (SHA-256 `7aa50930921c494e50803b037c4cc6ed9be880c3f795e4b171e16ab59cf61fb5`). Fourth are the three logical axioms `propext`, `Classical.choice`, and `Quot.sound`. Everything else (roughly 299,000 lines across 1,062 files for the sub-$n \log n$ reduction, about 185,000 lines across 949 files in our independent formalization of the two-dimensional multiplier of [MS], and Mathlib's proofs) is verified by the kernel and need not be read line by line. On the distributed snapshot ([Lean26]; comment-stripped sources byte-identical to commit `3480d37`), the following three checks passed (commands and logs in `replay/` of the archive):

(1) Every module of the development was built without errors by Lean 4.33.1 (`lake build`, exit 0), and every declaration was replayed through the Lean kernel with `leanchecker`: each of the 2,011 modules against its imports, and the full import closure of each headline module (874,850 and 555,268 constants, including the Mathlib it uses) into a fresh kernel environment.

(2) `#print axioms` confirmed that `intMul_challenge` and its corollaries depend only on the three standard axioms, with zero occurrences of `sorry`, `axiom`, `native_decide`, `implemented_by`, `extern`, or `unsafe`.

(3) `comparator`[2] verified that the exported theorem has exactly the type stated in `SubNLogN/Challenge.lean`, that its proof uses only `propext`, `Quot.sound` and `Classical.choice`, and replayed the proof both in Lean's kernel and in the independent kernel `nanoda`[3]. The same pipeline rejected seven negative controls: three altered statements (exponent 2^{45} instead of 2^{46}; the finite-alphabet conjunct dropped; $\le$ replaced by $<$) and four altered proofs (`sorry`; an added axiom; an extra axiom used through a lemma; a `native_decide` step).

Availability. The Lean 4 formalization [Lean26] (including `SubNLogN/Challenge.lean`, the axiom-check gate modules, and the scripts that check the comment-stripped sources against the recorded digests and scan for `sorry`, `axiom`, `native_decide`, `implemented_by`, `extern`, and `unsafe`), the exact-arithmetic Python reference implementation [Code26] (Appendix E), and the standalone

[2] https://github.com/leanprover/comparator
[3] https://github.com/ammkrn/nanoda_lib

C/GMP implementation [CGMP26] of the two-dimensional multiplier of [MS] (Section 1.4) are available in our github repository. In [Lean26], the distributed source snapshot `00b8a4d2b263` (first 12 hex digits of the SHA-256 of `MANIFEST.sha256`) has comment-stripped `.lean` sources byte-identical to commit `3480d37` and was rebuilt under Lean 4.33.1 and Mathlib `0df444a3` (`lake build`, exit 0). That byte-identity was checked by the authors with the comparison mode of `tools/check_semantic_freeze.py`, while the archive's standalone mode re-verifies the shipped tree against the recorded digests `MANIFEST_stripped.sha256` and `SPEC_LOCK.sha256`.

9 Discussion and open problems

9.1 Where the exponent comes from

The asymptotic saving κ is governed by fundamentally different mechanisms depending on how record-address bits are grouped into chunks (Section 7). In [OAI26] and on the Bluestein comparison route (Appendix C), the layout map L is executed by width-K chunk interchanges. Under that layout, the layer constraint $c < \beta(1-\tau)/\tau$ and the guard constraint $\varepsilon < 1/C_1$ force the chunk-exchange gap $\varepsilon c(1-\tau)$ below $\beta(1-\tau)^2/(C_1\tau)$, capping κ quadratically in the bit network's deficit $1-\tau$. Executing L and $(LB)^{-1}$ via exact Walsh-layer positional permutations (Theorem 5.8) eliminates this external chunk-exchange cost altogether. The binding constraint becomes the simultaneous butterfly-layer gap $\varepsilon(1-\lambda')$: since Proposition A.11 requires $\lambda' > \lambda > \sigma$ and $\varepsilon < 1/C_1$, every admissible choice satisfies

$$\kappa < \varepsilon(1-\lambda') < \frac{1-\sigma}{C_1},$$

which is *linear* in the complex network's deficit $1-\sigma$.

This inequality bounds κ in terms of the *selected* rational exponent σ; it is not yet the exact ceiling of the $h = 100$ complex network itself. For our certified dyadic choice $1-\sigma = 2^{-42}$ and guard exponent $C_1 = 14$, $(1-\sigma)/C_1 \approx 2^{-45.8}$, so $\kappa = 2^{-46}$ comes within a factor below 2 of the dyadic limit (and at $C_1 = 20$, $(1-\sigma)/C_1 \approx 2^{-46.3}$, approached by $\kappa = 2^{-47}$). The $h = 100$ complex network admits any σ with $s_c/W_c < m^\sigma$, that is, $1-\sigma < \gamma_c = -\ln(1-\eta_c)/\ln m \approx 2.654 \cdot 10^{-13}$ (compared with $2^{-42} \approx 2.274 \cdot 10^{-13}$). Already the rational test of Proposition A.2 at $L = 14$ certifies $\sigma = 1 - \frac{9}{8}\cdot 2^{-42}$, yielding non-dyadic exponents slightly above $2^{-42}/C_1$ up to the fixed-network ceiling γ_c/C_1. When the structural constants are instead fixed to the values $\beta = \frac{1}{2}$, $1-\tau = 1-\sigma = 2^{-50}$, and $C_1 = 20$ printed in [OAI26, Prop. 14], the leaf condition $\lambda' > \sigma + \beta(1-\sigma) = 1 - 2^{-51}$ imposes the tighter ceiling $\kappa < \varepsilon(1-\beta)(1-\sigma) < 2^{-51}/20 < 2^{-55}$, making $\kappa = 2^{-56}$ the largest exponent of the form 2^{-j} admitted by our analysis with those constants. Finally, our guard exponent $C_1 = 14$ matches the threshold $11 + 3 = 14$ noted in [OAI26, §5.4] and proved in Proposition A.11; we make no claim regarding smaller guard exponents.

9.2 Activation sizes, scope, and practical limitations

Theorem 1 is purely an asymptotic bit-complexity result and yields no practical speedup at any feasible input size. Even reaching dimension $d = \lfloor k^\varepsilon \rfloor \geq 2$ at $\varepsilon = 1/15$ requires transform length $R = 2^k$ with $k \geq 2^{15}$, while the chunk-width condition $K = \lfloor d^c \rfloor \geq 6$ at $\kappa = 2^{-46}$ requires $\log_2 k \geq 3840 \cdot 2^{42} \log_2 6$, before accounting for the guard constant $C_0 \approx 10^{139}$ or the unspecified tape-stream thresholds of Facts 3.3–3.4. Moreover, even if all leading constants were 1, the asymptotic saving factor $(\log_2 n)^\kappa$ at $\kappa = 2^{-46}$ would be smaller than $1 + 10^{-12}$ for all $n \leq 10^{12}$. Accordingly, Algorithm 1 itself is executed only at test scale in exact arithmetic (Appendix E).

By contrast, the C/GMP implementation (`algo=1`) of the two-dimensional [MS]-type $O(n \log n)$ multiplier uses a windowed sinc-Gaussian real kernel with additive precision schedule and Nyquist-column alternating-sum identity, and has modest enough constants to execute bit-exactly against GMP up to 10^9 bits. That benchmarked program is not Algorithm 1 and not the negative-carrier or sub-$n \log n$ algorithm. On one Zen 3 core (one vCPU of a lightly loaded KVM guest; measurement details in [CGMP26]) at 10^8 bits, it takes 92.5 s single-threaded vs. 0.606 s for GMP (153× excluding 5.5 s table setup, or 6.31 s and 10.4× on 64 threads). At 10^9 bits, it takes 2613 s single-threaded vs. 8.10 s for GMP (323× excluding 26.0 s table setup), with 67.3 GiB (about 72 GB) peak RAM [CGMP26] (at 10^9 bits the natural threshold selects a second recursion level, $2^{27} \to 2^{14} \to 128$, whose parameters lie outside the regime covered by the written analysis of [MS]; the output is nevertheless bit-exact).

We also emphasize the scope of Theorem 5.8: it permutes the m_0 bit *positions* of a complete binary record address carrying a long contiguous payload. It is not a sub-$n \log n$ routine for arbitrary permutations of 2^{m_0} records, nor does it replace the general field interchange of Lemma A.8 used for the bridge's $O(d)$ axis exposures (where lines carry no long retained payload). Because the gadget forms complex Walsh–Hadamard linear combinations between exact permutation boundaries, it operates outside pure data-movement models [HvdH25, §1.4].

9.3 What became simpler, and what it cost

Relative to [HvdH21] and [OAI26], the outer reduction discards every number-theoretic and polynomial-ring layer around the two $h = 100$ finite networks (Table 1). Working on one power-of-two length $R = t_1 \cdots t_d = 2^k$ eliminates the Chinese remainder map onto distinct odd primes, trial-division prime search, and coprime-length Gaussian resampling with its Fourier-grid expansion. Modulating the sinc-Gaussian kernel by a negative half-bin phase eliminates the Nyquist special case of [MS] and gives one two-term cyclic solver with $\|B_a^{-1}\|_\infty \le \frac{4}{7} t_a$ uniformly across all cycle lengths. Direct axis exposures replace $O(d^2)$ adjacent swaps by $O(d)$ interchanges. Exact Walsh-layer positional permutations replace $O(k/K)$ chunk interchanges, Bluestein's chirp, last-axis twists, $\mathbb{C}[y]/(y^r + 1)$ ring convolutions, M-fold error amplification, $\lg M$-bit output widening, and $\Theta(rP)$-bit packed products by a plain complex tensor FFT. That FFT delegates only $O(P^2)$-bit integer products under a single precision contract $P = O(p + \lg R)$. Correspondingly, the external dependency surface of the written proof shrinks to zero results of [HvdH21] and only the stream primitives and $h = 100$ network certificates of [OAI26, Lemmas 3–4, Props. 7–8] as black boxes (and to zero external results in the Lean 4 formalization).

These simplifications come with concrete costs of their own. The sub-$n \log n$ saving still originates entirely in the two $h = 100$ finite networks of [OAI26]. Turning the simultaneous butterfly layer into an exact positional-permutation engine requires six completed layer calls per slice-batched CNOT stage (at most $36 N_{\text{slice}}$ layer calls per positional map), together with a temporary precision lift of $8d$ fractional bits so intermediate dyadic denominators never truncate (Section 5). Reaching $\kappa = 2^{-46}$ beyond the $\kappa = 2^{-56}$ printed-constants bound also still requires the parameter-exposed re-derivation of the interchange, packed-addition, and butterfly-layer procedures in Appendix A. Once those network-backed tape primitives are in place, however, the entire reduction from integer multiplication to them consists of a fixed convolution kernel, a power-of-two FFT, and a two-term recurrence.

9.4 Open problems

(1) *Larger deficits and smaller guards.* Construct finite linear networks with relative rank deficit substantially larger than $\eta_c \approx 3.67 \cdot 10^{-12}$, or establish butterfly-layer guard bounds with exponent $C_1 < 14$. By Section 9.1, either advance translates linearly into a larger exponent κ.

(2) *Explicit numerical cutoff.* Extract explicit constants for all multi-stack or multitape stream primitives so as to give an explicit numerical upper bound on the threshold n_* in Theorem 1.

(3) *Other applications of Walsh-layer permutations.* Determine whether exact positional permutations via selective butterfly layers improve the exponents of other problems accelerated by the networks of [OAI26], notably matrix transposition [OAI26, §11] and the exact discrete Fourier transform [OAI26b].

(4) *Downstream arithmetic primitives.* Decide whether the CRT-free fractional-shift bridge also simplifies downstream operations such as integer division and square roots [OAI26, §10]. Since these reduce to multiplication by Newton iteration (and $\mathsf{M}(n)/n$ is nondecreasing for our bound), they inherit the $O(n(\log n)^{1-2^{-46}})$ complexity in the standard way, though we have not checked the details (nor those for base conversion).

References

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A Butterfly layer and interchanges with general parameters

Starting from the elementary tape operations (Fact 3.3, [OAI26, Lemmas 3–4]) and the finite linear network certificates (Fact 3.4, [OAI26, Props. 7–8]), this appendix establishes the exact network-exponent and guard certificates at $h = 100$ (Proposition A.2), the parameter-exposed

address interchanges (Proposition A.7 and Lemma A.8), and the packed selected-bit addition (Lemma A.9). From these it derives the recursive complex phase-frame reduction (Lemma A.10) and the simultaneous butterfly layer with general structural and guard parameters (Proposition A.11). In particular, we do not invoke [OAI26, Prop. 14] outside its printed parameter tuple; instead we follow the written arguments of [OAI26, §§4–5] step by step with all exponents and guard parameters exposed. The finite-network constructions themselves are those of [OAI26, §§3.1–3.3, Lemmas 5–6], used here solely through the finite algebraic data of Definition A.1 ([OAI26, Props. 7–8, §§3.4–3.5]).

Conventions. We use the machine model and array conventions of Section 3.2 and Appendix D ([OAI26, §2]). Arrays are stored in lexicographic order with the first field most significant, and addresses are implicit. A trailing payload of B bits may be viewed as a final field of range $[B]$, and an address permutation f moves the entry at x to $f(x)$. The logical volume V is the number of data bits of the current array; temporary copies and parked streams are charged in time but never replace V in a recurrence. Every machine has a fixed finite alphabet and a fixed finite number of tapes; implied constants may depend on the fixed networks, the fixed rational exponents and the fixed comparison constants, but not on field lengths or data. Every supplied shape is given by a fixed number of canonical binary lengths unless a longer list is explicitly charged to the caller, and products of adjacent spectator lengths are formed before a primitive is called. As in Sections 5 and 6, q denotes the dyadic grid precision at which the layer operates, and Q_q truncates real and imaginary components toward zero to $2^{-q}\mathbb{Z}$ (so that $\|Q_q z\|_\infty \le \|z\|_\infty$ and $\|Q_q z - z\|_\infty < \sqrt{2}\cdot 2^{-q}$). Finally, Γ is the self-delimiting header code (60) ([OAI26, (1)]), and q_{rad} is a fixed prime radix.

A.1 Fixed network data, certified exponents, and elementary procedures

Definition A.1 (Fixed network data)**.** The following finite data are fixed by Fact 3.4:

(a) *Bit network.* A finite pointwise XOR circuit on W_b roles, auxiliary roles included, whose completed scalar action is a permutation ρ_b of all inputs, for arbitrary initial auxiliary values. Each terminal and gate carries a rational $m \times m$ matrix, common to all incidences of a gate. With $A_e = M_{\mathrm{head}(e)} - M_{\mathrm{tail}(e)}$ and $s_b = \sum_e \operatorname{rank}_{\mathbb{Q}} A_e$,

$$M_{\mathrm{out}(\rho_b(w))} - M_{\mathrm{in}(w)} = I_m \quad \text{for every input role } w, \qquad s_b < W_b m.$$

(b) *Complex network.* A finite scalar linear circuit on W_c roles with coefficients in $\{0, \pm 1, \pm\frac{1}{2}\}$ and binary subspace labels as in [OAI26, Prop. 8, §3.5]. Adjacent labels are nested, the nonzero edge residuals have fixed orthonormal bases, and their total dimension is $s_c < W_c m$. The scalar circuit exchanges its two data banks with the prescribed signs and restores its auxiliary values for arbitrary initial values; its source, sink and bank corrections are those of [OAI26, proof of Prop. 8, §3.5, pp. 19–20]. We use only that this scalar circuit is fixed and finite; no bound on its number of gates is assumed (Remark A.13).

For the triple-network constructions at $h = 100$ ([OAI26, §3.1, (3)–(4); Props. 7–8, §§3.4–3.5; §5.4, p. 35]),

$$m = 10^6, \quad W_b = 177\,176\,569\,091\,445\,000\,000, \quad s_b = 177\,176\,569\,088\,785\,861\,287\,000\,000,$$

$$W_c = 1\,873\,807\,244\,643\,542\,670\,000, \quad s_c = 1\,873\,807\,244\,636\,671\,267\,308\,000\,000,$$

with exact relative deficits ([OAI26, §3.6])

$$\eta_b = \frac{W_b m - s_b}{W_b m} = \frac{339}{22\,587\,335\,000\,000}, \qquad \eta_c = \frac{W_c m - s_c}{W_c m} = \frac{73}{19\,906\,842\,167\,500}.$$

Following [OAI26, §5.4], we define from the complex-network counts (m, W_c, s_c) the guard constants

$$E_{\mathrm{net}} = 64(W_c + m + 1)^3, \quad B_{\mathrm{net}} = s_c + E_{\mathrm{net}}, \quad \nu_{\mathrm{net}} = \min\{j \geq 1 : m^j \geq B_{\mathrm{net}}\}, \quad C_0 = 128\, m\, B_{\mathrm{net}}^2 \tag{47}$$

(written ν in [OAI26, §5.4], subscripted here to avoid collision with the bridge alias index $\nu \in \mathbb{Z}$ of Section 4). At $h = 100$ the exact integer values are

$$\begin{aligned}
W_b m - s_b &= 2\,659\,138\,713\,000\,000, \qquad W_c m - s_c = 6\,871\,402\,692\,000\,000,\\
E_{\mathrm{net}} &= 421\,070\,402\,185\,313\,436\,016\,911\,541\,181\,936\,724\,563\,007\,566\,900\,236\,040\,789\,184\,640\,064,\\
B_{\mathrm{net}} &= 421\,070\,402\,185\,313\,436\,016\,911\,541\,181\,936\,724\,564\,881\,374\,144\,872\,712\,056\,492\,640\,064,\\
C_0 &= 22\,694\,436\,300\,352\,206\,084\,304\,065\,594\,397\,346\,702\,264\,594\,168\\
&\quad\ 261\,607\,007\,508\,662\,043\,469\,612\,184\,775\,064\,463\,128\,060\,707\,693\\
&\quad\ 386\,195\,338\,619\,369\,397\,054\,146\,227\,284\,214\,284\,288\,000\,000,
\end{aligned}$$

where the three lines of C_0 concatenate to a 140-digit integer ($C_0 \approx 2.2694 \times 10^{139}$, $\log_2 C_0 = 462.93\ldots$). Table 8 records the exact closed-form counts across representative ground-set sizes h, showing the thresholds at which each family first achieves a rank saving ($h = 39$ for the bit network and $h = 22$ for the complex network).

Table 8: Exact closed-form counts ($m = h^3$) for the triple networks of [OAI26, §3]: roles W, edge-rank sum s, signed saving $Wm - s$, and certified rational upper bound on $\ln m$. Only $h = 100$ is used in the theorems.

h	network	W	s	$Wm - s$	upper bound $\ln m$
21	bit	3 244 397 863 700	30 046 370 943 598 100	−2 327 872 400	9.133567313171
21	complex	6 145 204 591 400	56 910 739 919 072 200	−198 116 800	9.133567313171
22	bit	5 628 295 349 600	59 930 092 117 403 200	−3 234 862 400	9.273127360075
22	complex	11 249 136 760 400	119 780 808 120 388 800	104 350 400	9.273127360075
39	bit	4 329 455 044 049 525	256 818 943 756 887 996 302	1 085 777 173	10.990684938389
39	complex	16 598 761 236 611 294	984 621 917 049 702 208 108	744 843 140 678	10.990684938389
100	bit	177 176 569 091 445 000 000	177 176 569 088 785 861 287 000 000	2 659 138 713 000 000	13.815510557965
100	complex	1 873 807 244 643 542 670 000	1 873 807 244 636 671 267 308 000 000	6 871 402 692 000 000	13.815510557965

Proposition A.2 (Network exponents at $h = 100$)**.** *For the integers W_b, s_b, W_c, s_c of the triple networks of [OAI26, §3] at $h = 100$ ($m = 10^6$; Definition A.1),*

$$\frac{s_b}{W_b} < m^{1-2^{-40}}, \qquad \frac{s_c}{W_c} < m^{1-2^{-42}},$$

so $\tau = 1 - 2^{-40}$ and $\sigma = 1 - 2^{-42}$ are admissible in Proposition A.11. Moreover $\nu_{\mathrm{net}} = 11$, so $C_1 = 20$ and $C_1 = 14$ are both admissible. These are exact arithmetic properties of the displayed integers; that the integers are the counts of working networks is Fact 3.4.

Proof. By Definition A.1 ([OAI26, §3.6, (10)]) the relative deficits are $\eta_b = 339/22\,587\,335\,000\,000$ and $\eta_c = 73/19\,906\,842\,167\,500$, with $s_b/W_b = m(1 - \eta_b)$ and $s_c/W_c = m(1 - \eta_c)$. For $u > 0$ and any $L \geq \ln m$,

$$m^{1-u} = m\, e^{-u \ln m} > m(1 - u \ln m) \geq m(1 - Lu). \tag{48}$$

Hence $s/W < m^{1-u}$ whenever $\eta \geq Lu$; with $u = 2^{-e}$, it suffices to check $\eta \cdot 2^e \geq L$. Take $L = 14$: since $\sum_{j=0}^{11} 7^j/j! = 41\,438\,755\,442/39\,916\,800 > 1000$, we have $e^7 > 10^3$ and hence $\ln 10^6 < 14$. The

exact integer comparisons

$$339 \cdot 2^{40} = 372\,734\,441\,816\,064 > 316\,222\,690\,000\,000 = 14 \cdot 22\,587\,335\,000\,000,$$

$$73 \cdot 2^{42} = 321\,057\,395\,310\,592 > 278\,695\,790\,345\,000 = 14 \cdot 19\,906\,842\,167\,500$$

give $\eta_b > 14 \cdot 2^{-40}$ and $\eta_c > 14 \cdot 2^{-42}$. For the guard degree, $10^{21} < W_c + m + 1 < 2 \cdot 10^{21}$ and $s_c < 2 \cdot 10^{27}$, so $10^{60} < B_{\mathrm{net}} < 512 \cdot 10^{63} + 2 \cdot 10^{27} < 10^{66}$; since $m^{10} = 10^{60}$ and $m^{11} = 10^{66}$, we obtain $\nu_{\mathrm{net}} = 11$. □

Remark A.3 (Sharper logarithm enclosure and sharpness). At $h = 100$ the scaled deficits are $\eta_b 2^{40} = 16.50\ldots$ and $\eta_c 2^{42} = 16.12\ldots$, so the bound $\ln m < 20$ used in [OAI26, §3.6] to certify $1 - 2^{-50}$ does not certify $1 - 2^{-40}$ or $1 - 2^{-42}$. Any certified upper bound $L \le 16.125$ suffices, and $L = 14$ in Proposition A.2 is already elementary. A twelve-decimal rational enclosure follows from $\ln 10^6 = 36\,\mathrm{atanh}\,\frac{1}{3} + 12\,\mathrm{atanh}\,\frac{1}{9}$: with $S_N(z) = \sum_{j<N} z^{2j+1}/(2j+1)$ and $0 < \mathrm{atanh}\, z - S_N(z) < z^{2N+1}/((2N+1)(1-z^2))$, taking $N = 20$ yields

$$13.815510557964 < \ln 10^6 < 13.815510557965$$

by finite rational comparisons (giving margins $\eta_b 2^{40} - L = 2.6864\ldots$ and $\eta_c 2^{42} - L = 2.3124\ldots$). Moreover, the exponent $1 - 2^{-40}$ is false for the complex network at $h = 100$: with $u = 2^{-40}$ we have $\eta_c 2^{40} < 5$ and $10 < \ln m < 20$, so $1 - e^{-u \ln m} \ge u \ln m - \frac{1}{2} u^2 (\ln m)^2 > u(10 - 200u) > 5u > \eta_c$, whence $s_c / W_c = m(1 - \eta_c) > m^{1-2^{-40}}$ (and similarly for the bit network at $h = 39$).

Lemma A.4 (Elementary streams)**.** *Fix a radix $q_{\mathrm{rad}} \ge 2$. The following operations have linear bit cost on a fixed number of tapes:*

(1) any fixed permutation of the pair of values at two specified radix-q_{rad} digit positions;

(2) moving one specified radix-q_{rad} digit position to another position, preserving the order of the intervening positions;

(3) splitting at a specified radix-q_{rad} digit, applying a fixed pointwise map to the aligned streams, and merging;

(4) distributing complete rows cyclically among a fixed number W of streams and, when the number of rows in each prefix is a multiple of W (so that all streams have the same shape), merging them in a fixed order of the row classes;

(5) exact fixed-width addition, subtraction, sign changes, and binary shifts by a supplied integer exponent t, when the necessary guard bits are present; a shift of N words costs $O(N(w_{\mathrm{in}} + w_{\mathrm{out}}) + \lg(2 + |t|))$, linear in the input and output widths plus the length of the exponent's encoding, which is read once.

Moreover, let a target field of range $[Q]$, $Q = q_{\mathrm{rad}}^b$, follow its control fields, each relevant control having $O(b)$ digits. A cyclic shift of the target by an offset computable from the controls in $b^{O(1)}$ time costs $O(V)$, and so does multiplication of the target coordinate by a fixed rational unit modulo Q whose numerator and denominator are prime to q_{rad}.

Proof. Items (1)–(4). For (1), split the array among q_{rad}^2 tapes according to the two digit values; within every stream the remaining address fields keep their order, so merging with the permuted pair of labels gives the result. For (2) and (3), use q_{rad} streams and change the place in the nested merge

at which the selected digit is inserted. The number of streams is fixed, and complete spectator fields and trailing payloads are copied without decoding. For (4), send row r to stream $r \bmod W$; each stream keeps the row order, and a merge reads the streams in the prescribed order of row classes, copying every row a constant number of times.

Overheads and item (5). Binary counters are stored least significant bit first. An increment, including the return of the head over the changed prefix, has constant amortized cost, and wraparound and reset costs are charged to the preceding traversal. Initial shape preparation takes at most $C(\log_2(2V))^J$ steps for fixed C, J, which is at most $C'V$ for all $V \geq 1$. This accounting covers counters, head returns and clearing the regions visited by a call. Fixed-width coefficient arithmetic uses ordinary carry scans, and a bit reversal inside a coefficient, when needed, is a charged linear pass. For a shift by a supplied exponent t, partition the source and destination words into the retained interval and the intervals of known zeros or redundant sign bits; then copy the retained interval and fill the rest in one traversal. Exact right shifts discard only known zero low bits, and local counter preparation costs $O(\log(2w))$ per w-bit word, absorbed by its scan. Thus a shift by t is executed in one traversal rather than t one-bit shifts, as [OAI26, §5.4, p. 36] also notes. This gives (5).

Controlled shifts and units. Fix the preceding coordinates. The target fiber consists of Q consecutive suffix blocks, and a rotation splits it at the computed cut and concatenates the two intervals in the opposite order. The offset computation is polynomial in b, whereas the fiber contains at least q_{rad}^b bits, so the fiber scan absorbs the offset work, uniformly in b. Multiplication by a fixed positive integer a prime to Q splits the target coordinates into the at most a intervals on which $x \mapsto ax - kQ$ lies in $[Q]$; on each interval the outputs increase and lie in one residue class modulo a, and these classes are distinct since $\gcd(a, Q) = 1$. Merging the fixed number of streams in output order gives the product, and the inverse uses the reversed schedule. A negative multiplier adds the negation $y \mapsto -y \bmod Q$, which fixes the block at $y = 0$ and reverses the order of the other $Q - 1$ blocks while preserving each block's internal order: push the blocks onto a temporary tape and reverse each popped block once more, as in [OAI26, proof of Lemma 4, §2.1]. A rational unit is an integer multiplication composed with an inverse one. We use the fixed-tape details of [OAI26, Lemmas 3–4, §2.1] as summarized in Fact 3.3 (applied in [OAI26, proof of Lemma 10, proof of Prop. 11, §5.2, and §5.3 (25)]). □

The next lemma justifies the address-dependent Walsh diagonals S_U and CZ of Section 5 and the twiddle-index preparation of Section 6.

Lemma A.5 (Long-record address-dependent scans)**.** *Fix a recordwise address routine and payload routine and constants C_0, C_1, C_2, J. Suppose that, on an array of $M \geq 1$ records, the address preparation of every record takes at most $C_1 q^J$ steps, the payload processing of a record of ρ bits takes at most $C_2\rho$ steps, and a one-time setup takes at most $C_0 q^J$ steps. Then the total time is at most $C_0 q^J + M(C_1 q^J + C_2\rho_{\max}) + O(V)$; in particular, if every record has at least q^J bits (which holds for records of $\Theta(rq)$ bits once $r \geq q^J$, as in every family with $r/q^j \to \infty$ for each fixed j), the total time is $O(V)$ with a constant depending only on C_0, C_1, C_2. Consequently, an address-dependent phase in $\{1, -1, i, -i\}$ may be computed once per record and applied to all its coefficients in $O(V)$ time, and selecting a twiddle from a table of $O(rq)$ bits by one scan per record has $O(V)$ overhead (with general coefficient multiplications charged separately).*

Proof. Let the records have $\rho_1, \ldots, \rho_M$ bits, so that $V = \rho_1 + \cdots + \rho_M \leq M\rho_{\max}$. Summing over the records, setup, address preparation and payload processing take at most $C_0 q^J + MC_1 q^J + C_2 V$ steps, and traversing the array on a fixed number of tapes adds $O(V)$. When every record has at least q^J bits, $q^J \leq V$ and $Mq^J \leq V$, giving $O(V)$. For records of $\Theta(rq)$ bits, maintaining and decoding the

record address, scanning a polynomial-size mask or position list, and evaluating a fixed polynomial-time address routine cost at most Cq^J per record, so the total is $O(M(q^J + rq)) = O(Mrq) = O(V)$, and a table scan and rewind cost $O(rq)$ per record. The same long-record amortization appears in [OAI26, §5, p. 26 and §5.3, pp. 33–34]. □

A.2 Rational shears and power-width interchanges

For chunks of m^k digits, the bit network, with the shears of Lemma A.6 as frame changes, reduces one interchange to s_b interchanges of chunks m times shorter, each on $1/W_b$ of the volume, plus linear work (Proposition A.7).

Lemma A.6 (Lower triangular factorization and shear)**.** *Every rational $m \times m$ matrix A of rank a has a factorization $A = E_1 \Pi E_2$, where E_1, E_2 are invertible lower triangular rational matrices and Π is a partial permutation matrix with exactly a ones. For a fixed finite collection of such matrices there is a fixed odd prime q_{rad} such that, on two ordered groups $H = (H_1, \dots, H_m)$ and $D = (D_1, \dots, D_m)$ of fields of range $[q_{\mathrm{rad}}^b]$ with every H_i before every D_j and arbitrary spectator fields, the shear $(H, D) \mapsto (H + AD, D)$ uses exactly a interchanges of pairs of b-digit fields, each between an H field and a D field, and $O(V)$ further work.*

Proof. Factorization. If the active matrix is nonzero, choose its topmost nonzero row and the rightmost nonzero entry of that row as the pivot. Right multiplication by lower triangular elementary matrices clears entries to the left of the pivot, and left multiplication by lower triangular elementary matrices clears entries below it. Scale the pivot to one and delete its row and column from the active index sets, retaining their original orders. Subsequent operations remain lower triangular and preserve earlier pivots, so after a steps the result is a partial permutation matrix with a ones; inverting the accumulated triangular matrices gives $A = E_1 \Pi E_2$ ([OAI26, Lemma 9, §4.1]).

The shear. Fix the factorizations for the finite collection once and for all, and choose an odd prime q_{rad} avoiding all denominators of the factors and their inverses and all numerators of their diagonal entries by finite rational arithmetic and trial division ([OAI26, proof of Lemma 10, §4.1]). Every identity then specializes modulo q_{rad}^b with invertible triangular factors. A lower triangular map costs $O(V)$ by updating coordinates in descending physical order using only earlier, still-unmodified coordinates (Lemma A.4, since m is fixed). Transform D by E_2 and H by E_1^{-1}, add ΠD to H, and undo both triangular changes. Each one of Π at (i, j) is executed by

$$D_j \leftarrow D_j + H_i, \qquad (H_i, D_j) \leftarrow (D_j, H_i), \qquad D_j \leftarrow H_i - D_j,$$

which maps $(h, d) \mapsto (h + d, d)$ using one interchange and two linear updates to the later field ([OAI26, proof of Lemma 10, §4.1]). □

Proposition A.7 (Parameterized power-width interchange)**.** *Take the bit network of Definition A.1, and choose the prime q_{rad} of Lemma A.6 for the edge matrices A_e, also avoiding the denominators of the assigned terminal and gate matrices. For every fixed rational $\tau \in (0, 1)$ with $s_b/W_b < m^\tau$, two disjoint contiguous chunks of $e = m^k$ radix-q_{rad} digits can be interchanged in $O(Ve^\tau)$ time if a row field preceding both chunks has length divisible by W_b^k. All other fields are preserved, and the machine, tape count and time constant are uniform in all field lengths and data.*

Proof. Construction and frame invariant. Write $W = W_b$ and $s = s_b$. The case $k = 0$ is Lemma A.4(1). For $k > 0$, write each chunk as m fields of $b = e/m$ digits, and split whole rows cyclically among W role streams: writing the old row index as $Wg + (w - 1)$, role w has row index g, volume V/W, and row count divisible by W^{k-1}. For a rational matrix M let Φ_M move

(H, D) to $(H + MD, D)$, so $\Phi_{M'}\Phi_M = \Phi_{M'+M}$ and $\Phi_M^{-1} = \Phi_{-M}$. Apply Φ_{A_e} on each circuit edge by Lemma A.6 with recursive pivot interchanges, and apply each circuit gate pointwise to aligned role streams. Defining the logical array at a vertex with frame Φ_M as Φ_M^{-1} of the physical array, edges preserve the logical array and common address permutations commute with pointwise gates, so input role w reaches output role $\rho_b(w)$ with physical operator $\Phi_{M_{\mathrm{out}(\rho_b(w))}}\Phi_{-M_{\mathrm{in}(w)}} = \Phi_{I_m}$. Merging output role $\rho_b(w)$ back into row class w applies $H \leftarrow H + D$ on every original row, including auxiliary rows with arbitrary initial contents ([OAI26, proof of Prop. 11, §4.2]). Then $D \leftarrow D - H$, $H \leftarrow H + D$, $D \leftarrow H - D$ executes $(H, D) \mapsto (D, H)$ with one circuit call and two $O(V)$ updates ([OAI26, proof of Prop. 11, pp. 22–23]).

Recurrence, fixed tapes, and descriptors. There are s recursive calls on volume V/W plus $O(V)$ work, so $F_0 = O(1)$ and $F_k \le (s/W)F_{k-1} + O(1)$; since $a = s/W < m^\tau$, $\sum_{j=0}^{k} a^j = O(m^{k\tau})$ and $F_k = O(e^\tau)$ ([OAI26, proof of Prop. 11, (12)]). Depth-first scheduling uses W role tapes, fixed I/O and work tapes, a parking-stack tape and a descriptor-stack tape. Inactive streams are pushed before a child call, parent role areas are erased while traversed, and popped streams are written from the right end of their known destination intervals to restore direction without scanning ancestor parked data, adding $O(V)$ at the parent ([OAI26, §4.2, p. 23]). At depth j from root width $e_0 = m^{k_0}$, inactive digits remain spectators, so $V_j \ge q_{\mathrm{rad}}^{2e_0}$ and $\log V_0 = O(\log(2V_j))$, absorbing the $\mathrm{poly}(\log(2V_j))$ child-descriptor work into $O(V_j)$ ([OAI26, §4.2, pp. 23–24]). □

A.3 Arbitrary-width interchanges

Lemma A.8 (Parameterized arbitrary-width interchange)**.** *Under the hypotheses of Proposition A.7, for the same fixed τ, a bit array on $[A] \times [2^u] \times [G] \times [2^u] \times [B]$, $u \ge 1$, can be transformed by $(x, h, g, d, z) \mapsto (x, d, g, h, z)$ in $O(Vu^\tau)$ time, uniformly in all positive A, G, B and all data (including shape processing, padding, unpadding and cleanup, with $B = 1$ permitted). Two fields whose widths differ by one bit are interchanged across the intervening field by one such interchange and one single-bit move.*

Proof. Radix-q_{rad} chunks. First let the chunks have range $[q_{\mathrm{rad}}^e]$. Put $k = \lceil \log_m e \rceil$ and $k_{\mathrm{hi}} = \min\{j \ge 0 : q_{\mathrm{rad}}^{2j} \ge W_b^k\}$. For large e, $1 \le k_{\mathrm{hi}} < e$ with $k_{\mathrm{hi}} = O(\log(2e))$ (bounded widths use individual digit swaps). Split each chunk into its top k_{hi} digits and the rest, $h = (h_+, h_-)$ and $d = (d_+, d_-)$. Interchange h_+ and d_+ digit by digit and move h_+ immediately after d_+, transforming $(h_+, h_-, g, d_+, d_-) \mapsto (d_+, h_+, h_-, g, d_-)$ in $O(Vk_{\mathrm{hi}})$ steps so that (d_+, h_+) forms a row field of range $R_{\mathrm{row}} = q_{\mathrm{rad}}^{2k_{\mathrm{hi}}} \ge W_b^k$ before both remaining chunks. Pad this row range in each prefix to $R'_{\mathrm{row}} = W_b^k \lceil R_{\mathrm{row}}/W_b^k \rceil < 2R_{\mathrm{row}}$. Expand $e - k_{\mathrm{hi}} = \sum_{j=0}^{k} a_j m^j$ $(0 \le a_j < m)$ and interchange corresponding power-of-m pieces by Proposition A.7 in cost $O(V)\sum_j a_j m^{j\tau} = O(Ve^\tau)$. Delete the padded rows and reverse only the joining moves to reach (d_+, d_-, g, h_+, h_-), with descriptor and movement overhead $O(V \log(2e)) = O(Ve^\tau)$. These steps follow [OAI26, proof of Lemma 12, §4.3]; none uses the specific value $1 - 2^{-50}$ fixed in [OAI26, §4.2, p. 24].

Binary chunks. Choose the least e with $q_{\mathrm{rad}}^e \ge 2^u$ and pad both ranges numerically from $[2^u]$ to $[q_{\mathrm{rad}}^e]$ (increasing volume by less than q_{rad}^2 without per-address radix conversion, as numerical and lexicographic orders coincide). After the radix-q_{rad} interchange, scan away entries with either coordinate $\ge 2^u$; since $e = O(u)$, the time is $O(Vu^\tau)$ ([OAI26, proof of Lemma 12, p. 25]). Widths differing by one bit use one equal-width interchange and one single-bit move as in [OAI26, §4.3, p. 25]. □

A.4 Packed selected-bit addition

Lemma A.9 (Parameterized packed selected-bit addition)**.** *Fix τ as in Lemma A.8. Let $R_{\mathrm{rec}} \ge 1$, $K \ge 6$, $f \ge 1$, $L = fK$ and $0 \le \rho < K$, let $N_0, \dots, N_3$ be positive integers, and let (χ_1, χ_2, χ_3) be a permutation of the chunk names (x, y, z). Consider R_{rec}-bit records on the complete rectangle $[N_0] \times [2^L]_{\chi_1} \times [N_1] \times [2^L]_{\chi_2} \times [N_2] \times [2^L]_{\chi_3} \times [N_3]$, and put $M = 2^{3L} \prod_{j=0}^{3} N_j$, $V = MR_{\mathrm{rec}}$ and $A_{\mathrm{rect}} = \lceil \log_2(2V) \rceil$. One fixed-tape procedure performs $y_{\rho+iK} \leftarrow y_{\rho+iK} \oplus x_{\rho+iK}$ for $0 \le i < f$, preserves every other address bit and every payload, and restores the entire auxiliary chunk z for every initial address value. Its time is*

$$O(V(L^\tau + 1) + MA_{\mathrm{rect}}^3 + \delta_{\mathrm{bad}} MA_{\mathrm{rect}}(R_{\mathrm{rec}} + A_{\mathrm{rect}})), \qquad \delta_{\mathrm{bad}} = \min\{1, 80(f-1)2^{-K}\}. \tag{49}$$

In particular, if $A_{\mathrm{rect}}^3 \le R_{\mathrm{rec}}$ and $80(f-1)2^{-K} A_{\mathrm{rect}}(1 + A_{\mathrm{rect}}/R_{\mathrm{rec}}) \le 1$, the bound is $O(V(L^\tau + 1))$ with an absolute implied constant; both hold for all sufficiently large q, uniformly in shapes and payloads, whenever $A_{\mathrm{rect}} \le C_A q$, $f \le d \le q$, $R_{\mathrm{rec}} \ge q^4$ and $K \ge 3 \log_2 q + \log_2(160\, C_A)$.

Proof. Scalar program and packing. For $c \in \{0, 1\}$ and integers u, w, execute in chronological order (congruences modulo 2, using current values):

$$\begin{array}{llll} u \leftarrow u - c[w \equiv 0], & w \leftarrow w - c[u \equiv 0], & u \leftarrow u + c[w \equiv 0], & w \leftarrow w + c[u \equiv 0], \\ w \leftarrow w + c[u \equiv 1], & u \leftarrow u + c[w \equiv 1], & w \leftarrow w - c[u \equiv 1], & u \leftarrow u - c[w \equiv 1]. \end{array}$$

For $c = 0$ this is the identity; for $c = 1$ the two halves induce (00 01 10) and (01 10 11) on each square $\{2u_0, 2u_0+1\} \times \{2w_0, 2w_0+1\}$, composing to $(u', w') = (u + c(1 - 2(u \bmod 2)),\ w)$ with displacement at most four ([OAI26, proof of Lemma 13, (20)–(21)]). Omitting the top selected bit j_{f-1}, write the K-bit segments at $j_i = \rho + iK$ $(i < f-1)$ in y, z as $u_i = 2g_i + a_i$, $w_i = 2h_i + b_i$. Each program line across all $i < f-1$ is a single cyclic shift modulo 2^L (e.g. $y \leftarrow y + \epsilon \sum_{i=0}^{f-2} 2^{j_i}[x_{j_i} = 1][z_{j_i} = b] \bmod 2^L$), hence a bijection of the rectangle. Let B be the set of addresses where some guard g_i or h_i fails $10 \le g \le 2^{K-1} - 11$. Off B, every segment starts in $[20, 2^K - 21]$ and stays within $[0, 2^K - 1]$ without inter-segment carries, so the eight-shift composite S_0 agrees with the ideal XOR map T_0 on $i < f-1$. By the union bound over $2(f-1)$ guards with 20 excluded values out of 2^{K-1} each, $|B| \le \delta_{\mathrm{bad}} M$; since T_0 preserves guards, $T_0 S_0^{-1}$ permutes B and fixes B^c ([OAI26, proof of Lemma 13, (22)–(24)]).

Movement and deterministic repair. Moving each shift's target to the rightmost chunk position by Lemma A.8, applying the controlled rotation of Lemma A.4, and undoing the swap uses at most 16 interchanges and 8 rotations at cost $O(V(L^\tau + 1))$ ([OAI26, proof of Lemma 13, pp. 30–31]). Next scan with a record-rank counter, test membership in B in $O(A_{\mathrm{rect}}^3)$ steps per record, extract records $q_{\mathrm{addr}} \in B$ with destination rank of $T_0(S_0^{-1}(q_{\mathrm{addr}}))$, mark their holes, radix-sort the extracted bank in $O(|B| A_{\mathrm{rect}}(R_{\mathrm{rec}} + A_{\mathrm{rect}}))$ time, and reinsert in one forward scan; finally apply the omitted two-bit XOR at j_{f-1} by Lemma A.4(1) in $O(V)$ time ([OAI26, proof of Lemma 13, (19), p. 31]). Neither B nor the repair depends on τ. □

A.5 Complex frames and the simultaneous butterfly layer

Put $H_{\mathrm{u}} = \sqrt{2} H_0$, $S = \mathrm{diag}(1, i)$, $Z = \mathrm{diag}(1, -1)$ and $C = H_{\mathrm{u}} S H_{\mathrm{u}} = aI + bX$ with $a = \frac{1+i}{2}$, $b = \frac{1-i}{2}$ and $X = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Direct multiplication gives $H_0 = bSCS$, $C^{-1} = -iZCZ$ and $b^2 = -i/2$ ([OAI26, §5.1, (13)–(14) and §5.4, p. 36]).

Lemma A.10 (Exact recursive complex contract)**.** *For the complex network of Definition A.1, on a group of $e = mf$ consecutive K-bit chunks with one common selected offset, $C^{\otimes e}$ is obtained on every role, exactly over $\mathbb{Z}[i, 1/2]$, by*

(1) exactly s_c forward child calls $C^{\otimes f}$, each on one slot of f consecutive chunks;

(2) a fixed number of packed binary row additions and their inverses;

(3) a fixed number of coefficientwise Gaussian dyadic scalar gates, address signs and fourth-root phases.

No extra child call is needed for an inverse directional factor or for an endpoint correction.

Proof. For a binary subspace label U put $q_U(x) = \mathrm{wt}(P_U x) \bmod 4$ and $C_U = H_{\mathrm{u}}^{\otimes m} \operatorname{diag}_x(i^{q_U(x)}) H_{\mathrm{u}}^{\otimes m}$. Orthogonal binary vectors meet in an even number of coordinates, so weight is additive modulo 4 on orthogonal sums. On a nested edge with orthonormal residual basis $\mathcal{B}$, $q_V(x) - q_U(x) = \sum_{v\in\mathcal{B}} \mathrm{wt}(v)[v\cdot x] \bmod 4$ (negated on a decreasing edge), with each $\mathrm{wt}(v)$ odd. Since $i^{[v\cdot x]} = a + b(-1)^{v\cdot x}$ and conjugating $\operatorname{diag}_x((-1)^{v\cdot x})$ by $H_{\mathrm{u}}^{\otimes m}$ yields $(X_v f)(x) = f(x+v)$, each $v \in \mathcal{B}$ contributes $aI + bX_v$ or its inverse ([OAI26, proof of Prop. 8, §3.5]).

Extending $\mathcal{B}$ to $M \in \mathrm{GL}(m, \mathbb{F}_2)$ with first columns $\mathcal{B}$, the move permutation P_M satisfies $P_M X_v P_M^{-1} = X_{Mv}$, realizing the edge on f columns by P_M^{-1}, one coordinate kernel per $v \in \mathcal{B}$, and P_M, where inverse factors use $(C^{-1})^{\otimes f} = (-i)^f Z^{\otimes f} C^{\otimes f} Z^{\otimes f}$ without extra child calls ([OAI26, §5.1, (15)–(17)]). Each fixed $M^{\pm 1}$ factors into at most $m(m-1) + 3m$ row additions, each executed across two f-chunk slots by Lemma A.9 using a third slot ($m \geq 3$) as restored scratch ([OAI26, §5.1, p. 27 and §5.2, pp. 31–32]). Commutation of common address-linear frames with pointwise role gates, together with the source/sink signs $Z_{a,f}$, scalar i^{27f} and bank restoration, yields $C^{\otimes e}$ on every role from s_c child calls ([OAI26, proof of Prop. 8, §3.5 and §5.1, pp. 27–28]). Finally,

$$H_0^{\otimes D} = b^D \mathcal{S}_D C^{\otimes D} \mathcal{S}_D, \qquad (\mathcal{S}_D z)(x) = i^{\sum_{j=1}^{D} x_j} z(x), \qquad b^D = (-i/2)^{\lfloor D/2 \rfloor} b^{D \bmod 2}, \tag{50}$$

as in [OAI26, §5.1, (18)], with $\mathcal{S}_D$ executed in $O(V)$ time by Lemma A.5. □

Proposition A.11 (Simultaneous butterfly layer with general parameters)**.** *Let ν_{net} and C_0 be as in (47). Fix rationals $0 < \tau < 1$ and $0 \leq \sigma < 1$ with*

$$s_b/W_b < m^{\tau}, \qquad s_c/W_c < m^{\sigma},$$

and rational parameters $0 < \beta < 1$, $c > 0$, $0 < \varepsilon < 1$, λ, λ' with

$$\max\{\tau, \sigma\} < \lambda < 1, \qquad \tau(1 + c/\beta) < \lambda, \tag{51}$$

$$\max\{\lambda,\ \sigma + \beta(1-\sigma)\} < \lambda' < 1. \tag{52}$$

Fix an integer $C_1 \geq \nu_{\mathrm{net}} + 3$ with $\varepsilon C_1 < 1$, and positive constants $a_d \leq b_d$, $a_r \leq b_r$, C_M, C_{desc} and $C_w > 1$. Then there are a cutoff $q_{\min}$, one fixed finite-alphabet multitape procedure and a uniform time constant with the following property. Suppose that

$$q \geq q_{\min}, \qquad a_d q^{\varepsilon} \leq d \leq b_d q^{\varepsilon}, \qquad 2^{a_r q/d} \leq r \leq 2^{b_r q/d}, \qquad K = \lfloor d^c \rfloor, \tag{53}$$

and let $1 \leq D \leq d$, $A, S \geq 1$ and $0 \leq \rho < K$ be integers with

$$M = AS\, 2^{KD}, \qquad \lg(2M) \leq C_M q, \qquad q + 2 \leq w \leq C_w q.$$

The input is a disk-grid array in lexicographic order on $[A] \times [2^K]^D \times [S]$. Each address carries a record in $\mathcal{R}_r$, stored in coefficient order, and each component is 2^{-q} times a signed w-bit two's-complement integer, most significant bit first. The supplied header is $\Gamma(q)\Gamma(d)\Gamma(r)\Gamma(D)\Gamma(A)\Gamma(S)\Gamma(K)\Gamma(\rho+1)\Gamma(w)$, of length at most $C_{\mathrm{desc}} q$, with Γ as in (60). Let H apply H_0 to bit ρ, counted from the least

significant end, of each of the D chunks. The procedure returns $Q_q(Hz)$, in the original record and component format, in $O(Vd^{\lambda'})$ steps, where $V = 2Mrw$. It evaluates the entire layer exactly before this final truncation; the output is a disk array and $\|Q_q(Hz) - Hz\|_\infty < \sqrt{2} \cdot 2^{-q}$. The recursion stops when $e < d^\beta$ (with $e = 1$ always a leaf). With $\Delta = \lceil C_0 d^{C_1} \rceil$, the procedure uses at most $q + \Delta$ fractional and $\Delta + 3$ integer and sign bits per real component internally; these widths are $O(q)$. The time includes padding, deterministic repair, role-stream scheduling, unpadding and workspace cleanup.

Remark A.12 (Where the parameters enter). As printed, [OAI26, Prop. 14] fixes $\beta = \frac{1}{2}$, $\tau = \sigma = 1 - 2^{-50}$ and $C_1 = 20$. In the proof below, following [OAI26, §§5.1–5.4], the parameters enter only through the hypotheses of Proposition A.11. The parameter β enters the leaf test $e < d^\beta$ (evaluated for $\beta = \beta_1/\beta_2$ as the integer comparison $e^{\beta_2} < d^{\beta_1}$) and the bound $K \le e^{c/\beta}$. The exponents τ and σ enter the interchange costs of Lemmas A.8–A.9 and the time unrolling via $s_c/W_c < m^\sigma$, respectively. Finally, $C_1 \ge \nu_{\text{net}} + 3$ enters the actual guard allocation $\Delta = \lceil C_0 d^{C_1} \rceil$ together with $\varepsilon C_1 < 1$, which ensures $\Delta = o(q)$ (already noted in [OAI26, §5.4, p. 35] as requiring only exponent $14 = \nu_{\text{net}} + 3$).

A.6 Proof of Proposition A.11

Individual kernels. Applying $C(u, v) = (\frac{u+v+i(u-v)}{2}, \frac{u+v-i(u-v)}{2})$ at one selected bit splits on that bit, scans aligned coefficient pairs, and merges in $O(V)$ time at any $O(q)$ guarded width, so e individual kernels cost $O(Ve)$ ([OAI26, §5.3, (25)]).

Rows, pieces, and scheduling. Put $d_0 = \lceil \log_2 W_c \rceil \lceil \log_m(2d) \rceil$ and $k_0 = \lceil \log_m d \rceil$. If $D \le d_0$, use individual kernels. Otherwise apply individual kernels to the first d_0 selected bits; their complete consecutive chunks form a row field of range $R_{\text{row}} = 2^{d_0 K} \ge W_c^{k_0}$, padded in each prefix to the next multiple of $W_c^{k_0}$ (at most doubling volume), and the remaining $D - d_0$ chunks split into at most $(m-1)(1 + \lfloor \log_m d \rfloor)$ consecutive power-of-m pieces. At each internal node, cyclic row splitting gives W_c role streams of volume V/W_c with row count divisible by $W_c^{k_0 - j}$ at depth $j \le k_0$, supporting recursion down to one chunk regardless of β and preserving the three-slot rectangles of Lemma A.9 ([OAI26, §5.3, pp. 33–34]). Depth-first parking as in Proposition A.7 adds $O(V)$ overhead per node across the fixed number s_c of children, and completed invocations commute with the row-bit kernels and leave padded zero rows zero for removal ([OAI26, §5.3, pp. 33–34]).

Arithmetic depth and guard. Let E_* be the constant number of elementary coefficient operations (additions, subtractions, halvings, signs and multiplications by i) performed outside child calls at one internal node by the fixed finite scalar circuit and its fourth-root/sign corrections. Then the dependency depth satisfies $A(e) \le s_c A(e/m) + E_*$ at internal nodes and $A(e) \le 8e$ at leaves ([OAI26, §5.4, p. 35]). Over $j_{\max} \le \lceil \log_m d \rceil$ levels with $1 < s_c < m^5$, we have $s_c^{j_{\max}} \le s_c d^5$ and $A(e) \le C_{\text{dep}} d^6$ where $C_{\text{dep}} = s_c(8 + E_*/(s_c - 1))$. Across at most md pieces (for large d), at most d preprocessing kernels, the two outer diagonals $\mathcal{S}_D$ and the scalar b^D (whose shift by $t = \lfloor D/2 \rfloor$ is charged as t halvings), total denominator and magnitude growth obeys ([OAI26, §5.4, p. 36])

$$G(d) \le mC_{\text{dep}} d^7 + 8d + t + 9 \le (mC_{\text{dep}} + 18)\, d^7 \le C_0 d^{C_1} \qquad (d \ge d_*), \tag{54}$$

with $d_* = \lceil ((mC_{\text{dep}} + 18)/C_0)^{1/(C_1 - 7)} \rceil$ finite since $C_1 \ge \nu_{\text{net}} + 3 = 14 > 7$; the condition $d \ge d_*$ holds via $d \ge a_d q^\varepsilon$ once $q \ge q_{\min}$. Starting from disk inputs on $2^{-q}\mathbb{Z}[i]$, values at depth j lie in $2^{-(q+j)}\mathbb{Z}[i]$ with modulus at most 2^j, so $\Delta = \lceil C_0 d^{C_1} \rceil$ ($q + \Delta$ fractional and $\Delta + 3$ integer/sign bits) represents every intermediate value exactly, and temporary bit-level interchanges cause no coefficient rounding ([OAI26, §5.4, p. 36]). Since $d \le b_d q^\varepsilon$ and $\varepsilon C_1 < 1$, $\Delta = o(q)$, so $\Delta \le q$ for $q \ge q_{\min}$ and guarded records have width $R_g = 2r(q + 2\Delta + 3) \in [2rq, 12rq]$.

Uniform packed-call cost and time recurrence. Every packed child has $A_* = \lceil \log_2(2M_{\text{child}}R_g) \rceil \le C_A q$ with $C_A = C_M + b_r + 7$, while $K \ge \frac{1}{2} a_d^c q^{\varepsilon c}$ and $r \ge 2^{(a_r/b_d)q^{1-\varepsilon}}$, so both extra normalized terms of (49) tend to zero uniformly across all children and descriptors/masks are absorbed by Lemma A.5 ([OAI26, §5.3, pp. 33–34 and proof of Prop. 14, §5.4, pp. 37–38]). With $a_c = s_c/W_c$ and $F(e)$ the time divided by current logical volume,

$$F(e) \le a_c F(e/m) + O((eK)^\tau + 1), \qquad F(e) = O(e) \quad (e < d^\beta). \tag{55}$$

For $e \ge d^\beta$ we have $K \le d^c \le e^{c/\beta}$, so (51) bounds the node overhead by $O(e^\lambda)$ and internal nodes sum to $O(e^\lambda)$ since $a_c < m^\sigma < m^\lambda$, while leaves of size $u = e/m^j < d^\beta$ contribute $O(a_c^j u) \le O(e^\sigma u^{1-\sigma}) \le O(e^\sigma d^{\beta(1-\sigma)})$. Summing over the base-$m$ pieces and preprocessing yields

$$O\big(\log(2d)\,[d^\lambda + d^{\sigma+\beta(1-\sigma)} + 1]\big) = O(d^{\lambda'}) \tag{56}$$

by (52) ([OAI26, §5.3, p. 34]).

Order of operations and output. Re-encode into the guarded format (sign-extending or trimming redundant sign bits to $\Delta + 3$ integer/sign bits and appending Δ fractional zero bits, giving exact width $q + 2\Delta + 3$), apply $\mathcal{S}_D$ in the original layout, execute the row-bit kernels, row padding and power-of-m pieces for $C^{\otimes D}$, unpad, apply $\mathcal{S}_D$ again and b^D as in (50), and apply Q_q once. The chronological order matters: for $D = 1$, placing both $\mathcal{S}_1 = S$ factors after C would give $bS^2C(1,0)^\mathsf{T} = (\frac{1}{2}, \frac{i}{2})^\mathsf{T} \ne H_0(1,0)^\mathsf{T}$. The final truncation acts on the absolute numerator and then restores the sign, implementing componentwise truncation toward zero Q_q in two's complement; because each row of $H = H_0^{\otimes D}$ has absolute sum 1, H preserves the unit disk, and Q_q preserves the disk with error $< \sqrt{2} \cdot 2^{-q}$ in width $w \ge q + 2$ ([OAI26, proof of Prop. 14, p. 38]). Taking $q_{\min}$ above the finitely many thresholds proves Proposition A.11. □

Remark A.13 (Explicit gate-count threshold). The proof above uses only finiteness of the scalar circuit. Counting at most $12W_c$ scalar gates of at most $2W_c^2$ operations each, $4s_c$ inverse-child operations and $4W_c + 4$ endpoint operations as in the proof text of [OAI26, §5.4, pp. 35–36] gives $E_* \le 24W_c^3 + 4s_c + 4W_c + 4 < E_{\text{net}}$, whence $A(e) \le 8B_{\text{net}}^2 d^{\nu_{\text{net}}+1}$ and $G(d) < 128mB_{\text{net}}^2 d^{\nu_{\text{net}}+3} \le C_0 d^{C_1}$ as soon as the piece count is at most md, without any unspecified d_*. No theorem of this paper relies on that gate count.

Remark A.14 (Lifted precision). Proposition A.11 applies directly at the lifted precision $q_* = q + 8d$ used by the Walsh gadgets of Section 5 (with width $w_* = w + 8d$ and halved lower band constants $(a_d/2, b_d, a_r/2, b_r)$ once $q_* \le 2q$; Lemma 5.3).

B Deferred proofs for the fractional-shift bridge

This appendix supplies two deferred parts of Section 4. Section B.1 proves the two negative results justifying the negative half-bin complex carrier within our two-alias deconvolution framework, and Section B.2 gives the certified rational-series evaluation of the bridge's filter, solver, and phase tables (proving Lemma 4.6(iii)). Throughout we follow the notation of Sections 3 and 4: $a \in \{2, \dots, d\}$ indexes the spread axes, $\nu \in \mathbb{Z}$ indexes the frequency aliases, and $\alpha_\pi = \sqrt{\pi}\,\alpha$.

B.1 Limits of positive-carrier and real-kernel designs

On spread axis a of even power-of-two length $t = t_a \ge 2$, Lemma 4.4 retains the two aliases $\nu \in \{0, 1\}$ of (18), obtaining both an omitted-alias bound $< 0.120 \cdot 2^{-\omega}$ uniform over all $t \ge 2$ and an inverse norm $\|B_a^{-1}\|_\infty \le 4t_a/7$ independent of the cycle length $L = A_a$. The two propositions below show

why both the sign of the half-bin modulation and the complex nature of the kernel are needed within this fixed two-alias design.

First consider the *positive* half-bin carrier $g_a^+(s) = e^{+\pi i s/t_a}\,\mathrm{sinc}(s)\,e^{-\pi s^2/\alpha^2}$, whose Fourier transform is $\hat{g}_a^+(\xi) = G(\xi - \frac{1}{2t_a})$. For $u \in [t]$, its frequency samples lie at $y = (u - \frac{1}{2})/t$; in particular, the $u = 0$ sample lands at $y = -\frac{1}{2t} \notin (0,1)$, within distance $\frac{1}{2} - \frac{1}{2t}$ of the passband of the omitted alias $\nu = -1$.

Proposition B.1 (No uniform alias bound for the positive carrier with retained pair $\{0,1\}$)**.** *With the positive carrier g_a^+, Gaussian width $\alpha = 2\sqrt{\omega \ln 2/\pi}$ ($\omega \geq 8$), and retained alias pair $\nu \in \{0,1\}$, the omitted alias weight on an axis of length $t = t_a \geq 2$ at frequency digit $u = 0$ is at least $\frac{1}{2}\,\mathrm{erfc}(\alpha_\pi(\frac{1}{2} - \frac{1}{2t}))$. For $t = 2$ this equals $\frac{1}{2}\,\mathrm{erfc}(\frac{1}{2}\sqrt{\omega \ln 2}) = \Theta(\omega^{-1/2}2^{-\omega/4})$. Consequently there is no constant C such that the omitted alias weight is at most $C \cdot 2^{-\omega}$ for all $\omega \geq 8$ and all powers of two $t \geq 2$. This already occurs in dimension $d = 2$.*

Proof. At $u = 0$ we have $y = -\frac{1}{2t}$. As in the proof of Lemma 4.4(ii), removing the intervals $[y - \nu - \frac{1}{2}, y - \nu + \frac{1}{2}]$ of the retained aliases $\nu \in \{0,1\}$ leaves $(-\infty, y - \frac{3}{2}] \cup [y + \frac{1}{2}, \infty)$, so the omitted alias weight is

$$\Psi^+(y) = \tfrac{1}{2}\,\mathrm{erfc}(\alpha_\pi(y + \tfrac{1}{2})) + \tfrac{1}{2}\,\mathrm{erfc}(\alpha_\pi(\tfrac{3}{2} - y)) \geq \tfrac{1}{2}\,\mathrm{erfc}\Big(\alpha_\pi\Big(\frac{1}{2} - \frac{1}{2t}\Big)\Big).$$

With $\alpha_\pi = 2\sqrt{\omega \ln 2}$ and $t = 2$, the right-hand side is $\frac{1}{2}\,\mathrm{erfc}(z)$ where $z = \frac{1}{2}\sqrt{\omega \ln 2} \geq \sqrt{2 \ln 2} > 1$. Integrating $\frac{2}{\sqrt{\pi}}e^{-s^2}$ over $[z, z + 1/z]$ and using $(z + 1/z)^2 = z^2 + 2 + z^{-2} \leq z^2 + 3$ yields

$$\mathrm{erfc}(z) \geq \frac{2}{\sqrt{\pi}\,z}\,e^{-(z+1/z)^2} \geq \frac{2e^{-3}}{\sqrt{\pi}\,z}\,e^{-z^2},$$

so substituting $z = \frac{1}{2}\sqrt{\omega \ln 2}$ and $e^{-z^2} = 2^{-\omega/4}$ gives

$$\Psi^+\Big(-\frac{1}{4}\Big) \geq \frac{1}{2}\,\mathrm{erfc}(z) \geq \frac{2e^{-3}}{\sqrt{\pi\omega \ln 2}}\,2^{-\omega/4},$$

whose ratio to $2^{-\omega}$ grows without bound as $\omega \to \infty$. More generally, for any fixed $t \geq 2$ the omitted weight at $u = 0$ is $\Theta(\omega^{-1/2}2^{-\omega(1-1/t)^2})$, which exceeds $C \cdot 2^{-\omega}$ once ω is large relative to t. □

Three modifications restore uniform $O(2^{-\omega})$ decay: reversing the sign of the carrier (our choice in Definition 4.1), switching the retained pair at $u = 0$ to $\nu \in \{-1, 0\}$, or widening α by a factor $t/(t-1)$. Numerically, at $\omega = 64$ the worst-column omitted weight drops from $1.5 \times 10^7 \cdot 2^{-64}$ (positive carrier) to $5 \times 10^{-13} \cdot 2^{-64}$ (negative carrier) at $t = 4$, from $1564 \cdot 2^{-64}$ to $3 \times 10^{-7} \cdot 2^{-64}$ at $t = 8$, and from $0.17 \cdot 2^{-64}$ to $0.01 \cdot 2^{-64}$ at $t = 64$. (Note that Proposition B.1 concerns uniformity over all pairs (ω, t). When $t \geq \omega$, as along the root family of Lemma 7.1 where $t_a \geq r/2 \gg \omega$, the inequality $(1 - 1/t)^2 \geq 1 - 2/t$ still bounds the positive carrier's omitted weight by $O(2^{-\omega})$, whereas the negative carrier holds uniformly for every $t \geq 2$ with no coupling between t and ω.)

The second result shows why no real-valued kernel can give a two-alias cyclic operator whose inverse is bounded independently of the cycle length L.

Proposition B.2 (No real kernel gives a uniformly conditioned two-alias inverse)**.** *Let g be any real-valued Schwartz kernel and retain the two aliases $\nu \in \{0,1\}$ at the Nyquist sample $\xi = \frac{1}{2}$, with coefficients $c_0 = \hat{g}(\frac{1}{2})$ and $c_1 = \hat{g}(-\frac{1}{2})$. Then $|c_0| = |c_1| =: c$, and whenever the cyclic bidiagonal*

operator $B\colon Z_j \mapsto c_0 Z_j + c_1 Z_{(j-1) \bmod L}$ *on a cycle of length* $L \geq 1$ *is invertible, every row of* B^{-1} *has absolute sum*

$$\|B^{-1}\|_\infty = \frac{L}{c\,|1-(-c_1/c_0)^L|} \geq \frac{L}{2c}.$$

In particular, if g *is even, then* $c_0 = c_1$ *and* B *is singular for every even cycle length* L.

Proof. Because g is real-valued, $\hat{g}(-\xi) = \overline{\hat{g}(\xi)}$, so $|c_0| = |c_1| = c$. If $c = 0$, then $B = 0$ is never invertible. Assume $c > 0$ and write $B = c_0(I + \zeta T)$ where $\zeta = c_1/c_0$ satisfies $|\zeta| = 1$ and $(TZ)_j = Z_{(j-1) \bmod L}$. If $(-\zeta)^L = 1$, the nonzero vector $v_j = (-\zeta)^j$ $(0 \leq j < L)$ is well-defined on $\mathbb{Z}_L$ and satisfies $v_j + \zeta v_{(j-1) \bmod L} = 0$ for all $j \in [L]$, so B is singular. If $(-\zeta)^L \neq 1$, then

$$B^{-1} = c_0^{-1}\bigl(1-(-\zeta)^L\bigr)^{-1} \sum_{i=0}^{L-1} (-\zeta T)^i,$$

each row of which has L nonzero entries in distinct columns, every one of modulus $\bigl(c\,|1-(-\zeta)^L|\bigr)^{-1} \geq (2c)^{-1}$. When g is even, $\hat{g}$ is even, so $\zeta = 1$ and $(-\zeta)^L = 1$ for every even L. □

On the bridge's deconvolution cycles of even length $L = A_a = \prod_{i<a} t_i$, any real kernel therefore either fails to be invertible or amplifies errors by at least $A_a/(2c)$ rather than $O(t_a)$.

B.2 Certified evaluation of filter, solver, and phase tables

We now prove Lemma 4.6(iii) in full detail, showing that every transcendental constant used by the bridge can be generated deterministically on a multitape Turing machine in polynomial bit complexity from truncated rational power series, with certified error bounds and no exact-rounding oracle.

Lemma B.3 (Certified evaluation of bridge and phase tables)**.** *Let* $P \geq 20$, $q = P + 2$, $v = 2^{-q}$, *and* $8 \leq \omega \leq P$. *Let* $\alpha^2 = 4\lambda_G \omega \ln 2/\pi$ *with a fixed rational* $\lambda_G \in [1, 4]$, *and let* $R = t_1 \cdots t_d = 2^k$ *with each* $t_i \geq 2$ *a power of two and* $A_a = \prod_{i<a} t_i$.

(a) For every integer precision $b \geq \max(8, P)$, *each unfolded kernel sample* $g_a(e+\delta)$ *(*$2 \leq a \leq d$, $|e| \leq P$, $\delta \in A_a^{-1}\mathbb{Z} \cap [0,1)$*), each alias weight* $c_0 = G(y)$ *and* $c_1 = G(y-1)$ *at* $y = (u + \frac{1}{2})/t_a$ *(*$u \in [t_a]$*), the solver parameters* $c_{\max}^{-1}$ *and* $\rho = \min(c_0, c_1)/c_{\max}$, *and every unit phase* $e^{\pi i \vartheta}$ *(*$\vartheta \in \mathbb{Q}$ *with* $O(k + \lg P)$*-bit numerator and denominator) can be computed deterministically with certified error at most* 2^{-b} *in* $O((b+k)^3)$ *bit operations on a fixed number of tapes, without any transcendental oracle.*

(b) Consequently, at grid precision $q = P + 2$*:*

- *every truncated folded tap* $\sum_{s \in e + t_a\mathbb{Z},\, |s| \leq P} g_a(s+\delta)$ *(*$e \in [t_a]$*) is computed as an entry* $\hat{H}_\delta^{(a)}(e) \in \widetilde{\mathbb{C}}_q$ *with certified error below* $0.71\,v < v$ *relative to the truncated sum and below* $0.95\,v < v$ *relative to the full folded tap* $H_\delta^{(a)}(e)$, *in* $O(P(P+k)^3)$ *bit operations per tap;*
- *the solver parameters* $\hat{\mu}, \hat{\rho} \in 2^{-q}\mathbb{Z}$ *satisfy* $\hat{\rho} \geq 0$, $|\hat{\mu} - c_{\max}^{-1}| < v$, *and* $|\hat{\rho} - \rho| < v$*;*
- *every unit phase* $e = e^{\pi i \vartheta}$ *has a disk-grid approximation* $\hat{e} \in \mathbb{D}_q$ *with* $|\hat{e}| \leq 1$ *and* $|\hat{e} - e| < 2v$ *(with fourth roots of unity exact), and a* d*-factor tensor chirp formed by sequentially multiplying* d *one-axis disk phases with* Q_q *truncation after each product lies in* $\mathbb{D}_q$ *with error at most* $4dv$.

Proof. 1. Primitives and exact rational reductions. With $\alpha^2 = 4\lambda_G\omega\ln 2/\pi$ ($\lambda_G \in [1,4]$; Theorem 4.7 takes $\lambda_G = 1$), we have $\alpha_\pi^2 = \pi\alpha^2 = 4\lambda_G\omega\ln 2$, and the Gaussian spatial exponent is $x = \pi s^2/\alpha^2 = \pi^2 s^2/(4\lambda_G\omega\ln 2)$, which requires no square root of π. Every table entry is built from five scalar primitives evaluated at exact rational arguments:

(i) the constants π, $\ln 2$, $\sqrt{\pi}$, and $\alpha_\pi = 2\sqrt{\lambda_G\omega\ln 2}$;
(ii) $\cos(\pi\vartheta)$ and $\sin(\pi\vartheta)$ for $\vartheta \in \mathbb{Q}$, used for the carrier $e^{-\pi i(e+\delta)/t_a}$, for $\sin(\pi s)$, and for unit phases (twiddles, twists, and chirps);
(iii) $\operatorname{sinc}(s) = \sum_{n\ge 0}(-1)^n(\pi s)^{2n}/(2n+1)!$ for $|s| \le \frac{1}{2}$;
(iv) e^{-x} for $x \ge 0$;
(v) $\operatorname{erf}(z)$ for $z \in \{\alpha_\pi(y+\frac{1}{2}), \alpha_\pi(y-\frac{1}{2}), \alpha_\pi(y-\frac{3}{2})\}$ at $y = (u+\frac{1}{2})/t_a$, giving $c_0 = \frac{1}{2}[\operatorname{erf}(\alpha_\pi(y+\frac{1}{2})) - \operatorname{erf}(\alpha_\pi(y-\frac{1}{2}))]$ and $c_1 = \frac{1}{2}[\operatorname{erf}(\alpha_\pi(y-\frac{1}{2})) - \operatorname{erf}(\alpha_\pi(y-\frac{3}{2}))]$.

The unfolded offsets $s = e+\delta$ ($|e| \le P$, $\delta \in A_a^{-1}\mathbb{Z} \cap [0,1)$, $A_a \le R = 2^k$), the angles ϑ, and the half-bin coordinates y are rationals with $O(k+\lg P)$-bit numerators and denominators, manipulated in exact rational arithmetic. Each angle ϑ is reduced modulo 2 and then by exact octant symmetries of sin and cos into $[-\frac{1}{4}, \frac{1}{4}]$ (using $\sin(\pi s) = (-1)^e\sin(\pi\delta)$ for $s = e+\delta$), so only reduced angles $|\vartheta| \le \frac{1}{4}$ multiply our approximation to π in (ii). Because $y \in (0,1)$, we have $|y+\frac{1}{2}|, |y-\frac{1}{2}|, |y-\frac{3}{2}| \le \frac{3}{2}$, so every erf argument satisfies $z^2 \le \frac{9}{4}\alpha_\pi^2 = 9\lambda_G\omega\ln 2 \le 36\omega\ln 2 < 25P$, while $|s| < P+1$ and $\omega \ge 8$ give $x \le \pi^2(P+1)^2/(32\ln 2) < 0.45(P+1)^2$.

2. Working scale and fundamental constants. Given $b \ge \max(8, P) \ge 20$, set $S = 2b + 8\lceil\lg b\rceil + 64$, $N = 4S$, and $u_S = 2^{-S}$. All intermediate approximate values lie on the dyadic grid $2^{-S}\mathbb{Z}$: additions and subtractions are exact, and each multiplication or division is rounded once to $2^{-S}\mathbb{Z}$ (error at most u_S).

- Sum Machin's identity $\pi = 16\arctan\frac{1}{5} - 4\arctan\frac{1}{239}$ and the series $\ln 2 = 2\sum_{j\ge 0}(2j+1)^{-1}3^{-2j-1}$ to N terms ($0 \le j < N$), obtaining each signed term directly at scale 2^S by one rounded integer division. Their truncation tails are below $20\cdot 5^{-2N-1} < u_S$ and $3^{-2N} < u_S$, respectively, yielding $|\tilde{\pi} - \pi| \le 21Nu_S$ and $|\widetilde{\ln 2} - \ln 2| \le 3Nu_S$.
- Extracting integer square roots of the 2^{2S}-scaled approximations to π and $4\lambda_G\omega\ln 2$ gives $|\widetilde{\sqrt{\pi}} - \sqrt{\pi}| \le \frac{21Nu_S}{2\sqrt{3}} + u_S \le 7Nu_S$ and $|\tilde{\alpha}_\pi - \alpha_\pi| \le \sqrt{\frac{\lambda_G\omega}{\ln 2}}\,3Nu_S + u_S \le 8N\sqrt{P}\,u_S$ (since $\sqrt{4/\ln 2}\cdot 3 < 7.21$), whence $|\tilde{z} - z| \le \frac{3}{2}|\tilde{\alpha}_\pi - \alpha_\pi| + u_S \le 13N\sqrt{P}\,u_S$ for every erf argument z in (v).

3. Trigonometric, sinc, and Gaussian series. Primitives (ii)–(iv) reduce to alternating power series in a nonnegative argument of magnitude at most $\pi/4$ (for $\cos, \sin$), $\pi/2$ (for $\operatorname{sinc}(s)$ on $|s| \le \frac{1}{2}$), or $\ln 2$ (for e^{-r} after range reduction). Up to sign, their terms follow the recurrences $T_j = T_{j-1}w_0/((2j)(2j\pm 1))$ ($w_0 \le (\pi/2)^2$, multiplier $\le \pi^2/24 < 0.42$) or $T_n = T_{n-1}r/n$ ($r \le \widetilde{\ln 2} < 0.70$), so every step multiplier is strictly below 0.71. Rounding errors in the term recurrence are therefore geometrically damped at every step: every computed term lies within $4u_S$ of its exact value for the approximate argument, and the N-term sum incurs rounding error at most $4Nu_S$, while the series tail beyond $N = 4S$ is at most $2(1.6\,\mathrm{e}/N)^N \ll u_S$. All four functions are 1.01-Lipschitz.

For e^{-x} in (iv), if the computed exponent satisfies $\tilde{x} \ge (b+5)\widetilde{\ln 2}$, then $x \ge (b+4)\ln 2$ and $e^{-x} \le 2^{-b-4}$, so returning 0 incurs error at most 2^{-b-4}. Otherwise $x < (b+6)\ln 2$: set $q_0 = \lfloor \tilde{x}/\widetilde{\ln 2}\rfloor \le b+5$ and $\tilde{r} = \tilde{x} - q_0\widetilde{\ln 2} \in [0, \widetilde{\ln 2})$, evaluate the N-term Taylor series of $e^{-\tilde{r}}$, and shift right exactly by q_0 bits; the range-reduction error is at most $23N(b+6)u_S$. For $\operatorname{sinc}(s)$ when $|s| > \frac{1}{2}$, divide the value of $(-1)^e\sin(\pi\delta)$ from (ii) by $\tilde{\pi}s$, noting $|\tilde{\pi}s| > 1.5$. Combining argument, recurrence, and tail errors shows that every primitive in (ii)–(iv) and every unfolded kernel sample $g_a(s)$ formed from them has certified error at most $50N(b+P+8)u_S \le 2^{-b-4}$ (and at most 2^{-b-4} as well when the e^{-x} cutoff triggers).

4. Error-function series. Because erf is odd, assume $z \ge 0$. If $\tilde{z}^2 \ge (b+5)\widetilde{\ln 2}$, then $z^2 \ge$

$(b+4)\ln 2$, $z \ge 1/\sqrt{\pi}$, and $1-\operatorname{erf}(z) = \operatorname{erfc}(z) \le e^{-z^2}/(\sqrt{\pi}z) \le 2^{-b-4}$, so returning 1 has error at most 2^{-b-4}. Otherwise $w_z = \tilde{z}^2$ (formed exactly on $2S$ fractional bits) satisfies $w_z < (b+6)\ln 2 + u_S$, so $e^{w_z} < 2^{b+7}$ and $\mathrm{e}\, w_z/N \le \mathrm{e}/8$ (since $N = 4S > 8(b+6)\ln 2 + 8$). Evaluate $\frac{2}{\sqrt{\pi}}\sum_{n=0}^{N-1}(-1)^n T_n/(2n+1)$ with $T_0 = \tilde{z}$ and $T_n = T_{n-1}w_z/n$ (forming $T_{n-1}w_z$ on $3S$ fractional bits before rounding to u_S). A rounding error made at step n_0 is multiplied at step $n > n_0$ by $\prod_{n_0<j\le n}(w_z/j) \le w_z^{n-n_0}/(n-n_0)! \le e^{w_z}$, so the total recurrence and division error is below $2N^2 e^{w_z} u_S$; the tail beyond N is below $2\tilde{z}(\mathrm{e}/8)^N$, and $|\operatorname{erf}(\tilde{z}) - \operatorname{erf}(z)| \le \frac{2}{\sqrt{\pi}}|\tilde{z}-z|$. Summing these contributions bounds the total error in $\operatorname{erf}(z)$, and hence in c_0 and c_1, by $2N^2 2^{b+8} u_S \le 2^{-b-4}$ (since $S = 2b + 8\lceil \lg b\rceil + 64$ and $b \ge 20$).

5. Assembly of q-grid entries. Which of c_0, c_1 equals $c_{\max}$ is decided without numerical error ($c_0 > c_1$ if and only if $u < t_a/2$, by Lemma 4.4(iv)), and $c_{\max} > 0.49999999$ by Lemma 4.4(iii). Hence one guarded division each gives $c_{\max}^{-1}$ and $\rho = \min(c_0, c_1)/c_{\max}$ within $4.01\cdot 2^{-b-4} + u_S < 2^{-b-1} < 2^{-b}$, completing part (a). Setting $b = q+1$ and applying $\mathcal{Q}_q$ (followed by $\max(0,\cdot)$ for $\hat{\rho}$, which cannot increase the error since $\rho \ge 0$) gives $|\hat{\mu} - c_{\max}^{-1}| < 2^{-q-2} + 2^{-q-1} < v$ and $|\hat{\rho} - \rho| < v$.

For a truncated folded tap $\sum_{s\in e+t_a\mathbb{Z},\,|s|\le P} g_a(s+\delta)$, there are at most $2P+1$ terms. Evaluating each unfolded sample with $b = q+g$ where $g = \lceil \lg(2P+2)\rceil + 12$, summing the residue class exactly at scale 2^{-S}, and applying a single complex nearest rounding $\mathcal{Q}_q$ certifies the stored tap within $(2P+1)2^{-q-g-4} + \frac{\sqrt{2}}{2}v < 0.71\,v < v$ of the truncated folded tap; adding the spatial tail bound $0.06\cdot 2^{-P} = 0.24\,v$ from the proof of Lemma 4.6(i) places it within $0.95\,v < v$ of the full folded tap $H_\delta^{(a)}(e)$.

For a unit phase $e = e^{\pi i\vartheta}$, fourth roots of unity ($2\vartheta \in \mathbb{Z}$) are returned exactly; otherwise evaluating (ii) at $b = q+5$ gives e_0 with $|e_0 - e| < 2^{-q-9} < v/16$, and setting $\hat{e} = Q_q((1-v/8)e_0)$ satisfies $|(1-v/8)e_0| \le (1-v/8)(1+v/16) < 1$, so $\hat{e} \in \mathbb{D}_q$ and $|\hat{e} - e| < (\sqrt{2} + \frac{3}{16} + v)v < 2v$ by (7). Multiplying any disk operand z ($|z| \le 1$) by a stored phase $\hat{e} \in \mathbb{D}_q$ and truncating via Q_q preserves the closed unit disk and introduces error below $(2+\sqrt{2})v < 4v$; iterating across d one-axis factors therefore produces a tensor chirp in $\mathbb{D}_q$ within $2v + 4(d-1)v \le 4dv$ of the exact chirp, completing part (b).

6. Bit complexity. Each series evaluation uses $N = O(b)$ arithmetic operations on integers of $O(b)$ bits (the erf numerators remain below $2^{b+8}\sqrt{b}\cdot 2^S$), taking $O(b^3)$ bit operations even under schoolbook arithmetic, while the exact rational reductions on $O(k + \lg P)$-bit numerators and denominators take $O((k+\lg P)^2)$ bit operations. Each primitive or sample in (a) therefore costs $O((b+k)^3)$ bit operations on a fixed number of tapes, and each folded tap in (b) (with $b = q + O(\lg P) = P + O(\lg P)$) costs $O(P(P+k)^3)$ bit operations, using at most $3S + O(k)$ fractional bits for exact temporary products prior to rounding. □

C Comparison with a Bluestein middle transform and additional parameter rows

The main text computes the bridge's middle tensor DFT directly over $\mathbb{C}$ via Walsh-layer positional permutations (Sections 5–6), achieving $\kappa = 2^{-46}$. An alternative middle transform keeps the synthetic-ring framework of [OAI26, §§6–7] and reduces the power-of-two tensor DFT to a normalized cyclic convolution over $\mathcal{R}_r = \mathbb{C}[y]/(y^r+1)$ via Bluestein's chirp identity [Blu70] and a last-axis twist. This appendix records that Bluestein route in self-contained form, derives its quadratic layout bottleneck $\kappa \lesssim (1-\tau)^2/(C_1\tau)$, and tabulates the secondary Walsh and Bluestein parameter rows verified by `tools/h5_rows_check.py`.

C.1 Synthetic transforms and normalized ring convolution

For a power of two $r \geq 2$ let $\mathcal{R}_r = \mathbb{C}[y]/(y^r + 1)$ with coefficient norm $\|\sum_{j<r} z_j y^j\|_\infty = \max_j |z_j|$. For $t \mid 2r$ put $\omega_t = y^{2r/t}$, $(F_t^{\mp} z)_j = t^{-1} \sum_{u<t} \omega_t^{\mp ju} z_u$ on $\mathcal{R}_r^t$, and let B_t move index j to its $(\lg t)$-bit reversal. For $\mathbf{t} = (t_1, \ldots, t_{d-1})$ with $t_i \in \{r/2, r\}$ and $M = \prod_{i<d} t_i$, put $F_{\mathbf{t}}^{\pm} = \bigotimes_{i<d} F_{t_i}^{\pm}$ and $B = \bigotimes_{i<d} B_{t_i}$. Both routes share the named-slot layout map L of Lemma 6.3, which coincides with that constructed in [OAI26, proof of Lemma 16, pp. 41–42]; the difference is that Lemma C.1 executes L by chunk exchanges (Lemma A.8) rather than Walsh-layer positional permutations.

Lemma C.1 (Synthetic transform interface)**.** *Fix the rational parameters of Proposition A.11 and its band constants $a_d \leq b_d$, $a_r \leq b_r$. Fix constants $C_\ell > 0$ and $C_w > 1$, and instantiate Proposition A.11 with $C_M = C_\ell + 1$ and $C_{\mathrm{desc}} = 32(C_\ell + C_w + 1)$, as in [OAI26, Lemma 16, §6.2]. There are a cutoff q_1, fixed-tape procedures and a time constant with the following property. Let $q \geq q_1$ and d, r satisfy the bands (53), with $d \geq 2$, $r = 2^\ell$, $2 \leq \ell < q$, $d\ell \leq C_\ell q$ and $1 \leq K = \lfloor d^c \rfloor \leq \ell - 1$; let $t_i \in \{2^{\ell-1}, 2^\ell\}$ for $1 \leq i < d$, $M = \prod_{i<d} t_i$, $R = rM$ and $q + 2 \leq w \leq C_w q$. Given the header $\Gamma(q)\Gamma(d)\Gamma(r)\Gamma(K)\Gamma(w)\Gamma(t_1)\cdots\Gamma(t_{d-1})$ and M contiguous disk-grid records in $\mathcal{R}_r$ at grid precision q and width w, the forward procedure has exact counterpart $LBF_{\mathbf{t}}^-$ and the opposite procedure, which consumes that order, has exact counterpart $F_{\mathbf{t}}^+ B^{-1} L^{-1}$. Both return disk arrays on the q-grid in the same format, have error less than $\sqrt{2}\,\ell\, 2^{-q}$, and cost*

$$O(V[dK + qK^{\tau-1} + \ell d^{\lambda'}]), \qquad V = 2Rw.$$

Proof. With $b_0 = (\ell - 1) \bmod K$ and $J_{\mathrm{blk}} = (\ell - 1 - b_0)/K$ as in Lemma 6.3, moving the at most $(d-1)K$ prefix slots $\mathcal{P}$ to the front by single-bit moves (Lemma A.4(2)) costs $O(VdK)$. Rearranging the remaining $N = (d-1)J_{\mathrm{blk}} \leq C_\ell q/K$ consecutive K-bit chunks into chunk-level-major order $\mathcal{C}$ factors into at most $N - 1$ transpositions, each an equal-width chunk exchange of cost $O(VK^\tau)$ by Lemma A.8, executing L (and in reverse L^{-1}) in $O(V[dK + qK^{\tau-1}])$.

In layout L, run the normalized decimation-in-frequency recursion of [OAI26, §6.1, (27)] on all $d-1$ record axes in parallel at named slots $v = \ell - 1, \ldots, 0$: at slot v, each axis with $\lg t_i > v$ applies H_0 on slot v followed on branch $b_i = 1$ by $\omega_{2^{v+1}}^{-k_i} = y^{-(2r/2^{v+1})k_i}$, multiplying across axes to one recordwise monomial y^E (a signed cyclic coefficient shift of cost $O(rq)$ per record by Lemma A.5). The at most $1 + b_0 \leq K$ prefix rounds execute their at most $d - 1$ factors of H_0 exactly with $q + d$ fractional bits before one truncation Q_q in total cost $O(VdK)$; the at most ℓ chunk rounds ($b_0 \leq v \leq \ell - 2$) each invoke Proposition A.11 on $d-1$ consecutive K-bit chunks in total cost $O(V\ell d^{\lambda'})$. Since monomials are isometries commuting with Q_q and normalized butterfly groups are contractions, the per-round truncation errors $< \sqrt{2} \cdot 2^{-q}$ add to $< \sqrt{2}\,\ell\, 2^{-q}$. Reversing the schedule with H_0 in place of $2H_0$ and inverting each monomial before its butterfly gives $M^{-1}(LBF_{\mathbf{t}}^-)^{-1} = F_{\mathbf{t}}^+ B^{-1} L^{-1}$ because $F_{\mathbf{t}}^+ F_{\mathbf{t}}^- = M^{-1} I$ ([OAI26, Lemma 15, §6.1]). □

Lemma C.2 (Packed polynomial products)**.** *Let $r = 2^\ell < 2^q$ with $q \geq 2$, and let $f, g \in \mathcal{R}_r$ be disk-grid polynomials at precision q. For any independent integer multiplier of cost $\mathsf{M}(N)$ on operands of at most N bits, one can return $Q_q(fg/r)$ in time $O(\mathsf{M}(4rq + 2) + rq)$, using four signed integer products of at most $4rq + 2$ bits. The output is a disk-grid polynomial with error less than $\sqrt{2} \cdot 2^{-q}$ from fg/r.*

Proof. Pack each of the four signed real/imaginary numerator lists (entries in $[-2^q, 2^q]$) in base $B_0 = 2^{4q}$. Every ordinary convolution coefficient has modulus at most $r2^{2q} < 2^{3q} < B_0/2$, so blockwise balanced-base decoding in $O(rq)$ bit operations recovers the exact products, subtraction at lag r reduces modulo $y^r + 1$, and shifting by r and applying Q_q gives a disk-grid output within $\sqrt{2} \cdot 2^{-q}$ of fg/r (since $\|fg/r\|_\infty \leq 1$). □

Proposition C.3 (Normalized ring convolution)**.** *Under the hypotheses of Lemma C.1, the cyclic convolution in the first $d-1$ axes, normalized as $\mathcal{M}_r(f,g) = (f * g)/(rM)$, of two disk-grid arrays has an approximation with error less than $\sqrt{2}\,M(3\ell+1)2^{-q}$, computed in time*

$$O(V[dK + qK^{\tau-1} + \ell d^{\lambda'}] + M\,\mathsf{M}(O(rq))),$$

returned in the original array and coefficient order at component width $w + \lg M$. If one exact input has norm at most $\varrho < 1$ and $\sqrt{2}\,M(3\ell+1)2^{-q} \le 1-\varrho$, the output is a disk array.

Proof. Since $F_{\mathbf{t}}^{+}F_{\mathbf{t}}^{-} = M^{-1}I$ and $F_{\mathbf{t}}^{-}(f*g) = M(F_{\mathbf{t}}^{-}f)(F_{\mathbf{t}}^{-}g)$ ([OAI26, Lemma 15, §6.1]), applying the forward procedure of Lemma C.1 to f and g, multiplying corresponding records in the common retained order LB by Lemma C.2, applying the opposite procedure, and multiplying exactly by M (appending $\lg M$ low zero bits) cancels LB and has exact counterpart $(f*g)/(rM)$. Because $(u,v) \mapsto uv/r$ is a bilinear contraction on disk arrays, the two forward transforms, the record product and the opposite transform accumulate error $< (3\sqrt{2}\,\ell + \sqrt{2})2^{-q}$ before the final multiplication by M. □

C.2 The power-of-two Bluestein adapter

Lemma C.4 (Power-of-two Bluestein adapter)**.** *Let every t_i $(1 \le i \le d)$ be even and put $a_{\mathbf{j}} = \exp(\pi i \sum_{i=1}^{d} j_i^2/t_i)$. Then a is periodic in every axis and*

$$F_{\mathbf{t}}^{+}z = a \odot (\bar{a} * (a \odot z))/R, \tag{57}$$

where $*$ is cyclic tensor convolution over $\prod_{i=1}^{d}\mathbb{Z}/t_i\mathbb{Z}$. The twist $x_d \mapsto \zeta y$, $\zeta = e^{\pi i/r}$, of the last axis maps cyclic convolution along that axis to multiplication in $\mathcal{R}_r = \mathbb{C}[y]/(y^r+1)$, and untwisting recovers the cyclic coefficients. Assume that (q,d,r,K), $t_1,\dots,t_{d-1}$ and $w = q+2$ satisfy the hypotheses of Lemma C.1 and the disk margin $(\sqrt{2}\,M(3\ell+1) + 12d + 16)\,v \le \frac{7}{8}$, where $v = 2^{-q}$ and $M = R/r$. Evaluating (57) via Proposition C.3 satisfies the bridge middle-transform contract (Section 4) with

$$\mathrm{E}_{\mathrm{mid}} = \sqrt{2}\,M(3\ell+1) + 12d + 16, \qquad \delta_{\mathrm{mid}} = \lg M.$$

Proof. Since t_i is even, $a_{\mathbf{j}+t_i\mathbf{e}_i}/a_{\mathbf{j}} = e^{2\pi i j_i + \pi i t_i} = 1$, and $R^{-1}a_{\mathbf{j}}\overline{a_{\mathbf{j}-\mathbf{k}}}a_{\mathbf{k}} = R^{-1}e^{2\pi i\sum_i j_i k_i/t_i}$ proves (57). For $k'+l' = j + hr$, $\zeta^{k'+l'}(-1)^h = \zeta^j$, verifying the negacyclic twist. Forming a from its d one-dimensional phase tables (certified to $2v$ in the disk) with componentwise truncations gives $\|\hat{a}-a\|_\infty \le 4dv$ and $\|\hat{a}\|_\infty \le 1$. On bridge-normalized input ($\|z\|_\infty \le 1/8$), the twisted data operand has error at most $(4d+6)v$ and the twisted kernel $\bar{a}$ at most $(4d+4)v$. By bilinear contraction of $\mathcal{M}_r$ (with norms $\le 1/8$ and ≤ 1) and Proposition C.3, the convolution output at width $w + \lg M$ has error $< (\sqrt{2}M(3\ell+1) + 8d + 10)v \le \frac{7}{8}$ from a vector of norm $\le 1/8$, hence lies in the disk; its redundant sign bits are removed, and untwisting plus the output chirp add $< (4d+6)v$, totaling $\mathrm{E}_{\mathrm{mid}}\,v$. Finally, $7\sqrt{2}Mv \le \frac{7}{8}$ implies $\lg M \le q-4 \le P$. □

Remark C.5 (Automatic disk margin under the bridge schedule). Since $M \le R$, $\ell \le k$, $d \le k$ and $R \ge 4$, we have $\sqrt{2}\,M(3\ell+1) + 12d + 16 \le 16R(k+1)$. Under the bridge precision schedule of Theorem 4.7, $16R(k+1)v < \frac{1}{100} \le \frac{7}{8}$, so the disk margin of Lemma C.4 holds automatically along the root family. (In [OAI26, Lemma 21], the chirp is written with the opposite sign $e^{-\pi i\sum_i s_i j_i^2/t_i}$, placing conjugation bars on the outer factors instead of the convolution kernel; for $s_i = 1$ both forms coincide.)

C.3 Bluestein cost table and the quadratic layout bottleneck

Combining Theorem 4.7, Lemma 4.8, Proposition C.3 and Lemma C.4 yields Table 9 for one bridge call on the Bluestein route, with total cost

$$\mathsf{T}_{\mathrm{B}}(R,P) = O\big(RP\,[dK + PK^{\tau-1} + \ell d^{\lambda'} + d(1+\ell^{\tau}) + d(\mathsf{m}(P) + \mathsf{m}(P^2)) + \mathsf{m}(rP)]\big). \tag{58}$$

Notice that the packed ring products of Lemma C.2 have $\Theta(rq) = \Theta(rP)$ bits, where $\lg(rP) = \Theta(P^{1-\varepsilon})$, so the row $\mathsf{m}(rP)$ achieves gap ε (up to logarithms) only when the inner multiplier satisfies $\mathsf{m}(N) = O(\lg N\,(\lg\lg N)^{O(1)})$ (for instance Schönhage–Strassen [SS71]). By contrast, the Walsh route needs only $\mathsf{m}(N) = (\lg N)^{O(1)}$ on $O(P^2)$-bit integers.

Table 9: Cost of one bridge call on the Bluestein route, divided by $V = RP$, for $d = \Theta(P^{\varepsilon})$, $\ell = \Theta(P^{1-\varepsilon})$ and $K = \Theta(P^{\varepsilon c})$ ($q = \Theta(P)$).

operation	normalized cost	gap below P^1
prefix-slot moves and individual rounds	dK	$1-\varepsilon(1+c)$
synthetic transform layout (chunk exchanges)	$PK^{\tau-1}$	$\varepsilon c(1-\tau)$
simultaneous butterfly rounds	$\ell d^{\lambda'}$	$\varepsilon(1-\lambda')$
axis exposures and restorations	$d(1+\ell^{\tau})$	$(1-\varepsilon)(1-\tau)$
spreading and cyclic solves	$d(\mathsf{m}(P)+\mathsf{m}(P^2))$	$1-\varepsilon$, up to logarithms
packed polynomial products	$\mathsf{m}(rP)$	ε, up to logarithms
chirps, twists, and scalar products	$d\,\mathsf{m}(P)$	$1-\varepsilon$, up to logarithms

Why Bluestein is quadratically capped. On the Bluestein route, executing the layout permutation L by chunk exchanges introduces the gap $g_{\mathrm{chunk}} = \varepsilon c(1-\tau)$. Within Proposition A.11, the internal basis-change constraint $\tau(1+c/\beta) < \lambda < 1$ forces $c < \beta(1-\tau)/\tau$, while the guard constraint $\varepsilon C_1 < 1$ forces $\varepsilon < 1/C_1$. Multiplying these three factors yields

$$\kappa < g_{\mathrm{chunk}} = \varepsilon c(1-\tau) < \frac{\beta(1-\tau)^2}{C_1\tau} < \frac{(1-\tau)^2}{C_1\tau}, \tag{59}$$

which is *quadratic* in the bit-network deficit $1-\tau$: even with the certified deficit $1-\tau = 2^{-40}$, this caps κ below $(1-\tau)^2/(20\tau) \approx 2^{-84.3}$ at $C_1 = 20$ and below $(1-\tau)^2/(14\tau) \approx 2^{-83.8}$ at $C_1 = 14$. On the Walsh route, Theorem 5.8 executes both L and $U = (LB)^{-1}$ in $O(V\ell d^{\lambda'})$ steps, eliminating g_{chunk} completely; the binding constraint becomes $g_{\mathrm{lay}} = \varepsilon(1-\lambda') < (1-\sigma)/C_1$, which is *linear* in the complex-network deficit $1-\sigma$.

C.4 Additional parameter rows and full exponent comparison

Let $\xi = 2^{-42}$, $\zeta = 2^{-40}$ and $u = 2^{-51}$. Table 10 lists all seven verified parameter rows across the Walsh and Bluestein routes, alongside the front-end/Bluestein baseline of [OAI26, (38)].

Verification of the secondary rows.

- *Walsh with printed structural constants ($\kappa = 2^{-56}$).* With $\beta = \frac{1}{2}$, $\tau = \sigma = 1-2u$ and $C_1 = 20$, we have $\sigma + \beta(1-\sigma) = 1-u$ and $\tau(1+c/\beta) = (1-2u)(1+u/2) = 1 - \frac{3}{2}u - u^2$, so the layer margins are $\lambda - \tau = \frac{127}{128}u$, $\lambda - \tau(1+c/\beta) = \frac{63}{128}u + u^2$, $\lambda' - (1-u) = \frac{u}{64}$, $\lambda' - \lambda = \frac{3u}{128}$, and $1 - \varepsilon C_1 = \frac{1}{21}$, with binding gap $g_{\mathrm{lay}} = \frac{3u}{64} = 3\cdot 2^{-57}$ against $\kappa = 2^{-56}$ (and $g_{\mathrm{axis}} = \frac{40}{21}u$).

Table 10: Full comparison of parameter rows ($\xi = 2^{-42}$, $\zeta = 2^{-40}$, $u = 2^{-51}$). "Printed" rows keep the structural constants $\beta = \frac{1}{2}$, $\tau = \sigma = 1 - 2^{-50}$, $C_1 = 20$ of [OAI26, Prop. 14]; "exposed" rows use Proposition A.11 and Proposition A.2. The last row is the main theorem (Section 7, formalized in Lean 4).

Route	Layer	C_1	β	ε	c	(τ, σ)	(λ, λ')	Binding gap $g_{\min}$	κ (slack)
Bluestein	printed	20	$\frac{1}{2}$	2^{-75}	2^{-56}	$(1-2^{-50}, 1-2^{-50})$	$(1-2^{-52}, 1-2^{-54})$	$g_{\text{chunk}} = 2^{-181}$	2^{-182} (2^{-182})
Bluestein	printed	20	$\frac{1}{2}$	$\frac{1}{32}$	2^{-52}	$(1-2^{-50}, 1-2^{-50})$	$(1-2^{-51}, 1-2^{-52})$	$g_{\text{chunk}} = 2^{-107}$	2^{-108} (2^{-108})
Bluestein	exposed	20	$\frac{15}{16}$	$\frac{1}{21}$	$\frac{7}{8}\zeta$	$(1-\zeta, 1-\frac{\zeta}{4})$	$(1-\frac{\zeta}{32}, 1-\frac{\zeta}{128})$	$g_{\text{chunk}} = \frac{\zeta^2}{24}$	2^{-85} $(\frac{\zeta^2}{96})$
Bluestein	exposed	14	$\frac{15}{16}$	$\frac{9}{128}$	$\frac{29}{32}\zeta$	$(1-\zeta, 1-\frac{\zeta}{4})$	$(1-\frac{\zeta}{32}, 1-\frac{\zeta}{128})$	$g_{\text{chunk}} = \frac{261}{4096}\zeta^2$	2^{-84} $(5{\cdot}2^{-92})$
Walsh	printed	20	$\frac{1}{2}$	$\frac{1}{21}$	$\frac{u}{4}$	$(1-2u, 1-2u)$	$(1-\frac{129}{128}u, 1-\frac{63}{64}u)$	$g_{\text{lay}} = \frac{3u}{64}$	2^{-56} (2^{-57})
Walsh	exposed	20	$\frac{1}{8}$	$\frac{1}{21}$	$\frac{\xi}{4}$	$(1-4\xi, 1-\xi)$	$(1-\frac{7}{8}\xi, 1-\frac{3}{4}\xi)$	$g_{\text{lay}} = \frac{\xi}{28}$	2^{-47} $(\frac{\xi}{224})$
Walsh	exposed	14	$\frac{1}{64}$	$\frac{1}{15}$	$\frac{\xi}{256}$	$(1-4\xi, 1-\xi)$	$(1-\frac{63}{64}\xi, 1-\frac{31}{32}\xi)$	$g_{\text{lay}} = \frac{31\xi}{480}$	2^{-46} $(\frac{\xi}{480})$

Moreover, *any* admissible choice of free parameters with these fixed structural constants satisfies $\lambda' > \sigma + \beta(1-\sigma) = 1 - u$ and $\varepsilon < 1/C_1 = 1/20$, so $\kappa < g_{\text{lay}} < \frac{u}{20} = \frac{8}{5}\,2^{-56} < 2^{-55}$; thus 2^{-56} is the largest exponent of the form 2^{-j} that our Walsh cost analysis admits with the printed structural constants of [OAI26, Prop. 14].

- *Walsh with $C_1 = 20$ ($\kappa = 2^{-47}$).* Using $\tau = 1 - 4\xi$ and $\sigma = 1 - \xi$ from Proposition A.2, $\tau(1 + c/\beta) = (1-4\xi)(1+2\xi) = 1 - 2\xi - 8\xi^2$, giving margins $\lambda - \sigma = \frac{\xi}{8}$, $\lambda - \tau(1+c/\beta) = \frac{9}{8}\xi + 8\xi^2$, $\lambda' - \lambda = \frac{\xi}{8}$ (with $\sigma + \beta(1-\sigma) = \lambda$), $1 - \varepsilon C_1 = \frac{1}{21}$, and binding gap $g_{\text{lay}} = \frac{\xi}{28}$ against $\kappa = \frac{\xi}{32} = 2^{-47}$.
- *Bluestein rows ($\kappa = 2^{-182}, 2^{-108}, 2^{-85}, 2^{-84}$).* At the parameters of [OAI26, (38)], the bridge with Bluestein middle reproduces $\kappa = 2^{-182}$ ($g_{\text{chunk}} = 2^{-181}$, showing that replacing the CRT front end alone does not alter κ). Retuning (ε, c) under the printed structural constants gives $\kappa = 2^{-108}$ ($g_{\text{chunk}} = 2^{-107}$), using the $O(d)$ axis exposures of Lemma 4.8 in place of the $O(d^2)$ adjacent swaps of [OAI26, §9]. Exposing (β, τ, σ) via Propositions A.11 and A.2 gives $\kappa = 2^{-85}$ at $C_1 = 20$ ($\tau(1 + c/\beta) = (1-\zeta)(1 + \frac{14}{15}\zeta) < 1 - \frac{\zeta}{15} < \lambda$, $g_{\text{chunk}} = \frac{\zeta^2}{24}$) and $\kappa = 2^{-84}$ at $C_1 = 14$ ($\lambda - \tau(1 + c/\beta) = \frac{\zeta}{480} + \frac{29}{30}\zeta^2 > 0$, $1 - 14\varepsilon = \frac{1}{64}$, $g_{\text{chunk}} = \frac{261}{4096}\zeta^2$).

At equal guard exponents, removing the chunk-layout bottleneck via Walsh-layer permutations replaces the quadratic cap (59) in $1 - \tau$ by a binding gap linear in the complex-network deficit $1 - \sigma$. This raises κ from 2^{-85} to 2^{-47} at $C_1 = 20$, from 2^{-84} to 2^{-46} at $C_1 = 14$, and from 2^{-108} to 2^{-56} at the printed structural constants.

D Statements and notation from [OAI26]

This appendix records the symbol translation between [OAI26] (version of 23 September 2026) and the present paper, and restates the results of [OAI26] cited as black boxes in Section 3.4 (Facts 3.3–3.4) together with the printed form of [OAI26, Prop. 14] for comparison with Proposition A.11.

Notation map. Table 11 lists every symbol of [OAI26] that is renamed here to avoid collision with the bridge parameters. Headers that concatenate positive integer parameters use the self-delimiting code [OAI26, (1)]:

$$b_v = \lfloor \log_2 v \rfloor + 1, \qquad \Gamma(v) = 0^{\,b_v - 1}\,\mathrm{bin}_{b_v}(v), \qquad |\Gamma(v)| = 2b_v - 1 \qquad (v \geq 1). \tag{60}$$

Table 11: Symbol correspondence between [OAI26] and this paper.

[OAI26]	This paper	Meaning
P (§2, Lemmas 3, 12, Prop. 14)	A	collapsed prefix address range (P is our bridge working precision)
q (§2, Lemmas 3–4; §4, Lemmas 10–12)	q_{rad}	fixed integer or odd prime radix (q is our layer grid precision)
p (§§5–9, Prop. 14)	q	dyadic grid precision of layers and middle transforms ($q = P + 2$ in §4)
R (§2, Lemma 3; §5.2, Lemma 13)	R_{row}, R_{rec}	row-field range; record bit width ($R = t_1 \cdots t_d$ is our DFT length)
E, B, ν, k (§5.4)	E_{net}, B_{net}, ν_{net}, $j_{\max}$	complex-network guard constants, guard degree, and recursion depth ($k = \lg R$)
T, $L = \log_2 T$ (§§6–9)	R, $k = \lg R$	total power-of-two transform box size and its binary logarithm
ϵ, log	ε, ln	dimension exponent; natural logarithm ($\lg = \log_2$)

D.1 Machine model and elementary stream lemmas

In [OAI26, §2], a multitape Turing machine has a fixed finite alphabet and a fixed finite number of one-dimensional tapes; an array on $[a_1] \times \cdots \times [a_t]$ is stored in lexicographic order (first coordinate most significant) with implicit addresses, logical bit volume V, and move convention $(fA)(f(x)) = A(x)$. The following two lemmas of [OAI26, §2.1] constitute Fact 3.3.

Proposition D.1 ([OAI26, Lemma 3, §2.1])**.** *Fix a radix $q_{\mathrm{rad}} \geq 2$. On an array of volume V, each of the following takes $O(V)$ steps on a fixed number of one-dimensional tapes (including shape preparation, counter initialization/resets, and workspace cleanup): (1) any fixed permutation of the pair of digit values at two specified radix-q_{rad} positions; (2) moving one specified radix-q_{rad} digit to another position, preserving the order of intervening positions; (3) splitting on a specified radix-q_{rad} digit, applying a fixed pointwise map to the aligned streams, and merging; and (4) within each prefix of an array on $[A] \times [R_{\mathrm{row}}] \times [B]$, splitting whole rows cyclically into a fixed number W of streams and, when $W \mid R_{\mathrm{row}}$, merging them in any fixed order of the row classes.*

Proposition D.2 ([OAI26, Lemma 4, §2.1])**.** *Fix a prime q_{rad}, a fixed number of control fields, and finitely many rational coefficients with denominators prime to q_{rad} (and numerators prime to q_{rad} when used as units). Put $Q = q_{\mathrm{rad}}^b$ ($b \geq 1$), and let a target field $y \in [Q]$ follow its control fields, each control used in an offset having $O(b)$ digits, with arbitrary spectator fields and suffix records. An update $y \mapsto y + a(\mathit{controls}) \pmod{Q}$ costs $O(V)$ whenever the offset is computable in $b^{O(1)}$ time from the controls (in particular for fixed rational linear combinations modulo Q), and so does multiplication of y by any of the specified rational units or its inverse. Fixed-width signed fixed-point addition, subtraction, sign change, and shifts by fixed powers of two (with sufficient guard bits), as well as intra-record bit reversal, also have $O(V)$ bit cost.*

D.2 Finite-network certificates and printed butterfly layer

At $h = 100$, $v = \binom{h}{3} = 161\,700$, $N = v^3$, and $m = h^3 = 10^6$, [OAI26, §3.1] constructs a three-stage bit network over $\mathbb{F}_2$ (W_b wires, neighbor rule $|S \cap T| = 1$) and complex network over $\mathbb{Z}[i, 1/2]$ (W_c wires, distinct triples with $|S \cap T|$ even) on two data banks X_a, Y_a ($a \in \binom{[h]}{3}^3$) and per-invocation scratch wires. By [OAI26, Lemmas 5–6, §§3.1–3.3], the forward-inverse-forward schedule sends

$(X_a, Y_a) \mapsto (-Y_a, X_a)$, restores all scratch wires from arbitrary initial values, and attaches nested nondegenerate subspaces of $\mathcal{F} = F^{\otimes 3}$ ($\dim \mathcal{F} = m$) whose only decreasing edges lose dimension h on central wires ($L_b/N = 100/539$, $L_c/N = 101/539$). Here $F = \mathbb{Q}^h$ under $x^{\mathsf{T}}(I_h - J_h/9)y$ for the bit network (where I_h and J_h are the $h \times h$ identity and all-ones matrices), and $F = \mathbb{F}_2^h$ under the dot product for the complex network. The resulting certificates form Fact 3.4:

Proposition D.3 ([OAI26, Props. 7–8, §§3.4–3.5])**.** *(a)* Bit network ([OAI26, Prop. 7]). *Assigning orthogonal projections P_U at gates/sinks, $-P_{U_a}$ at sources X_a and 0 at other sources, the bit network uses only pointwise XOR gates on W_b roles, induces the role permutation ρ_b exchanging X_a and Y_a while fixing scratch roles, satisfies $M_{\mathrm{sink}(\rho_b(w))} - M_{\mathrm{source}(w)} = I_m$ for every input role w, and has total rational edge-rank sum $s_b = W_b m - N + 2L_b < W_b m$ with the exact integers of Definition A.1.*

(b) Complex network ([OAI26, Prop. 8]). *With $H_{\mathrm{u}} = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, $C = H_{\mathrm{u}} \operatorname{diag}(1, i) H_{\mathrm{u}} = aI + bX$ ($a = \frac{1+i}{2}$, $b = \frac{1-i}{2}$), $q_U(x) = \mathrm{wt}(P_U x) \bmod 4$ over $\mathbb{F}_2$ and $C_U = H_{\mathrm{u}}^{\otimes m} \operatorname{diag}_x(i^{q_U(x)}) H_{\mathrm{u}}^{\otimes m}$, each edge frame difference $C_V C_U^{-1}$ is a product of one translation kernel $aI + bX_v$ or its inverse per vector in an orthonormal basis of the edge residual, with total basis length $s_c = W_c m - 2N + 2L_c < W_c m$ as in Definition A.1. After source/sink diagonal sign corrections ($Z_{a,f}$ before X_a and $i^{27} f Z_{a,f}$ after Y_a on f columns) and undoing the signed bank exchange, telescoping yields $(C^{\otimes m})^{\otimes f}$ on every role, including scratch roles with arbitrary initial values.*

By [OAI26, (18), §5.1], $C^{\otimes D}$ converts to the normalized butterfly layer $H_0^{\otimes D}$ via

$$H_0^{\otimes D} = b^D \, \mathcal{S}_D \, C^{\otimes D} \, \mathcal{S}_D, \qquad (\mathcal{S}_D z)(x) = i^{x_1 + \cdots + x_D} z(x), \qquad b^D = (-i/2)^{\lfloor D/2 \rfloor} b^{D \bmod 2}. \tag{61}$$

Finally, for comparison with Proposition A.11, we record [OAI26, Prop. 14] as printed:

Proposition D.4 ([OAI26, Prop. 14, §5.4], printed constants)**.** *Proposition A.11 holds as stated in [OAI26, Prop. 14] at the fixed structural constants $\beta = \frac{1}{2}$, $\tau = \sigma = 1 - 2^{-50}$ and $C_1 = 20$ (with q written p and A written P there), for any fixed positive rationals $c, \varepsilon, \lambda, \lambda'$ satisfying (51)–(52) and $\varepsilon C_1 < 1$.*

E Reference implementation

To complement the written proofs and the Lean 4 formalization of Section 8, we wrote an executable Python 3.13 reference implementation of every algorithmic component in this paper. Its role is to make the data layouts, rounding rules, error budgets, and finite-network certificates concrete and testable at small sizes; it is not a fast multiplier and does not simulate multitape Turing-machine head steps; some address permutations inside the engines are realized as index maps, and built-in integer multiplication is used for inner products unless the hierarchy mode is selected.

Exact-arithmetic design and implemented components. No floating-point arithmetic occurs on any authoritative data path. Every intermediate and stored value is represented in exact integer or rational arithmetic, and every rounding step is an explicit integer operation. On the dyadic grids of the fractional-shift bridge (Section 4) this operation is rounding to nearest, whereas on the disk-valued records of the butterfly layers and middle transforms (Sections 5–6 and Appendix A) it is componentwise truncation toward zero. All transcendental tables (sinc-Gaussian spreading taps, cyclic deconvolution coefficients, and roots of unity) are computed from certified rational interval enclosures (with Ziv precision refinement), guaranteeing faithful rounding to the target grid.

The package implements the complete pipeline in modular form:

- The d-dimensional fractional-shift bridge with negative half-bin modulation, overlap-add spreading in axis order $d, \dots, 2$, chronological cyclic deconvolution $B_2^{-1}, \dots, B_d^{-1}$, and two-call root packing and recovery.
- The direct complex tensor FFT of Section 6, including the three-layer selective H_0 gadget, slice-batched CNOT flips, two-involution positional permutations (Section 5), and combined per-record twiddles, as well as the Bluestein middle transform and normalized $\mathcal{R}_r$-convolution of Appendix C.
- The recursive power-width and arbitrary-width chunk interchanges, the packed selected-bit addition with its eight-shift program and deterministic radix-sort repair on the exceptional set, and the recursive $C^{\otimes e}$ phase-frame engine of Appendix A.
- A non-recursive inner-multiplier hierarchy so that end-to-end runs can route every packed and word-scale integer product above 64 bits through the internal multiplier rather than built-in big-integer multiplication.

Test-scale execution and asymptotic caveat. As noted in Section 9, the sub-$n \log n$ regime activates only at astronomically large input lengths ($d \geq 2$ already requires $k \geq 2^{15}$ at $\varepsilon = 1/15$, and the guard allocation $\Delta = \lceil C_0 d^{14} \rceil$ involves $C_0 \approx 2.27 \times 10^{139}$), while the $h = 100$ triple networks have $m = 10^6$ and $W_c \approx 1.87 \times 10^{21}$ roles. Consequently, the unforced theorem schedule falls back to a two-dimensional $O(N \log N)$ fractional-shift multiplier with negative half-bin carrier (the $d = 2$ specialization of Section 4; cf. [MS] for the real-kernel two-dimensional multiplier) at every feasible input size. To exercise the sub-$n \log n$ machinery end to end, test policies force small dimensions and transform shapes ($d \in \{2, \dots, 10\}$, lengths up to $R = 4096$, chunk widths $K \in \{1, 2\}$). They then run the recursive interchange and butterfly-layer engines on small test-scale network fixtures ($W = 3$, $m \in \{2, 3\}$) with guards compiled from the active fixture's depth recurrence. These small fixtures satisfy all structural, scratch-restoration, frame, and endpoint identities of Definition A.1, exercising every recursive, basis-change, and exceptional-repair path. They do, however, have $s > Wm$ and therefore no rank saving.

What the test suites verify.

(1) *Bit-exact products and component equivalence.* Across more than 2,000 unit and integration tests (and, separately, one full-cascade hierarchy run on each of the Walsh and Bluestein middle transforms), every integer product returned by the multiplier is verified bit for bit against Python's independent integer product. Component suites likewise verify bit-identical agreement between the network-driven interchange and butterfly-layer engines and direct dense/reference operators, and check runtime invariants (exact zero-bit halvings, guarded-width bounds, disk containment, scratch restoration, and hole-key bijection in the sorting repair).

(2) *Measured errors against proved bounds.* Across 71 transform configurations spanning dimensions $d \in \{2, \dots, 5\}$ and lengths up to $R = 1024$, the measured final rounded error is at most $2^{-2.51} \cdot 2^{-p}$ (within the final rounding term $(\sqrt{2}/8)\, 2^{-p}$ of Theorem 4.7). On the 15 non-leaf runs of the two-dimensional fallback path (17 configurations in total, including 2 base-FFT leaves), pre-rounding errors lie at least 12.9 bits below the nominal node certificate. That certificate is nominal because the fallback rounds twist roots and leaf twiddles to nearest rather than into the unit disk. Pre-rounding errors lie at least 21.5 bits below the proved analytic budget on the d-dimensional bridge with a direct reference middle transform (24 configurations), and at

least 23.9 bits below on the Walsh and Bluestein middle-transform routes (30 configurations). Direct tests of the Walsh middle transform against multiprecision reference DFTs also confirm its $12\ell \cdot 2^{-q}$ bound (Theorem 6.5).

(3) *Exact symbolic network certificates.* For the $h = 100$ triple networks of [OAI26, §3], a symbolic certifier exploits $(S_h)^3$ orbit symmetry and exact rational cell calculus in exact arithmetic without constructing $10^6 \times 10^6$ matrices. In about two seconds per network, it verifies the closed-form role counts W_b, W_c, edge-rank sums s_b, s_c, endpoint identities, guard constants ($\nu_{\text{net}} = 11$ and the 140-digit integer C_0), and relative deficits η_b, η_c. An exact rational interval series for $\ln m$ produces the width-2^{-80} enclosure rounding outward to $13.815510557964 \le \ln 10^6 \le 13.815510557965$ (Table 8 and Remark A.3), independently confirming the certified exponents $\tau = 1 - 2^{-40}$ and $\sigma = 1 - 2^{-42}$ of Proposition A.2. (For the bit network's odd prime radix q_{rad}, whose existence follows by finite avoidance in Lemma A.6, a symbolic grammar certificate and brute-force elimination over all 96,172 ordered one-factor primitive matrices at $h \in \{7, 8, 10, 11, 12\}$ find denominator and pivot-numerator prime support $\{2, 3\}$. This provides computational evidence for $q_{\text{rad}} = 5$; no theorem of this paper relies on this value.)

Availability. The exact rational parameter-verification script `tools/h5_rows_check.py` is included in the source bundle of this paper. The companion software artifact [Code26] comprises the full Python 3.13 exact-arithmetic reference package described in this appendix, while the separate directory [CGMP26] (`companion/impl/`) contains the standalone C/GMP/MPFR implementation (`intmul_fastdft.c`), benchmark runner, and raw logs (`bench/BENCHMARK.md`, `bench/logs/`) of the two-dimensional $O(n \log n)$ multiplier of [MS] (Sections 1.4 and 9.2). Together with the Lean 4 formalization [Lean26] (Section 8), both software directories are available in our github repository.